

SCHEME AND SYLLABUS

Bachelor of Engineering

(ELECTRONICS AND COMMUNICATION ENGINEERING)

2023-2027



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)



THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

PATIALA, PUNJAB, INDIA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

COURSE SCHEME AND SYLLABUS

FOR

**B.E. (ELECTRONICS AND
COMMUNICATION ENGINEERING)**



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
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2023

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SEMESTER-I

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UPH013	PHYSICS	BSC	3	1	2	4.5	6
2.	UES101	ENGINEERING DRAWING	ESC	2	4	0	4.0	6
3.	UHU003	PROFESSIONAL COMMUNICATION	HSS	2	0	2	3.0	4
4.	UES102	MANUFACTURING PROCESSES	ESC	2	0	2	3.0	4
5.	UMA010	MATHEMATICS-I	BSC	3	1	0	3.5	4
		TOTAL					18.0	24

SEMESTER-II

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UCB009	CHEMISTRY	BSC	3	0	2	4.0	5
2.	UES103	PROGRAMMING FOR PROBLEM SOLVING	ESC	3	0	2	4.0	5
3.	UES013	ELECTRICAL & ELECTRONICS ENGINEERING	ESC	3	1	2	4.5	6
4.	UEN008	ENERGY AND ENVIRONMENT	BSC	2	0	0	2.0	2
5.	UMA004	MATHEMATICS-II	BSC	3	1	0	3.5	4
		TOTAL					18.0	22

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SEMESTER-III

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UES012	ENGINEERING MATERIALS	BSC	3	1	2	4.5	6
2.	UMA035	OPTIMIZATION TECHNIQUES	BSC	3	0	2	4.0	5
3.	UEC311	CIRCUIT ANALYSIS AND SYNTHESIS	PCC	2	1	0	2.5	3
4.	UEC612	DIGITAL SYSTEM DESIGN	PCC	3	1	2	4.5	6
5.	UEC513	EMBEDDED SYSTEMS	PCC	3	0	2	4.0	5
6.	UTD003	APTITUDE SKILLS BUILDING	HSS	2	0	0	2.0	2
7.	UTA016	ENGINEERING DESIGN PROJECT – I (2 SELF EFFORT HOURS)	ESC	1	0	2	3.0	3
		TOTAL					24.5	30

SEMESTER-IV

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UNC402	DATA STRUCTURES	PCC	3	0	2	4.0	5
2.	UMA033	NUMERICAL AND STATISTICAL METHODS	BSC	3	0	2	4.0	5
3.	UEC307	ELECTROMAGNETIC FIELD THEORY AND TRANSMISSION LINES	PCC	3	1	0	3.5	4
4.	UEC404	SIGNALS AND SYSTEMS	PCC	3	1	2	4.5	6
5.	UEC310	INFORMATION AND COMMUNICATION THEORY	PCC	3	1	0	3.5	4
6.	UEC301	ANALOG ELECTRONIC CIRCUITS	PCC	3	1	2	4.5	6
7.	UTA024	ENGINEERING DESIGN PROJECT-II	ESC	1	0	2*	3.0	5
8.	UHU050	EVOLUTIONARY PSYCHOLOGY (1 SELF EFFORT HOUR)	HSS	1*	0	0	1.0	0.5
		TOTAL					28.0	35.5

(*Alternate week & 2 hours as self-effort hours)

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SEMESTER-V

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UEC519	ANALOG and DIGITAL COMMUNICATION	PCC	3	1	2	4.5	6
2.	UEC502	DIGITAL SIGNAL PROCESSING	PCC	3	1	2	4.5	6
3.	UEC512	LINEAR INTEGRATED CIRCUITS AND APPLICATIONS	PCC	3	0	2	4.0	5
4.	UEC610	COMPUTER ARCHITECTURE	PCC	3	0	0	3.0	3
5.	UEC750	MOS CIRCUIT DESIGN	PCC	3	1	2	4.5	6
6.	UTA025	INNOVATION & ENTREPRENEURSHIP	HSS	1	0	2*	3.0	2
7.		ELECTIVE-I	PEC	-	-	-	3.0	3/4
		TOTAL					26.5	31/32

SEMESTER-VI

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UEC867	MODERN CONTROL THEORY	PCC	2	1	0	2.5	3
2.	UEC640	IOT BASED SYSTEMS	PCC	2	0	2	3.0	4
3.	UNC504	ARTIFICIAL INTELLIGENCE	ESC	3	0	2	4.0	5
4.	UHU005	HUMANITIES FOR ENGINEERS	HSS	2	0	2	3.0	4
5.		ELECTIVE - II	PEC	-	-	-	3.0	3/4
6.		ELECTIVE-III	PEC	-	-	-	3.0	3/4
7.	UEC797	CAPSTONE PROJECT (STARTS)	PRJ	1*	0	2	-	3
		TOTAL					18.5	25-27

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SEMESTER-VII

S. No.	Course Code	Course Name	CODE	L	T	P	Cr	Contact Hours
1.	UEC719	5G WIRELESS COMMUNICATION	PCC	3	0	2	4	5
2.	UEC720	ANTENNA AND WAVE PROPAGATION	PCC	2	0	2	3	4
3.		ELECTIVE - IV	PEC	-	-	-	3.0	3/4
4.		GENERIC ELECTIVE	OEC	2	0	0	2.0	2
5.	UEC797	CAPSTONE PROJECT	PRJ	0	0	2	8.0	2
		TOTAL					20.0	16/17

SEMESTER-VIII

S. No.	Course Code	Course Name	CODE	L	T	P	Cr
1.	UEC898	PROJECT SEMESTER	PRJ	0	0	0	15
		Or					
1.	UNC801	DATA SCIENCE for ENGINEERS	PCC	3	0	0	3
2.	UEC714	FIBER OPTIC COMMUNICATION	PCC	2	0	2	3
3.	UEC897	PROJECT	PRJ	0	0	0	9
		TOTAL					15
		Or					
1.	UEC899	START-UP SEMESTER	PRJ	0	0	0	15

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ELECTIVES

ELECTIVE-I

Sr. No.	Course No.	TITLE	L	T	P	Cr	Contact Hours
1	UEC520	MACHINE LEARNING	2	0	2	3.0	4
2	UNC515	OPERATING SYSTEMS	2	0	2	3.0	4
3	UEC521	DESIGN USING SYSTEM VERILOG	2	0	2	3.0	4
4	UEC522	DATA COMMUNICATION	2	0	2	3.0	4

ELECTIVE-II

Sr. No.	Course No.	TITLE	L	T	P	Cr	Contact Hours
1	UEC754	IMAGE PROCESSING AND COMPUTER VISION	2	0	2	3.0	4
2	UEC832	RADAR AND REMOTE SENSING	3	0	0	3.0	3
3	UEC638	VLSI TESTING AND VERIFICATION	2	0	2	3.0	4
4	UEC707	NETWORK VIRTUALIZATION AND SOFTWARE DEFINED NETWORKING	2	0	2	3.0	4

ELECTIVE-III

Sr. No.	Course No.	TITLE	L	T	P	Cr	Contact Hours
1	UEC643	AUDIO AND VIDEO PROCESSING	2	0	2	3.0	4
2	UNC602	DATABASE MANAGEMENT SYSTEMS	2	0	2	3.0	4
3	UEC644	ANALOG IC DESIGN	2	0	2	3.0	4
4	UEC645	MICROWAVE ENGINEERING	2	0	2	3.0	4

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ELECTIVE-IV

Sr. No.	Course No.	TITLE	L	T	P	Cr	Contact Hours
1	UEC751	DSP PROCESSORS	3	0	0	3.0	3
2	UEC630	DEEP LEARNING AND APPLICATIONS	2	0	2	3.0	4
3	UEC752	IC FABRICATION TECHNOLOGY	2	0	2	3.0	4
4	UEC812	WIRELESS SENSOR NETWORKS	3	0	0	3.0	3

Name of Basket: **Smart Autonomous Electric vehicle System (SAEV)** Elective Focus Basket (EFB) in collaboration with Volvo

List of Courses to be offered (ECE)

S.No	Semester	Name of Course	Elective
1	5 th	UEC646: Network and Communications for Connected Vehicles	Elective-1
2	6 th	Intelligent Transportation Systems	Elective-2
3	6 th	Vehicle Diagnostics System/Model Based Mobility System Design.	Elective-3
4	7 th	Cyber Security for Mobility Systems	Elective-4

Industry collaborative Elective Basket

I. Financial Derivative (Future First Collaboration)
II. DevOps and Continuous Delivery (Xebia Collaboration)
III. Full stack Development (Xebia Collaboration)
IV. Conversational AI (Nvidia Collaboration)
V. Edge AI & Robotics (Nvidia Collaboration)

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	Financial Derivative	DevOps and Continuous Delivery	Full stack Development	Conversational AI	Edge AI and Robotics
Elective-I Credit: 3 Sem: V	UCS539: Finance, Accounting and Valuation	UCS537: Source Code Management	UCS542: UI & UX Specialist	UCS551: Conv AI: Accelerated Data Science	UCS668: Edge AI and Robotics: Data Centre Vision
Elective-II Credit: 3 Sem: VI	UCS675: Financial Markets and Portfolio Theory	UCS659: Build and Release Management	UCS677: Data Engineering	UCS664: Conv AI: Natural Language Processing	UCS547: Edge AI and Robotics: Accelerated Data Science
Elective-III Credit: 3 Sem: VI	UCS658: Derivatives Pricing, Trading and Strategies	UCS660: Continuous Integration and Continuous Deployment	UCS662: Test Automation	UCS749: Conv AI: Speech Processing & Synthesis	UCS671: Edge AI and Robotics: Embedded Vision
Elective-IV Credit: 3 Sem: VII	UMC743: Quantitative and Statistical Methods for Finance	UCS758: System Provisioning and Configuration Management	UCS745: Cloud & DevOps	UCS748: Conv AI: Generative AI	UCS760: Edge AI and Robotics: Reinforcement Learning & Conversational AI

GENERIC ELECTIVE

S No.	CODE	TITLE	L	T	P	Cr	Contact Hours
1	UHU016	INTRODUCTORY COURSE IN FRENCH	2	0	0	2.0	2
2	UCS002	INTRODUCTION TO CYBER SECURITY	2	0	0	2.0	2
3	UTD004	'CAMPUS 2 CORPORATE'	2	0	0	2.0	2
4	UEN006	TECHNOLOGIES FOR SUSTAINABLE DEVELOPMENT	2	0	0	2.0	2
5	UHU018	INTRODUCTION TO CORPORATE FINANCE	2	0	0	2.0	2
6	UHU017	INTRODUCTION TO COGNITIVE SCIENCE	2	0	0	2.0	2
7	UPH064	NANO SCIENCE AND NANO- MATERIALS	2	0	0	2.0	2
8	UMA069	GRAPH THEORY AND APPLICATIONS	2	0	0	2.0	2
9.	UMA070	ADVANCED NUMERICAL METHODS	2	0	0	2.0	2
10.	UBT510	BIOLOGY FOR ENGINEERS	2	0	0	2.0	2

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SEMESTER-I and II

UES101: Engineering Drawing				
	L	T	P	Cr
	2	4	0	4.0
Course Objective: This module is dedicated to graphics and includes two sections: 2D drafting and 3D modelling of solid objects. This course is aimed at making the student understand the concepts of projection systems, learn how to create projections of solid objects using first and third angle orthographic projection as well as isometric and auxiliary projection, concept of sectioning, to interpret the meaning and intent of toleranced dimensions and to create/edit drawings using drafting software. In addition, this course shall give an insight on the basic 3D modelling concepts like extrude, revolve, sweep, construction of complex solids.				
Syllabus Engineering Drawing Concepts 1. Introduction to Engineering Drawing 2. Projection systems: First angle and third angle projection system 3. Orthographic Projection: Points, Lines, Solid objects 4. Isometric Projections 5. Auxiliary Projections 6. Development of surfaces 7. Section of solids 8. Limits, fits and tolerances 2D Drafting 1. Management of screen menus commands 2. Creating basic drawing entities 3. Co-ordinate systems: Cartesian, polar and relative coordinates 4. Drawing limits, units of measurement and scale 5. Layering: organizing and maintaining the integrity of drawings 6. Design of prototype drawings as templates. 7. Editing/modifying drawing entities: selection of objects, object snap modes, editing commands, 8. Dimensioning: use of annotations, dimension types, properties and placement, adding text to drawing 3D Modelling 1. Management of screen menus commands 2. Introduction to basic 3D modelling commands such as extrude, revolve, sweep etc. 3. Creation of 2D drawings from a 3D model Micro Projects /Assignments: 1. Completing the views - Identification and drawing of missing lines and views in the projection of objects				

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2. Projects related to orthographic and isometric projections Using wax blocks/soap bars/any soft material to develop three dimensional object from given orthographic projections
3.
 - a. 3D modelling of complex machine components
 - b. Development of production drawings of individual components from the model

Course Learning Objectives (CLO)

The students will be able to:

1. creatively comprehend the geometrical details of common engineering objects
2. draw dimensioned orthographic and isometric projections of simple engineering objects
3. interpret the meaning and intent of limits, fits and tolerances in the drawing
4. create/edit the engineering drawings for simple engineering objects using 2D drafting software
5. create/edit 3D models of engineering components using 3D modelling software

Text Books

1. Jolhe, D.A., *Engineering Drawing*, Tata McGraw Hill, 2008
2. Davies, B. L., Yarwood, A., *Engineering Drawing and Computer Graphics*, Van Nostrand Reinhold (UK), 1986

Reference Books

1. Gill, P.S., *Geometrical Drawings*, S.K. Kataria & Sons, Delhi (2008).
2. Gill, P.S., *Machine Drawings*, S.K. Kataria & Sons, Delhi (2013).
3. Mohan, K.R., *Engineering Graphics*, Dhanpat Rai Publishing Company (P) Ltd, Delhi (2002).
4. French, T. E., Vierck, C. J. and Foster, R. J., *Fundamental of Engineering Drawing & Graphics Technology*, McGraw Hill Book Company, New Delhi (1986).
5. Rowan, J. and Sidwell, E. H., *Graphics for Engineers*, Edward Arnold, London (1968).
6. *Mastering AutoCAD 2021 and AutoCAD LT 2021*, Brian C. Benton, George Omura, Sybex - John Wiley and Sons, Indiana (2021).

Evaluation Scheme

Course Component	Weightage
AutoCAD tutorials/SolidWorks/Project work*	35
MST (1.5 hours-CAD based)**	20
EST (2 hours-CAD based)**	45

***Students are required to bring their personal computers for the tutorial work.**

***Availability of institute server resources for sharing the software licences with the student community.**

****Institute computational resources in collaboration with other academic units / departments for conducting the mid semester and end semester test.**

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UES102: Manufacturing Processes				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course introduces the basic concepts of manufacturing via machining, forming, casting and joining, enabling the students to develop a basic knowledge of the mechanics, operation and limitations of basic machining tools along with metrology and measurement of parts. The course also introduces the concept of smart manufacturing.				
Syllabus Machining Processes: Principles of metal cutting, Cutting tools, Cutting tool materials and applications, Geometry of single point cutting tool, Introduction to computerized numerical control (CNC) machines, G and M code programming for simple turning and milling operations, introduction of canned cycles. Metal Casting: Introduction & Principles of sand casting, Requisites of a sound casting, Permanent mold casting processes, casting defects Metal Forming: Hot & cold metal working, Forging, Rolling, Sheet Metal operations. Joining Processes: Method of joining, type of electric arc welding processes, Methods of shielding, Power source characteristics, Resistance welding, Soldering, Brazing. Smart Manufacturing: IoT and ML in manufacturing, Introduction to Additive Manufacturing, Robotics and Automation in manufacturing.				
Laboratory Work Relevant shop floor exercises involving practices in Sand casting, Machining, Welding, Sheet metal fabrication techniques, CNC turning and milling exercises, Experiments on basic engineering metrology and measurements to include measurements for circularity, ovality, linear dimensions, profiles, radius, angular measurements, measurement of threads, surface roughness. Basic knowledge and derivations related to above measurements, uncertainties, statistical approaches to estimate uncertainties, Line fitting, static and dynamic characteristics of instruments will be discussed in laboratory classes.				
Assignments: Assignments for this course will include the topics: Manufacturing of micro-chips used in IT and electronics industry and use of touch screens. Another assignment will be given to practice numerical exercises on topics listed in the syllabus. Case study related to smart manufacturing.				
Micro Project: Fabrication of multi-operational jobs using the above processes as per requirement by teams consisting of 4 -6 members. Quality check should be using the equipment available in metrology lab.				
Course Learning Objectives (CLO) The students will be able to: 1. Identify & analyse various machining processes/operations for manufacturing of				

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industrial components 2. apply the basic principle of bulk and sheet metal forming operations 3. apply the knowledge of metal casting for different requirements. 4. identify and analyse the requirements to for achieving a sound welded joint apply the concept of smart manufacturing														
Text Books 1. Degarmo, E. P., Kohser, Ronald A. and Black, J. T., <i>Materials and Processes in Manufacturing</i> , Prentice Hall of India (2008) 8 th ed. 2. Kalpakjian, S. and Schmid, S. R., <i>Manufacturing Processes for Engineering Materials</i> , Dorling Kingsley (2006) 4 th ed.														
Reference Books 1. Martin, S.I., Chapman, W.A.J., <i>Workshop Technology</i> , Vol.1 & II, Viva Books (2006) 4 th ed. 2. Zimmer, E.W. and Groover, M.P., <i>CAD/CAM - Computer Aided Designing and Manufacturing</i> , Dorling Kingsley (2008). 3. Pandey, P.C. and Shan, H. S., <i>Modern Machining Processes</i> , Tata McGraw Hill (2008). 4. Mishra, P. K., <i>Non-Conventional Machining</i> , Narosa Publications (2006). 5. Campbell, J.S., <i>Principles of Manufacturing, Materials and Processes</i> , Tata McGraw Hill Company (1999). 6. Lindberg, Roy A., <i>Processes and Materials of Manufacture</i> , Prentice Hall of India (2008) 4 th ed.														
Evaluation Scheme														
<table border="1"> <thead> <tr> <th>Sr. No.</th><th>Evaluation elements</th><th>Weightage (%)</th></tr> </thead> <tbody> <tr> <td>1</td><td>MST</td><td>25-30</td></tr> <tr> <td>2</td><td>EST</td><td>40-45</td></tr> <tr> <td>3</td><td>Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)</td><td>30</td></tr> </tbody> </table>			Sr. No.	Evaluation elements	Weightage (%)	1	MST	25-30	2	EST	40-45	3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30
Sr. No.	Evaluation elements	Weightage (%)												
1	MST	25-30												
2	EST	40-45												
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30												

UHU003: Professional Communication				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The course is designed to develop the interpersonal, written, and oral as well as the non-verbal communication skills of the students. The course begins by building up on the theoretical concepts and then practicing on the applicability of the various elements. Since the course has very high applicability content, the students are advised to practice in class as well as off class. A very high level of interaction is expected of the students in the class.				
Syllabus Fundamentals of Communication: Meaning, Types and Characteristics of communication, Applicability of Transactional Analysis and Johari Window for enhancing interpersonal communication skills. Seven Cs of Effective Communication, Barriers to Effective Communication. Effective Oral Communication: Understanding Principles of Oral communication, Formal and Informal Oral Communication, Oral Communication and Behavioral Patterns, Advantages and Disadvantages of Oral Communication. Effective Listening: Listening vs Hearing, Active Listening techniques, Barriers to Listening. Effective non-verbal communication: Meaning and Importance of Non-Verbal Communication, Different Types of Non-verbal Communication, Interpretation of Non-verbal Cues. Effective written Communication: Characteristics of Good Writing, Choice of Words, Sentence Construction, Paragraph development, Forms of writing. Business Communication: Technical Report Writing, Designing Resumes and Cover Letters for effective job application, E-mail writing and e-mail etiquette. Organizational Communication: Directional communication: Downward, Upward and Horizontal Communication, Grapevine. Reading: The following texts (one from each of the two categories listed below) are required to be read by the students in the semester: Category 1: Animal Farm by George Orwell, Lord of the Flies by William Golding, Life of Pi by Yann Martel Category 2: The Namesake by Jhumpa Lahiri, The God of Small Things by Arundhati Roy, Q&A by Vikas Swarup				
Laboratory Work 1. Needs-assessment of spoken and written communication with feedback. 2. Training for Group Discussions through simulations and role plays. 3. Technical report writing on survey-based projects. 4. Project-based team presentations.				

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Course Learning Objectives (CLO)

The students will be able to:

1. Apply communication concepts for effective interpersonal communication.
2. Speak assertively and effectively.
3. Interpret non-verbal cues in professional communication.
4. Write objectively, purposefully and effectively.
5. Design effective resumes and reports.

Text Books

1. Mukherjee H.S., *Business Communication: Connecting at Work*. Oxford University Press.(2013)
2. Lesikar R.V, and Flatley M.E., *Basic Business Communication Skills for empowering the internet generation*.(2006)
3. Raman, M.,and Singh ,P, *Business Communication* . Oxford . University Press (2008).

Reference Books

1. Riordan, G.R. *Technical Communication*. Cengage Learning India Private Ltd. (2012)
2. Butterfield, Jeff., *Soft Skills for everyone*, Cengage Learning New Delhi, (2013).
3. Robbins, S.P., & Hunsaker, P.L., *Training in Interpersonal Skills*, Prentice Hall of India, New Delhi, (2008).
4. Orwell, G., *Animal Farm*, Fingerprint Publishing, New Delhi, (2017).
5. Golding, W, *Lord of the Flies*, Faber & Faber; Export edition (1999)
6. Martel,Y., *Life of Pi*, RHC, New Delhi, (2012).
7. Lahiri,J., *The Namesake*, Harpercollins (2007)
8. Arundhati Roy,A., *The God of Small Things*, Penguin India, (2002).
9. Swarup,V., *Q&A*, Black Swan,(2009).

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UES103: Programming for Problem Solving				
	L	T	P	Cr
	3	0	2	4.0
Course Objectives: This course is designed to solve and explore the problems using the art of computer programming with the help of C Language. Students will be able to apply these problem solving concepts in real life applications.				
Syllabus Introduction to Computer Fundamentals- Computer Memory Hierarchy, Types of Software Binary number system, Algorithm, Flowchart, Formulate simple algorithms for logical and arithmetic problems. Basics of C Programming: Structure and Life cycle of a C Program, Data types, Identifiers, Variables, Keywords, Constants, input/output statements, Operators, Type conversion and type casting. Translate the algorithms to code snippets. Decision Making and Iterative Statements- Decision making- if, if-else, Nested if-else, Multiple if, else if, switch, Ternary Operator, Loops- (while, do-while, for), Nesting of Loops, break, continue and goto. Implement the switch () to solve the basic functions of scientific calculator. Functions: Function prototype, Definition and Call, Type of Functions, Scope of variables in (Block, Function, Program, File), Storage classes (Auto, Register, Static and Extern), Recursion (with the introduction of Stack), Implementation of recursion to solve the problem of Tower of Hanoi. Arrays and Strings- One-dimensional array its operations (Traversal, Linear Search, Insertion, Deletion, Bubble Sort), Two-dimensional and its operations (Addition, Transpose and Multiplication), Passing of array into a function (row and entire array), Input and output of a string, string inbuilt functions, 2-D Character array. Pointers: Introduction to Pointers, Pointer arithmetic, Passing arguments to a function using pointer (understanding of call by value and call by reference), Accessing arrays using pointers Dynamic memory allocation (malloc(), calloc(), realloc() and free()), Pointer and Functions. Structures and Union: Structure declaration, Initialization of structures, Structure variables, Accessing structure elements using (.) operator, Array of structure variables, Passing structure variable to a function (individual and entire structure), Structure pointer, Comparison of Structure and Union. File Handling: Introduction of Files (streams in C), using File (Declaring, Opening and Closing), Operations on File (Reading, Writing and appending), and Random Access of				

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a file, command line argument.

Laboratory Work

To implement programs for various kinds of real life applications in C Language.

Course Learning Outcomes (CLOs) /Course Objectives (COs):

On completion of this course, the students will be able to:

1. Comprehend and analyze the concepts of number system, memory, compilation and debugging of the programs in C language.
2. Analyze the control & iterative statements to solve the problems with C language source codes.
3. Design and create programs for problem solving involving arrays, strings and pointers.
4. Evaluate and analyze the programming concepts based on user define data types and filehandling using C language.

Text Books

1. *C Programming Language*, Brian W. Kernighan Dennis M. Ritchie, 2nd ed, 2012.
2. *Programming in ANSI C*, Balagurusamy G., 8th ed., 2019

Reference Books

1. *Let Us C*, Kanetkar Y., 16th ed., 2017
2. *Programming with C*, Byron S Gottfried, McGraw Hill Education, Forth edition, 2018

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UPH013: Physics				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To introduce the student to the basic physical laws of oscillators, acoustics of buildings, ultrasonics, electromagnetic waves, wave optics, lasers, and quantum mechanics and demonstrate their applications in technology. To introduce the student to measurement principles and their application to investigate physical phenomena				
Syllabus Oscillations and Waves: Oscillatory motion and damping, Applications - Electromagnetic damping – eddy current; Acoustics: Reverberation time, absorption coefficient, Sabine's and Eyring's formulae (Qualitative idea), Applications - Designing of hall for speech, concert, and opera; Ultrasonics: Production and Detection of Ultrasonic waves, Applications - green energy, sound signaling, dispersion of fog, remote sensing, Car's airbag sensor. Electromagnetic Waves: Scalar and vector fields; Gradient, divergence, and curl; Stokes' and Green's theorems; Concept of Displacement current; Maxwell's equations; Electromagnetic wave equations in free space and conducting media, Application - skin depth. Optics: Interference: Parallel and wedge-shaped thin films, Newton rings, Applications as Non-reflecting coatings, Measurement of wavelength and refractive index. Diffraction: Single and Double slit diffraction, and Diffraction grating, Applications - Dispersive and Resolving Powers. Polarization: Production, detection, Applications – Anti-glare automobile headlights, Adjustable tint windows. Lasers: Basic concepts, Laser properties, Ruby, HeNe, and Semiconductor lasers, Applications – Optical communication and Optical alignment. Quantum Mechanics: Wave function, Steady State Schrodinger wave equation, Expectation value, Infinite potential well, Tunneling effect (Qualitative idea), Application - Quantum computing.				
Laboratory Work 1. Determination of damping effect on oscillatory motion due to various media. 2. Determination of velocity of ultrasonic waves in liquids by stationary wave method. 3. Determination of wavelength of sodium light using Newton's rings method. 4. Determination of dispersive power of sodium-D lines using diffraction grating. 5. Determination of specific rotation of cane sugar solution. 6. Study and proof of Malus' law in polarization. 7. Determination of beam divergence and beam intensity of a given laser. 8. Determination of displacement and conducting currents through a dielectric. 9. Determination of Planck's constant.				
Micro Project:				

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Students will be given physics-based projects/assignments using computer simulations, etc.

Course Learning Objectives (CLO)

The students will be able to:

1. understand damped and simple harmonic motion, the role of reverberation in designing a hall and generation and detection of ultrasonic waves.
2. use Maxwell's equations to describe propagation of EM waves in a medium.
3. demonstrate interference, diffraction and polarization of light.
4. explain the working principle of Lasers.
5. use the concept of wave function to find probability of a particle confined in a box.
6. perform an experiment, collect data, tabulate and report them and interpret the results with error analysis.

Text Books

1. Beiser, A., *Concept of Modern Physics*, Tata McGraw Hill (2007) 6th ed.
2. Griffiths, D.J., *Introduction to Electrodynamics*, Prentice Hall of India (1999) 3rd ed.
3. Jenkins, F.A. and White, H.E., *Fundamentals of Optics*, McGraw Hill (2001) 4th ed.

Reference Books

1. Wehr, M.R, Richards, J.A., Adair, T.W., *Physics of The Atom*, Narosa Publishing House (1990) 4th ed.
2. Verma, N.K., *Physics for Engineers*, Prentice Hall of India (2014) 1st ed.
3. Pedrotti, Frank L., Pedrotti, Leno S., and Pedrotti, Leno M., *Introduction to Optics*, Pearson Prentice HallTM (2008) 3rd ed.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UCB009: Chemistry				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The course aims at elucidating principles of applied chemistry in industrial systems, water treatment, engineering materials, computational and analytical techniques.				
Syllabus Atomic and Molecular spectroscopy: Introduction to spectroscopy, principles of atomic absorption, flame emission spectrophotometry and ICP-AES (Inductively Coupled Plasma- Atomic Emission Spectroscopy), Quantification by calibration method, Jablonski diagram, fluorescence and phosphorescence, Beer-Lambert's Law, principle and applications of UV-Vis and IR spectroscopy. Electrochemistry: Background of electrochemistry, Ionic mobility, Conductometric titrations, Modern Batteries: Pb-acid and Li ion battery, Corrosion and its protection. Water Treatment and Analysis: Physiochemical parameters of water quality, External and internal methods of Softening of water: carbonate, phosphate, calgon and colloidal conditioning, Zeolite process, Ion exchange process, treatment of water for domestic use, Desalination of brackish water: Reverse osmosis & Electrodialysis. Fuels: Classification of fuels, Calorific value, Cetane and Octane number, alternative fuels: biodiesel, Power alcohol, synthetic petrol, Fuel cells: H₂ production and storage, Water splitting, Rocket propellant. Chemistry of Polymers: Classification of polymers, tacticity of polymers, molecular weight calculations, Polymers in daily life, conducting, inorganic and biodegradable polymers. Computers in Chemistry: Introduction to SMILES (Simplified Molecular Input Line-Entry System): Methodology and encoding rules, SMILES notation-chemical structure interconversions and its applications.				
Laboratory Work Electrochemical measurements: Experiments involving use of pH meter, conductivity meter, potentiometer, Spectroscopic technique, Volumetric titrations: Determination of mixture of bases, hardness, alkalinity, chloride and iron content, Application of polymers and SMILES Language.				
Course Learning Objectives (CLO) The students will be able to: 1. recognize principles and applications of atomic and molecular spectroscopy. 2. explain the concepts of conductometric titrations, modern batteries and corrosion. 3. apply and execute water quality parameter and treatment methods.				

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4. discuss the concept of alternative fuels, application of polymers and SMILES.
5. execute laboratory techniques like pH metry, potentiometry, spectrophotometry, conductometry and volumetry.

Text Books

1. *Engineering Chemistry*, S. Vairam and S. Ramesh, Wiley India 1st ed, 2014.
2. *Engineering Chemistry*, K. S. Maheswaramma, and M. Chugh. Pearson, 2016.

Reference Books

1. *Engineering Chemistry*, B. Sivasankar, Tata McGraw-Hill Pub. Co. Ltd, New Delhi, 2008.
2. *Engineering Chemistry*, M.J. Shulz, Cengage Learnings, 2007.
3. *J. Chem. Inf. Comput. Sci.*, D. Weininger, Vol. 28, 1988, 31-36.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UMA010: Mathematics-I				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To provide students with skills and knowledge in sequence and series, advanced calculus, calculus of several variables and complex analysis which would enable them to devise solutions for given situations they may encounter in their engineering profession.				
Syllabus Sequences and Series: Introduction to sequences and infinite series, Tests for convergence/divergence, Limit comparison test, Ratio test, Root test, Cauchy integral test, Alternating series, Absolute convergence, and conditional convergence. Series Expansions: Power series, Taylor series, Convergence of Taylor series, Error estimates, Term by term differentiation and integration. Partial Differentiation: Functions of several variables, Limits and continuity, Chain rule, Change of variables, Partial differentiation of implicit functions, Directional derivatives and its properties, Maxima and minima by using second order derivatives. Multiple Integrals: Double integral (Cartesian), Change of order of integration in double integral, Polar coordinates, Graphing of polar curves, Change of variables (Cartesian to polar), Applications of double integrals to areas and volumes, Evaluation of triple integral (Cartesian). Complex analysis: Introduction to complex numbers, Geometrical interpretation, Functions of complex variables, Examples of elementary functions like exponential, trigonometric and hyperbolic functions, Elementary calculus on the complex plane (limits, continuity, differentiability), Cauchy – Riemann equations, Analytic functions, Harmonic functions.				
Course Learning Objectives (CLO) The students will be able to: 1. determine the convergence/divergence of infinite series, approximation of functions using power and Taylor's series expansion and error estimation. 2. examine functions of several variables, define and compute partial derivatives, directional derivatives, and their use in finding maxima and minima in some engineering problems. 3. evaluate multiple integrals in Cartesian and Polar coordinates, and their applications to engineering problems. 4. represent complex numbers in Cartesian and Polar forms and test the analyticity of complex functions by using Cauchy – Riemann equations.				
Text Books 1. Thomas, G.B. and Finney, R.L., <i>Calculus and Analytic Geometry</i>, Pearson Education				

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- (2007), 9th ed.
2. Stewart James, *Essential Calculus*; Thomson Publishers (2007), 6th ed.
 3. Kasana, H.S., *Complex Variables: Theory and Applications*, Prentice Hall India, 2005 (2nd edition).

Reference Books

1. Wider David V, *Advanced Calculus: Early Transcendentals*, Cengage Learning (2007).
2. Apostol Tom M, *Calculus, Vol I and II*, John Wiley (2003).
3. Brown J.W and Churchill R.V, *Complex variables and applications*, McGraw Hill, (7th edition)

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UMA004: Mathematics - II				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To introduce students the theory and concepts of differential equations, linear algebra, Laplace transformations and Fourier series which will equip them with adequate knowledge of mathematics to formulate and solve problems analytically.				
Syllabus Ordinary Differential Equations: Review of first order differential equations, Exact differential equations, Second and higher order differential equations, Solution techniques using one known solution, Cauchy - Euler equation, Method of undetermined coefficients, Variation of parameters method, Engineering applications of differential equations. Laplace Transform: Definition and existence of Laplace transforms and its inverse, Properties of the Laplace transforms, Unit step function, Impulse function, Applications to solve initial and boundary value problems. Fourier Series: Introduction, Fourier series on arbitrary intervals, Half range expansions, Applications of Fourier series to solve wave equation and heat equation. Linear Algebra: Row reduced echelon form, Solution of system of linear equations, Matrix inversion, Linear spaces, Subspaces, Basis and dimension, Linear transformation and its matrix representation, Eigen-values, Eigen-vectors and Diagonalisation, Inner product spaces and Gram-Schmidt orthogonalisation process.				
Course Learning Objectives (CLO) The students will be able to: 1. solve the differential equations of first and 2nd order and basic application problems described by these equations. 2. find the Laplace transformations and inverse Laplace transformations for various functions. Using the concept of Laplace transform students will be able to solve the initial value and boundary value problems. 3. find the Fourier series expansions of periodic functions and subsequently will be able to solve heat and wave equations. 4. solve systems of linear equations by using elementary row operations. 5. identify the vector spaces/subspaces and to compute their bases/orthonormal bases. Further, students will be able to express linear transformation in terms of matrix and find the eigenvalues and eigenvectors.				
Text Books 1. Simmons, G.F., <i>Differential Equations (With Applications and Historical Notes)</i>, Tata McGraw Hill (2009). 2. Krishnamurthy, V.K., Mainra, V.P. and Arora, J.L., <i>An introduction to Linear Algebra</i>, Affiliated East West Press (1976).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Reference Books

1. Kreyszig Erwin, *Advanced Engineering Mathematics*, John Wiley (2006), 8th edition.
2. Jain, R.K. and Iyenger, S.R.K., *Advanced Engineering Mathematics*, Narosa Publishing House (2011), 4th edition.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

UES013: Electrical and Electronics Engineering

L	T	P	Cr
3	1	2	4.5

Course Objective: To introduce the basic concepts of electrical and electronics engineering.

Syllabus

DC Circuits: Introduction to circuit elements; rms and average values for different wave shapes, independent and dependent current and voltage sources; Kirchhoff's laws; mesh and node analysis; source transformations; network theorems: Superposition theorem, Thevenin's and Norton's theorem, Maximum power transfer theorem; star-delta transformation; steady state and transient response of R-L and R-C and R-L-C circuits.

AC Circuits: Concept of phasor, phasor representation of circuit elements; analysis of series and parallel AC circuits; concept of real, reactive and apparent powers; resonance in RLC series and parallel circuits; balanced three phase circuits: voltage, current and power relations for star and delta arrangement; analysis of balanced and unbalanced circuits; three phase power measurement using two-wattmeter and one-wattmeter methods.

Magnetic circuits: analogy between electric and magnetic circuits; series and parallel magnetic circuits; operating principles of electrical appliances: single-phase transformer and rotating machines; tests and performance of single-phase transformer.

Digital Logic Design: Digital signals, Number systems, Positive and negative representation of numbers, Signed-number representation, Binary arithmetic, Postulates and theorems of Boolean Algebra, Algebraic simplification, Sum of products and product of sums formulations (SOP and POS), Gate primitives, Logic Gates and Universal Gates, Minimization of logic functions, Karnaugh Maps, Logic implementation using Gates, Decoder, MUX, Flip-Flops, Asynchronous up/down counters.

Electronic Devices: p- n junction diode: V-I characteristics of diode, Operation of Bipolar Junction Transistor, CB and CE configuration, Transistor as a switch, Operation of SCR, DIAC and TRIAC.

Operational Amplifier Circuits: The ideal operational amplifier, the inverting, non-inverting amplifiers, Op-Amp Characteristics, Applications of Op-amp: summing amplifier, differentiator and integrator.

Laboratory Work: Kirchhoff's laws, network theorems, ac series and parallel circuit, three phase power measurement, magnetic circuit, tests on transformer, resonance in AC circuit, combinational circuits, flip flops, shift register and binary counters, asynchronous and synchronous up/down counters, BJT characteristics.

Course Learning Objectives (CLO)

The students will be able to:

1. Apply various networks laws and theorems to solve dc circuits
2. Compute different ac quantities with phasor representation
3. Comprehend the operation in magnetic circuits, single phase transformer and

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rotating machines 4. Recognize and apply the number systems and Boolean algebra. 5. Reduce and simplify Boolean expressions and implement them with logic gates. 6. Discuss and explain the working of diode, transistor and operational amplifier, their configurations and applications.														
Text Books 1. Hughes, E., Smith, I.M., Hiley, J. and Brown, K., <i>Electrical and Electronic Technology</i> , Prentice Hall (2008) 10th ed. 2. Nagrath, I.J. and Kothari, D.P., <i>Basic Electrical Engineering</i> , Tata McGraw Hill (2002). 3. Boylestad, R.L. and Nashelsky, L., <i>Electronic Devices & Circuit Theory</i> , Perason (2009). 4. Mano M. M. and Ciletti, M.D., <i>Digital Design</i> , Pearson, Prentice Hall, (2013).														
Reference Books 1. Chakraborti, A., <i>Basic Electrical Engineering</i> , Tata McGraw–Hill (2008). 2. Del Toro, V., <i>Electrical Engineering Fundamentals</i> , Prentice–Hall of India Private Limited (2004). 3. David Bell, <i>Electronics Devices and Circuits</i> , Oxford Publications (2009).														
Evaluation Scheme														
<table border="1"> <thead> <tr> <th>Sr. No.</th><th>Evaluation elements</th><th>Weightage (%)</th></tr> </thead> <tbody> <tr> <td>1</td><td>MST</td><td>25-30</td></tr> <tr> <td>2</td><td>EST</td><td>40-45</td></tr> <tr> <td>3</td><td>Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)</td><td>30</td></tr> </tbody> </table>			Sr. No.	Evaluation elements	Weightage (%)	1	MST	25-30	2	EST	40-45	3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30
Sr. No.	Evaluation elements	Weightage (%)												
1	MST	25-30												
2	EST	40-45												
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30												

UEN008: Energy and Environment				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: The exposure to this course would facilitate the students in understanding the terms, definitions and scope of environmental and energy issues pertaining to current global scenario; understanding the need of sustainability in addressing the current environmental & energy challenges.				
Syllabus Introduction: Concept of sustainability and sustainable use of natural resources, Climate Change & its related aspects. Air Pollution: Origin, Sources and effects of air pollution; Primary and secondary meteorological parameters; wind roses; Atmospheric stability; Source reduction and Air Pollution Control Devices for particulates and gaseous pollutants in stationary sources. Water Pollution: Origin, Sources of water pollution, Category of water pollutants, Physicochemical characteristics, Components of wastewater treatment systems. Solid waste management: Introduction to solid waste management, Sources, characteristics of municipal solid waste, Solid waste management methods: Incineration, composting, landfilling. Energy Resources: Classification of Energy Resources; Non-conventional energy resources- Biomass energy, Thermo-chemical conversion and biochemical conversion route; Solar energy-active and passive solar energy absorption systems; Type of collectors; Thermal and photo conversion applications.				
Course Learning Objectives (CLO) The students will be able to: 1. comprehend the interdisciplinary context of environmental issues with reference to sustainability 2. assess the impact of anthropogenic activities on the various elements of environment and apply suitable techniques to mitigate their impact. 3. demonstrate the application of technology in real time assessment and control of pollutants. 4. correlate environmental concerns with the conventional energy sources associated and assess the uses and limitations of non-conventional energy technologies				
Text Books 1. Moaveni, S., <i>Energy, Environment and Sustainability</i>, Cengage (2018) 2. Rajagopalan, R., <i>Environmental Studies</i>, Oxford University Press (2018) 3. O'Callagan, P.W., <i>Energy Management</i>, McGraw Hill Book Co. Ltd. (1993).				

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Reference Books

1. Peavy H.S., Rowe D.S., and Tchobanoglous, G. (2013) *Environmental Engineering*, McGraw Hill.
2. Rao, M.N. and Rao, H.V.N. (2014) *Air Pollution*, McGraw Hill.
3. Metcalf and Eddy. (2003) *Wastewater Engineering: Treatment and Reuse*, Fourth Edition, McGraw Hill.
4. Rai, G.D. (2014) *Non-conventional Energy Resources*, Khanna Publishers.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25-30
2	EST	40-45
3	Sessional: (May include the following) Assignment, Sessional (Includes Regular Lab assessment and Quizzes Project (Including report, presentation etc.)	30

SEMESTER-III

UES012 : ENGINEERING MATERIALS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To provide basic understanding of engineering materials, their structure and the influence of structure on mechanical, chemical, electrical and magnetic properties.				
Syllabus Structure of solids: Classification of engineering materials, Structure-property relationship in engineering materials, Crystalline and non-crystalline materials, Miller Indices, Crystal planes and directions, Determination of crystal structure using X-rays, Inorganic solids, Silicate structures and their applications. Defects; Point, line and surface defects. Mechanical properties of materials: Elastic, An elastic and Visco elastic behaviour, Engineering stress and engineering strain relationship, True stress - true strain relationship, review of mechanical properties, Plastic deformation by twinning and slip, Movement of dislocations, Critical shear stress, Strengthening mechanism, and Creep. Equilibrium diagram: Solids solutions and alloys, Gibbs phase rule, Unary and binary eutectic phase diagram, Examples and applications of phase diagrams like Iron - Iron carbide phase diagram. Electrical and magnetic materials: Conducting and resistor materials, and their engineering application; Semiconducting materials, their properties and applications; Magnetic materials, Soft and hard magnetic materials and applications; Superconductors; Dielectric materials, their properties and applications. Smart materials: Sensors and actuators, piezoelectric, magnetostrictive and electrostrictive materials. Corrosion process: Corrosion, Cause of corrosion, Types of corrosion, Protection against corrosion. Materials selection: Overview of properties of engineering materials, Selection of materials for different engineering applications.				
Laboratory Work: The micro-project will be assigned to the group(s) of students at the beginning of the semester. Based on the topic of the project the student will perform any of the six experiments from the following list: 1. To determine Curie temperature of a ferrite sample and to study temperature dependence of permeability in the vicinity of Curie temperature. 2. To study cooling curve of a binary alloy. 3. Determination of the elastic modulus and ultimate strength of a given fiber strand. 4. To determine the dielectric constant of a PCB laminate. 5. Detection of flaws using ultrasonic flaw detector (UFD). 6. To determine fiber and void fraction of a glass fiber reinforced composite specimen. 7. To investigate creep of a given wire at room temperature. 8. To estimate the Hall coefficient, carrier concentration and mobility in a semiconductor crystal. 9. To estimate the band-gap energy of a semiconductor using four probe technique. 10. To measure grain size and study the effect of grain size on hardness of the given				

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metallic specimens.
Course Learning Objectives (CLO) The students will be able to: - Classify engineering materials based on its structure. - Draw crystallographic planes and directions. - Distinguish between elastic and plastic behavior of materials. - Distinguish between isomorphous and eutectic phase diagram. - Classify materials based on their electrical and magnetic properties. - Propose a solution to prevent corrosion.
Text Books - W.D. Callister , <i>Materials Science and Engineering</i>; John Wiley & Sons, Singapore, 2002. - W.F. Smith, <i>Principles of Materials Science and Engineering: An Introduction</i>; Tata Mc-Graw Hill, 2008. V. Raghavan, <i>Introduction to Materials Science and Engineering</i>; PHI, Delhi, 2005. - Name of the Book, Author/s name, Publisher, Edition, Year
Reference Books S. O. Kasap, <i>Principles of Electronic Engineering Materials</i>; Tata Mc-Graw Hill, 2007. L. H. Van Vlack, <i>Elements of Material Science and Engineering</i>; Thomas Press, India, 1998. K. G. Budinski, <i>Engineering Materials – Properties and selection</i>, Prentice Hall India, 1996 - Name of the Book, Author/s name, Publisher, Edition, Year

UMA035: OPTIMIZATION TECHNIQUES				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The main objective of the course is to formulate mathematical models and to understand solution methods for real life optimal decision problems. The emphasis will be on basic study of linear and non-linear programming problems, Integer programming problem, Transportation problem, Two person zero sum games with economic applications and project management techniques using CPM.				
Syllabus Scope of Operations Research: Introduction to linear and non-linear programming formulation of different models. Linear Programming: Geometry of linear programming, Graphical method, Linear programming (LP) in standard form, Solution of LP by simplex method, Exceptional cases in LP, Duality theory, Dual simplex method, Sensitivity analysis. Integer Programming: Branch and bound technique, Gomory's Cutting plane method. Network Models: Construction of networks, Network computations, Free Floats, Critical path method (CPM), optimal scheduling (crashing). Initial basic feasible solutions of balanced and unbalanced transportation problems, optimal solutions, assignment problem. Multiobjective Programming: Introduction to multiobjective linear programming, efficient solution, efficient frontier. Nonlinear Programming: Unconstrained Optimization: unimodal functions, Fibonacci search method, Steepest Descent method, Conjugate Gradient method Constrained Optimization: Concept of convexity and concavity, Maxima and minima of functions of n-variables, Lagrange multipliers, Karush-Kuhn-Tucker conditions for constrained optimization				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon Completion of this course, the students would be able to: 1) formulate the linear and nonlinear programming problems. 2) solve linear programming problems using Simplex method and its variants. 3) construct and optimize various network models. 4) solve multiobjective linear programming problems. 5) solve nonlinear programming problems.				

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Text Books

1. Chandra, S., Jayadeva, Mehra, A., *Numerical Optimization and Applications*, Narosa Publishing House, (2013).
2. Taha H.A., *Operations Research-An Introduction*, PHI (2007).

Reference Books

1. Pant J. C., *Introduction to optimization: Operations Research*, Jain Brothers (2004)
2. BazaarraMokhtar S., Jarvis John J. and ShiraliHanif D., *Linear Programming and Network flows*, John Wiley and Sons (1990)
3. Swarup, K., Gupta, P. K., Mammohan, *Operations Research*, Sultan Chand & Sons, (2010).
4. H.S. Kasana and K.D. Kumar, *Introductory Operations research*, Springer publication, (2004)
5. Ravindran, D. T. Phillips and James J. Solberg: *Operations Research- Principles and Practice*, John Wiley & Sons, Second edn. (2005).

UEC311: CIRCUIT ANALYSIS AND SYNTHESIS				
	L	T	P	Cr
	2	1	0	2.5
Course Objective: To acquaint the students with the fundamental principles of circuit theory and network synthesis.				
Syllabus Introduction: Review of KCL, KVL, Source transformations, Network theorems– Superposition, Reciprocity, Thevenin, Norton, Maximum power transfer, Tellengen’s theorem. Two Port Network Descriptions: Two-port description in terms of Z-, Y-, ABCD-parameters, Image parameters, Inter-connection of two-port network. Network Functions: Definition of Laplace transform (LT) & its properties, Concept of complex frequency & s-plane, Introduction to basic functions and their LT, Transform impedances, Network functions of one-port. Time Domain Analysis: Solution of networks – RL & RC series using Laplace transform and their Steady state analysis. Network Synthesis: Concept of Poles & Zero, Pole-Zero diagram, Stability, Hurwitz polynomials, Positive real functions (PRF), Synthesis of One-port networks with two kinds of circuit elements, Synthesis of Driving-point functions, basic synthesis procedure, Methods of Synthesis – Foster and Cauer forms.				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Understand and analyze two-port networks. 2. Apply Laplace transformation techniques to analyze the transient behavior of the electrical circuits. 3. Check the stability of given network function. 4. Synthesize one port electrical network.				
Text Books 1. D Roy Choudhary, <i>Network Analysis & Synthesis</i>, NewAge International, (2013), 2nd ed. 2. P. Ramesh Babu, <i>Electric Circuit Analysis</i>, SCITECH, (2018), 2nd ed. 3. Sudhakar, A., <i>Circuits and Networks</i>, Tata McGraw Hill (2010) 4th ed.				
Reference Books 1. Vanvalkenberg, M.E., <i>Networks Analysis</i>, Prentice Hall of India (2007) 3rd ed. 2. Kuo, F., <i>Network Analysis and Synthesis</i>, John Wiley (2003) 2nd ed. 3. Anderson, B.D.O., Vongpanitlerd, S., <i>Network Analysis and Synthesis</i>, Dover Publications (2006) 3rd ed.				

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UEC612: DIGITAL SYSTEM DESIGN				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To familiarize the student with the analysis, design and evaluation of digital systems of medium complexity based on SSI, MSI and Programmable logic devices. To familiarize the students with the issues in the design of iterative networks, timing analysis of synchronous and asynchronous systems. To introduce Hardware description language (VHDL or Verilog) and familiarize students to design combinational and sequential circuits using HDL and simulators.				
Syllabus Binary Codes: Review of special binary codes, Error detection and correction codes. Combinational Circuits: Q. M. Method, Variable Map Method, Ripple carry adder, BCD adder, High speed adder, Subtractor, Code conversion, Magnitude comparators, Applications of Encoders, Decoders, MUX, DEMUX, Implementations using ROM, PLA, PAL. Standard ICs and their applications. Using combinational modules to design digital systems, Iterative networks. Sequential Circuits: Various types of latches and flip-flops and their conversions, Universal Shift Registers, Counters – Ring, Johnson, Design of Counters, Timing issues, Setup and hold times, operating frequency limitations, Static Timing Analysis, Standard ICs for their applications, Finite State Machines – Moore and Mealy, Design of Synchronous and Asynchronous sequential circuits, Races and hazards, hazard free design. Logic Circuits: TTL, MOS, CMOS logic families their comparison, Detailed study of TTL & CMOS logic families and their characteristics i.e. Fan-in, Fan-out, Unit load, Propagation delay, Power dissipation, Current & voltage parameters, Tristate Logic, Interfacing of TTL & CMOS logic families, reading and analyzing Datasheets, Performance estimation of digital systems. HDL: Introduction, Structure of HDL Module, Operators, Data types, Types of Descriptions, Simulation and synthesis, Brief comparison of VHDL and Verilog. Data-Flow Descriptions: Highlights of Data flow descriptions, Structure of data-flow description, Data type-vectors. Familiarization with standards: IEEE 91a-1991 and IEEE 91-1984.				
Laboratory Work: To study standard ICs and their usage, latches and Flip-flops, Design of registers and asynchronous/synchronous up/down counters, Variable modulus counters, Design of Finite State Machines, Study of timing waveforms, Usage of IC tester. Programming of combinational and sequential systems using VHDL or Verilog.				
Course Learning Objectives (CLO) The students will be able to: 1. Design the basic logic functions after simplification of expressions. 2. Design the combinational circuits using basic modules, iterative networks. 3. Design flip flops and sequential systems. 4. Compare the performance of a given digital circuits/systems with respect to their speed, power consumption, number of ICs, and cost. 5. Design, model and simulate the digital systems using VHDL or Verilog.				
Text Books				

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| <ol style="list-style-type: none">1. Fletcher, W.I., <i>Engineering Approach to Digital Design</i>, Prentice Hall of India (2007) 4thed.2. Mano, M.M. and Ciletti M. D., <i>Digital Design</i>, Prentice Hall (2001) 3rd ed. |
| Reference Books <ol style="list-style-type: none">1. Givone D. D., <i>Digital Principles and Design</i>, Tata McGraw Hill (2007) 2nded.2. Tocci, R.J., <i>Digital Systems: Principles and Applications</i>, Prentice-Hall (2006) 10thed.3. Wakerly, J.F., <i>Digital Design Principles and Practices</i>, Prentice Hall of India (2013) 5thed.4. Bhaskar J., <i>A VHDL Primer</i>, Prentice Hall of India (1999) 3rd ed.5. Palnitkar S., <i>Verilog HDL - A Guide to Digital Design and Synthesis</i>, (2003) 2nd ed. |

UEC513: EMBEDDED SYSTEMS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The objective of this course is to equip students with the necessary fundamental knowledge and skills that enable them to design basic embedded systems. It covers architecture, programming of ARM processor and its interfacing with peripheral devices.				
Syllabus Introduction to Embedded Systems: Definition, Embedded Systems Vs General Computing Systems, Classification of Embedded Systems, Major application areas. General purpose processor architecture and organization, Von-Neumann and Harvard architectures, CISC and RISC architectures, Big- and Little-endian processors, Processor design trade-offs, Processor cores: soft and hard. Introduction to ARM Processor: The ARM design philosophy, ARM core data flow model, Architecture, Register set, ARM7TDMI Interface signals, General Purpose Input Output Registers, Memory Interface, Bus Cycle types, Operational Modes, Pipeline: ARM 3 stage Pipeline, ARM family attribute comparison. ARM 5 stage Pipeline, Pipeline Hazards, Data forwarding. Programming based on ARM7TDMI: ARM Instruction set, condition codes, Addressing modes, Interrupts, Exceptions and Vector Table. Instruction Format Assembly Language Programming, Thumb state, Thumb Programmers model, Thumb Instruction format, Thumb Applications, ARM coprocessor interface and Instructions. ARM Tools and Interfacing of Peripherals: ARM Development Environment, Arm Procedure Call Standard (APCS), Example C/C++ programs, Embedded software development, Image structure, linker inputs and outputs, Protocols (I2C, SPI), Memory Protection Unit (MPU). Physical Vs Virtual Memory, Paging, Segmentation. The Advanced Microcontroller Bus Architecture (AMBA), DMA, Peripherals, Interfacing of peripherals with ARM. Familiarization with Standards: IEEE 1275.1-1994 and IEEE 1754.				
Laboratory Work: Introduction to Kiel Software, Introduction to ARM processor kit, Programming examples of ARM processor. Interfacing of LED, Seven Segment Display, Stepper Motor, LCD with ARM7TDMI processor.				
Course Learning Objectives (CLO) The students will be able to: 1. Explain embedded system, its processor architecture and distinguish it from general computing system. 2. Describe ARM processor internal architecture, assembly instructions, their format and Develop ARM processor-based assembly language program for a given statement. 3. Describe how thumb mode operations are designed and various coprocessors are interfaced in an embedded system. 4. Understand 3 stage and 5 stage Pipeline and its differences, Hazards, and Data forwarding techniques. 5. Interface various hardware peripherals in embedded systems.				

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6. Recognize issues to be handled in any processor software tool chain for embedded system development especially using C/C++.
Text Books 1. <i>Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian, "COMPUTER ORGANIZATION AND EMBEDDED SYSTEMS, Sixth Edition, McGraw Hill, 2012.</i> 2. <i>Steve Furber, "ARM System-on-Chip Architecture, Second Edition, PEARSON, 2013.</i>
Reference Books 1. <i>Stephen Welsh, Peter Knaggs, "ARM: Assembly Language Programming", Bourne Mouth University Publication, 2003.</i> 2. <i>Andrew N. Sloss, Dominic Symes, Chris Wright "ARM System Developers Guide, Designing and Optimizing System Software", Elsevier Publication.</i>

UTA013:ENGINEERINGDESIGNPROJECT-I (2 Self Hours)				
	L	T	P	Cr
	1	0	2	3.0
Course Objective: The project will introduce students to the challenge of electronic systems design & integration. The project is an example of ‘hardware and software co-design’ and the scale of the task is such that it will require teamwork as a coordinated effort.				
Syllabus To provide a basis for the technical aspects of the project a small number of lectures are incorporated into the module. As the students would have received little in the way of formal engineering instruction at this early stage in the degree course, the level of the lectures is to be introductory with an emphasis on the physical aspects of the subject matter as applied to the ‘Mangonel’ project. The lecture series include subject areas such as Materials, Structures, Dynamics and Digital Electronics delivered by experts in the field. This module is delivered using a combination of introductory lectures and participation by the students in 15 “activities”. The activities are executed to support the syllabus of the course and might take place in specialised laboratories or on the open ground used for firing the Mangonel. Students work in groups throughout the semester to encourage teamwork, cooperation and to avail of the different skills of its members. In the end the students work in sub-groups to do the Mangonel throwing arm redesign project. They assemble and operate a Mangonel, based on the lectures and tutorials assignments of mechanical engineering they experiment with the working, critically analyse the effect of design changes and implement the final project in a competition. Presentation of the group assembly, redesign and individual reflection of the project is assessed in the end.				
Break up of lecture details to be taken up by MED:				
LecNo.	Topic	Contents		
Lec1	Introduction	The Mangonel Project. History. Spreadsheet.		
Lec2	PROJECTILE MOTION	no DRAG, Design spread sheet simulator for it.		
Lec3	PROJECTILE MOTION	with DRAG, Design spread sheet simulator for it.		
Lec4	STRUCTURES FAILURE	STATIC LOADS		
Lec5	STRUCTURES FAILURE	DYNAMIC LOADS		
Lec6	REDESIGNING THE MANGONEL	Design constraints and limitations of materials for redesigning the Mangonel for competition as a group.		
Lec7	MANUFACTURING	Manufacturing and assembling the Mangonel.		
Lec8	SIMULATION IN ENGINEERING DESIGN	Simulation as an Analysis Tool in Engineering Design.		
Lec9	ROLE OF MODELLING & PROTOTYPING	The Role of Modelling in Engineering Design.		

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Breakup of lecture details to be taken up by ECED:

LecNo.	Topic	Contents
Lec1-5	Digital Electronics	Prototype, Architecture, Using the integrated development Environment(IDE)to prepare an arduino sketch, structuring an arduino program, using simple primitive types(Variables), Simple programming examples. Definition of a sensor and actuator.

Laboratory Work:

Associated Laboratory/Project Program: T-Mechanical Tutorial, L-Electronics Laboratory, W-Mechanical Workshop of “Mangonel” assembly, redesign, operation and reflection.

Title for the weekly work in 15 weeks	Code
Using a spread sheet to develop a simulator	T1
Dynamics of projectile launched by a Mangonel-No Drag	T2
Dynamics of projectile launched by a Mangonel-With Drag	T3
Design against failure under static actions	T4
Design against failure under dynamic actions	T5
Electronics hardware and Arduino controller	L1
Electronics hardware and Arduino controller	L2
Programming the Arduino Controller	L3
Programming the Arduino Controller	L4
Final project of sensors, electronics hardware and programmed Arduino controller based measurement of angular velocity of the “Mangonel” throwing arm.	L5
Assembly of the Mangonel by group	W1
Assembly of the Mangonel by group	W2
Innovative redesign of the Mangonel and its testing by group	W3
Innovative redesign of the Mangonel and its testing by group	W4
Final intergroup competition to assess best redesign and understanding of the “Mangonel”.	W5

Project: The Project will facilitate the design, construction and analysis of a “Mangonel”. In addition to some introductory lectures, the content of the students’ work during the semester will consist of:

1. the assembly of a Mangonel from a Bill Of Materials (BOM), detailed engineering drawings of parts, assembly instructions, and few pre-fabricated parts;
2. The development of a software tool to allow the trajectory of a “missile” to be studied as a function of various operating parameters in conditions of no-drag and drag due to air;
3. A structural analysis of certain key components of the mangonel for static and dynamic stresses using values of material properties which will be experimentally determined;
4. The development of a micro-electronic system to allow the angular velocity of the throwing arm to be determined;

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5. Testing the mangonel;
6. Redesigning the throwing arm of the mangonel to optimise for distance without compromising its structural integrity;
7. An inter-group competition at the end of the semester with evaluation of the group redesign strategies.

Course Learning Objectives (CLO)

The students will be able to:

1. Simulate trajectories of a mass with and without aerodynamic drag using a spreadsheet based software to allow trajectories be optimized;
2. Perform a test to acquire an engineering material property of strength in bending and analyze the throwing arm of the “Mangonel” under conditions of static and dynamic loading;
3. Develop and test software code to process sensor data;
4. design, construct and test an electronic hardware solution to process sensor data;
5. construct and operate a Roman catapult “Mangonel” using tools, materials and assembly instructions, in a group, for a competition;
6. operate and evaluate the innovative redesign of elements of the “Mangonel” for functional and structural performance;

Text Books

1. *Michael Mc Roberts, Beginning Arduino, Technology in action publications, 2nd Edition.*
2. *Alan G. Smith, Introduction to Arduino: A piece of cake, Create Space Independent Publishing Platform (2011).*

Reference Books

1. *John Boxall, Arduino Workshop - a Hands-On Introduction with 65 Projects, No Starch Press; 1 edition (2013).*

SEMESTER IV

UNC402: Data Structures				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: To gain understanding of object-oriented programming (OOP) principles, explore various data structures and their practical uses, and acquire knowledge of diverse algorithmic methods and approaches.				
Syllabus Basics of Object Oriented Programming (OOP): Introduction to OOP, Classes, Objects and Methods, Data Abstraction, Polymorphism, Inheritance, Abstract Data Type. Linear data structures: arrays, linked lists, strategies for choosing the appropriate data structure, abstract data types and their implementation: stacks, queues, priority queues. Basic Analysis: Asymptotic analysis, Big O notation, big omega and big theta notation, Time and space trade-offs in algorithms, Recurrence relations, Analysis of iterative and recursive algorithms, Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Heap Sort, Merge Sort. Algorithmic Strategies with examples and problem solving: Brute-force algorithms with examples, Greedy algorithms with examples, Recursive backtracking, Dynamic Programming with examples, Branch-and-bound with examples, Heuristics, Reduction: transform-and-conquer with examples. Non-Linear Data Structures and Sorting Algorithms: Hash tables, including strategies for avoiding and resolving collisions, Binary search trees, Graphs and graph algorithms, Representations of graphs, Depth- and breadth-first traversals, Heaps, Shortest-path algorithms (Dijkstra and Floyd), Minimum spanning tree (Prim and Kruskal).				
Laboratory Work: Implementation of OOP concepts, Arrays, Recursion, Stacks, Queues, Lists, Binary trees, Sorting techniques, Searching techniques. Implementation of all the algorithmic techniques.				
Course Learning Objectives (CLO) The students will be able to: 1. Implement the basic data structures and solve problems using fundamental algorithms. 2. Implement various searching and sorting techniques. 3. Analyze the complexity of algorithms, to provide justification for that selection, and to implement the algorithm in a particular context. 4. Apply appropriate data structure and algorithmic technique to solve real-world problems.				
Text Books 1. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C++", 4th Edition, Addison-Wesley, 2014. 2. Goodrich, Michael T., Roberto Tamassia, David Mount, "Data Structures and Algorithms in C++", 7th Edition, Wiley. 2004. 3. Narasimha Karumanchi, "Data Structures and Algorithms Made Easy" (2014), 2nd Ed.				

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Reference Books

1. *Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", Second Edition, McGraw Hill, 2002.*
2. *Bjarne Stroustrup, "The C++ Programming Language", 4th Edition, Pearson Education, 2013.*
3. *Ellis Horowitz, Sartaj Sahni and Dinesh Mehta, "Fundamentals of Data Structures in C++", Galgotia Publications, 2007.*

UMA033: NUMERICAL AND STATISTICAL METHODS				
	L	T	P	Cr
	3	0	2	4
Course Objective: The main objective of this course is to understand and implement various numerical and statistical methods to solve engineering, physical and real life problems.				
Syllabus Basic of Errors: Floating-point representation, rounding and chopping errors. Non-Linear Equations: Bisection, fixed point iteration, Newton – Raphson’s method for simple and multiple roots and order of convergence. Linear Systems and Eigen-Values: Gauss elimination method using partial pivoting, Gauss--Seidel method, Rayleigh’s power method for eigen-values and eigen-vectors. Interpolation and Approximations: Newton’s forward and backward differences, Lagrange (with error analysis), Newton’s divided difference interpolation formulas. Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis, Gauss-Legendre quadrature formulae. Differential Equations: Solution of initial value problems using Euler's, Modified Euler’s and Runge-Kutta methods (fourth-order). Curve Fitting: Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves. Probability Distribution: Mathematical expectations, Definition of probability distribution (Probability Mass Function and Probability Density Function), Poisson, Geometric, Binomial, Uniform and Normal distributions. Correlation and Regression: Bivariate distribution, correlation coefficients, regression lines, formula for regression coefficients.				
Laboratory Work: Lab experiments will be set in consonance with materials covered in the theory using MATLAB.				
Course Learning Objectives (CLO) The students will be able to: 1. Learn how to obtain numerical solution of nonlinear equations using bisection, Newton and fixed-point iteration methods. 2. Solve system of linear equations numerically using direct and iterative methods. 3. Analyze the correlated data using the least square and regression curves. 4. Solve integration and initial value problems numerically. 5. Solve real life problems using various probability distributions. 6. Approximate the data and functions using interpolating polynomials.				
Text Books 1. K. Atkinson and W. Han, <i>Elementary Numerical Analysis</i>, 3rd edition, John Wiley & Sons, 2004. 2. Brian Bradie, <i>A friendly Introduction to Numerical Analysis</i>, prentice Hall, 2007. 3. Burden L. R., Faires D. J. and Burden A.M., <i>Numerical Analysis</i>, Brooks Cole, 8th edition, 2004. 4. Richards A. Johnson, <i>Probability and Statistics for Engineers</i>, 8th Edition, PHI Learning, 2011. 5. Meyer, P.L., <i>Introductory Probability and Statistical applications</i>, 2nd edition,				

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Oxford, 1970

Reference Books

1. *Curtis, F. Gerald and Patrick O. Wheatley, Applied Numerical Analysis, 7th edition Pearson Education, 2003.*
2. *Walpole, Ronald E., Myers, Raymond H. Myers, and Sharon L. Myers, Probability and Statistics for Engineers and Scientists, 8th edition Pearson Education, 2007.*
3. *Steven C. Chapra, Applied Numerical Methods with MATLAB for Engineers and Scientists, McGraw-Hill Publishing; 2nd edition, 2007.*

UEC307: ELECTROMAGNETIC FIELD THEORY AND TRANSMISSION LINES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To enhance student's comprehensive capabilities in electromagnetic field theory by studying the behaviour of static and time varying electric and magnetic fields in a medium and transmission line.				
Syllabus Vector Analysis: Coordinate system: Cartesian, Cylindrical and spherical coordinate systems, Vector algebra: dot product, cross product, gradient, divergence and curl. Electrostatic fields: Introduction to coulomb's law, Gaussian law and its applications in determination of field of spherical and cylindrical geometries, Laplace's and Poisson's equation in various coordinate systems. Boundary conditions at electric interfaces, Method of images and its applications. Magnetostatics: Introduction to ampere's law, Magnetic vector potential, Magnetic forces, Boundary conditions at magnetic interfaces. Time Varying Fields and Maxwell's Equations: Maxwell's equation in integral and differential form: For static and time varying fields, For free space, For good conductors, For harmonically varying fields, Continuity of charge, Concept of displacement current. Electromagnetic wave propagation: Waves in General, Wave Propagation in Lossy Dielectrics, Plane Waves in Free Space, Plane Waves in Good Conductors, Wave Polarization, Poynting theorem and power flow: Energy stored and radiated power, Reflection of a Plane Wave at Normal Incidence/Oblique Incidence Transmission Lines and Matching Networks: Introduction, Transmission Line Parameters, Transmission Line Equations, Input Impedance, Characteristic impedance, Standing Wave Ratio, and Power, The Smith Chart, transmission line as circuit and matching elements (quarter wave, single stub and double stub) Guided waves: Waves between parallel plates, transverse electric waves and magnetic waves, characteristics of TE and TM waves, TEM waves, velocity of propagation, Characteristic impedance at radio frequencies, Propagation constant, Attenuation constant and phase constant Familiarization with standards: IEEE 148-1959, IEEE 1128-1998 and IEEE 1302-1998.				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Apply the concepts of co-ordinate systems and vector calculus. 2. Analyze the concepts of static electric and magnetic fields. 3. Analyze Maxwell's equations in various forms and apply those to engineering problems, while examining wave propagation phenomena across different media and interfaces.				

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4. Analyze the concepts of basic transmission media.

Text Books

1. Hayt, W.H., *Engineering Electromagnetics*, Tata McGraw., *Eth ed.*
2. Kraus, J.D., *Electromagnetics*, McGraw., *Eth ed.*

Reference Books

1. Sadiku, M.N.O, *Elements of Electromagnetics*, Oxford University Press (2009) 4th ed.

UEC404: SIGNALS AND SYSTEMS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The aim of this subject is to develop analytical capability of students, by which they would be able to handle real-time signal processing related problems and projects. The knowledge of various transforms will help students to work in multi-disciplinary fields of engineering in group activities.				
Syllabus Representation of Signals and Systems: Signals, Basic Continuous and discrete Time signals and systems, Energy and power signals, System modeling concepts, Linear time invariant systems, Representation of signals in terms of impulses, Discrete time LTI systems continuous time LTI systems, Properties of LTI systems, Systems described by differential and difference equations, Sampling theorem, Quantization. Fourier Analysis: Continuous and discrete time Fourier series, Trigonometric and exponential Fourier series, Properties of Fourier series, Parseval's theorem, Line spectrum, Continuous and discrete time Fourier transforms and its properties, Analysis of discrete time signals and systems, Correlation, Autocorrelation, Relation to Laplace transform. Z-Transform: Definition of Z-transform and Properties of Z-transform, Inverse Z-transform - Power series, partial fraction expansion, residue method and their comparison, Relation between Z.T. and F.T, Transfer function, Discrete time convolution, Stability considerations, Time domain and frequency domain analysis, Solution of difference equation, Applications of Z-transforms. Introduction to Fast Fourier Transforms: Discrete Fourier transform, Properties of DFT, Fast Fourier transforms, Divide and Conquer Approach, Decimation in time and decimation in frequency, Radix-2 FFT, Radix-4 FFT algorithms, Linear Convolution, Circular Convolution, Power spectrum and correlation with FFT. Other transforms: Discrete Sine Transform, Discrete Cosine Transform and its types. Familiarization with Standard: IEEE 1641-2004, IEEE 1139-1999.				
Laboratory Work: Signal generation, Solving difference equation, Calculating Z-transform, Linear and Circular convolution, Correlation, DFT / IDFT, FFT algorithms using MATLAB.				
Course Learning Objectives (CLO) The students will be able to: 1. Analyze the properties of signals and systems in the continuous-time-domain. 2. Analyze the properties of signals and systems in the discrete-time-domain. 3. Analyze the continuous-time signals and systems in the frequency-domain. 4. Analyze the discrete-time signals and systems in the frequency-domain.				
Text Books 1. Oppenheim, A.V. and Willsky, A.S., <i>Signal & Systems</i>, Prentice Hall of India (1997). 2. Kani, A.N. <i>Signals and Systems</i>, McGraw Hill Higher Education, (2011) 3. Proakis, J.G. and Manolakis, D.G., <i>Digital Signal Processing Principles Algorithm</i>				

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& Applications, Prentice Hall, (2007).

Reference Books

1. Lathi,B.P.,*Modern Digital and Analog Communication Systems, Oxford Univ. Press, 1998*
2. Papoulis,A., *Probability Random Variables and Stochastic Processes, McGraw Hill, 2008*

UEC310: INFORMATION AND COMMUNICATION THEORY				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To gain knowledge and understand the concepts of probability theory, random variables, stochastic processes and Information theory. To familiarize the students with the applications of probabilistic/stochastic techniques/methods in communication engineering and information theory.				
Syllabus Random Variable: Review of Probability; Bernoulli Trials; Concepts of Random Variables, Distribution and Probability Density Functions, Binomial Random variables; Conditional Distributions, Bayes' Theorem; Function of One Random Variable; Mean and Variance, Moments, Characteristic Functions; Joint distribution, Joint density, Marginal Density; Functions of Two Random Variables: Mean, Variance; Correlation and Covariance; Central Limit Theorem. Stochastic Processes and Noise Modelling: Stochastic Processes: Discrete And Continuous Time Processes; Probabilistic Structure of a Random Process; Mean, Autocorrelation and Autocovariance Functions; Strict-Sense Stationary and Wide-Sense Stationary (WSS) Processes; Ergodicity; Spectral Representation of a Real WSS Process-Power Spectral Density; Examples of Random Processes: White Noise and Poisson Process; Noise Statistics in Linear Time-Invariant Systems, Noise Power Spectral Densities, Signal-to-Noise-Ratio in Presence of AWGN and Interference; Probability Density of a Jointly-Gaussian Random Vector, Markov Chains. Estimation & Hypothesis Testing: Point Estimation; Estimation Methods: Minimum Mean-Square Error (MMSE) and Linear MMSE Estimators, Maximum Likelihood Estimation; Simple Binary Hypothesis Tests: Decision Criteria, Error in Hypothesis Testing, Neyman Pearson Tests, Z-Test, and P-Value Test. Information Theory: Information Measure and Entropy, Information Source, Markov Source, Adjoint of an Information Source, Joint and Conditional Information Measure, Instantaneous Codes, Kraft-McMillan Equality, Shannon First Theorem, Coding Strategies and Huffman Coding, Introduction to Information Channels, Mutual Information and Channel Capacity, Channel Capacity Calculations for Different Channels, Shannon Second Theorem. Familiarization with Standard: IEEE P1933.1.				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Apply the probabilistic concepts as well as properties of the random variables 2. Analyze engineering problems in the presence of randomness and interference by applying the concepts of stochastic processes, stationary properties, spectral representation, and noise statistics. 3. Incorporate the estimation and hypothesis testing principles to find remedial solutions 4. Employ information theory and coding concepts, to improve information symbol transmission rate, and also use it for data compression.				

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Text Books

1. *H. Stark and J. W. Woods, Probability and Random Processes with Applications to Signal Processing, 4th edition, Pearson, 2011*
2. *R. V. Hogg and A. Craig, Introduction to Mathematical Statistics, 5th edition, Pearson Education.*
3. *Bernard Sklar, Digital Communications, Fundamentals and Applications, Prentice Hall (2001).*

Reference Books

1. *Athanasios Papoulis, Probability Random Variables and Stochastic Processes, McGraw-Hill (1984).*
2. *P.Z. Peebles, Probability, Random Variables, and Random Signal Principles, McGraw-Hill (1980)*
3. *Dimitri P. Bertsekas, Robert G. Gallager, Data Networks, Prentice-Hall (1987)*
4. *A. Larson and B.O. Schubert, Stochastic Processes, vol. I and II, Holden-Day (1979)*
5. *W. Gardener, Stochastic Processes, McGraw Hill (1986)*
6. *IEEE Transactions on Information Theory*
7. *David J. C. Mackay, "Information Theory, Inference and Learning Algorithms", Cambridge University Press, 2003*

UEC301: ANALOG ELECTRONIC CIRCUITS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The aim of this course is to familiarize the student with the analysis and design of basic transistor amplifier circuits, oscillators and wave shaping circuits.				
Syllabus Introduction: p-n junction and contact potential, Fermi levels, Reverse and Forward bias, Zener and Avalanche breakdown. Capacitance of p-n junction, application of diodes Transistor Biasing and Thermal Stabilization: Operating Point, Biasing techniques, Stability, Stabilization against Variations in I_{CO} , V_{BE} , and β , Bias Compensation, Thermal Stabilization Transistor at Low and High Frequencies: Low frequency h-parameter model of BJT, The Hybrid-pi (II) Common-emitter Transistor Model, Hybrid-II conductances, Hybrid-II Capacitances, CE short-circuit current gain, gain-bandwidth product. Multistage Amplifiers: Classification of amplifiers, Distortion in amplifiers, Frequency response of an amplifier, Step Response of an amplifier, RC-coupled amplifier, Low-frequency response of an RC-coupled stage, Effect of an emitter Bypass capacitor on low-frequency response. Power Amplifiers: Class A, B, AB, Push pull & Class C amplifiers, Comparison of their Efficiencies, Types of distortion. Feedback Amplifiers and oscillators: Feedback concept, General characteristics of negative-feedback amplifiers, Types of feedback amplifiers, Analysis of a Feedback Amplifiers: Input resistance, Output resistance, Sinusoidal Oscillator, RC Phase-shift oscillator, A General form of oscillator circuit, Wien Bridge oscillator Wave shaping circuits: Multivibrators (Astable, Mono-stable, Bi-Stable), High pass and low pass filters using R-C Circuits & their response to step input, Pulse input, Square input and Ramp Input, Schmitt Trigger. Familiarization with standards: IEEE 218-1956, IEEE/AIEE 425-1957, IEEE Std 181-2011.				
Laboratory Work: Frequency response analysis of RC coupled amplifier, Tuned amplifiers, Push-pull amplifier, Feedback amplifier. Hartley and Colpitts Oscillator. RC Phase shift oscillator. Study of Multi-vibrators (Astable, Mono-stable, Bi-stable Multi-vibrator). Clipper and Clamper circuit, Schmitt Trigger.				
Course Learning Objectives (CLO) The students will be able to: 1. Evaluate operating point and various stability factors of transistor. 2. Analyse low and high frequency transistor model. 3. Analyse the performance of multistage, feedback and power amplifiers. 4. Design oscillator circuits and analyse its performance. 5. Analyse various filters and multi-vibrators circuits.				
Text Books 1. Milliman, J. and Halkias, C.C., <i>Intergrated Electronics</i> , Tata McGraw Hill (2007).				

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| 2. Milliman, J. & Taub, H., <i>Pulse, Digital and switching waveforms</i> , Tata McGraw Hill (2007). |
| Reference Books
1. Malvino, L., <i>Electronic principles</i> , Tata McGraw Hill (1998).
2. Cathey, J. J., <i>2000 Solved Examples in Electronics</i> , McGraw Hill (1991). |

UTA 024 Engineering Design Project-II (Buggy Lab)				
	L	T	P	Cr
	1	0	2	3.0
Course Objective: The project will introduce students to the challenge of electronic systems design & integration. The project is an example of ‘hardware and software co-design’ and the scale of the task is such that it will require teamwork as a co-ordinated effort.				
Syllabus Hardware overview of Arduino: • Introduction to Arduino Board: Technical specifications, accessories and applications. • Introduction to Eagle (PCB layout tool) software. Sensors and selection criterion: • Concepts of sensors, their technical specifications, selection criterion, working principle and applications such as IR sensors, ultrasonic sensors. Active and passive components: • Familiarization with hardware components, input and output devices, their technical specifications, selection criterion, working principle and applications such as- • Active and passive components: Transistor (MOSFET), diode (LED), LCD, potentiometer, capacitors, DC motor, Breadboard, general PCB etc. • Instruments: CRO, multimeter, Logic probe, solder iron, desolder iron • Serial communication: Concept of RS232 communication ,Xbee • Introduction of ATtiny microcontroller based PWM circuit programming. Programming of Arduino: • Introduction to Arduino: Setting up the programming environment and basic introduction to the Arduino micro-controller • Programming Concepts: Understanding and Using Variables, If-Else Statement, Comparison Operators and Conditions, For Loop Iteration, Arrays, Switch Case Statement and Using a Keyboard for Data Collection, While Statement, Using Buttons, Reading Analog and Digital Pins, Serial Port Communication, Introduction programming of different type of sensors and communication modules, DC Motors controlling. Basics of C#: • Introduction: MS.NET Framework Introduction, Visual Studio Overview and Installation • Programming Basics: Console programming, Variables and Expressions, Arithmetic Operators, Relational Operators, Logical Operators, Bitwise Operators, Assignment Operators, Expressions, Control Structures, Characters, Strings, String Input, serial port communication: Read and write data using serial port. • Software code optimization, software version control. Familiarization with Standards: IEEE 1451.2-1997 and IEEE 1212-2001.				
Laboratory Work: Schematic circuit drawing and PCB layout design on CAD tools, implementing hardware				

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module of IR sensor, Transmitter and Receiver circuit on PCB.

BronzeChallenge: Single buggy around track twice in clockwise direction, under full supervisory control. Able to detect an obstacle. Parks safely. Able to communicate state of the track and buggy at each gantry stop to the console.

Silver Challenge: Two buggies, both one loop around, track in opposite directions under full supervisory, control. Able to detect an obstacle. Both park safely. Able to communicate state of the track and buggy at each gantry stop with console.

Gold Challenge: Same as silver but user must be able to enter the number of loops around the track beforehand to make the code generalized.

Course Learning Objectives (CLO)

The students will be able to:

1. Recognize issues to be addressed in a combined hardware and software system design.
2. Draw the schematic diagram of an electronic circuit and design its PCB layout using CAD Tools.
3. Apply hands-on experience in electronic circuit implementation and its testing.
4. Demonstrate programming skills by integrating coding, optimization and debugging for different challenges.
5. Demonstrate teamwork skills by assigning tasks and integrating the contributions of each team member.

Text Books

1. *Michael McRoberts, Beginning Arduino, Technology in action publications, 2nd Edition.*
2. *Alan G. Smith, Introduction to Arduino: A piece of cake, CreateSpace Independent Publishing Platform (2011).*

Reference Books

1. *John Boxall, Arduino Workshop - a Hands-On Introduction with 65 Projects, No Starch Press; 1 edition (2013).*

SEMESTER-V

UEC519: ANALOG AND DIGITAL COMMUNICATION				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The aim of this course is to build the foundation for communication system design focusing on the challenges of digital communication. The intended objective is to impart knowledge to the engineering students about the transmission/reception of data over physical layer through any channel. They will be able to identify the physical interpretation of mathematical expressions/modelling, while dealing with communication systems in the presence of noise, interference and fading				
Syllabus Introduction to Analog communications: Amplitude modulation and demodulation, Angle modulation and demodulation, spectra of AM and FM, Pre-emphasis and De-emphasis, direct and indirect methods of Frequency demodulation, Super heterodyne receivers. Introduction to Pulse Modulation Systems: Basic model of digital communication system, Bandpass and low pass signal and system representations, low pass equivalent of bandpass signals and system, Pulse modulation. Sampling process. Pulse Amplitude and Pulse code modulation (PCM), Differential pulse code modulation. Delta modulation (DM), Adaptive Delta modulation (ADM), noise considerations in PCM, Time Division multiplexing. Digital Formats and Baseband Shaping for Data Transmission: NRZ, RZ, Manchester formats, Power spectra of discrete-PAM signals, ISI, Nyquist's criterion for distortion less baseband transmission with ideal and practical solutions, Correlative coding Fundamentals of Detection and Estimation: Gram-Schmidt orthogonalization procedure, MAP criterion, maximum likelihood (ML) decision rule, Correlator and Matched filter receiver structures, ML estimation procedure. Digital Modulation Schemes: Details about Binary-ASK, BFSK, BPSK, QPSK, M-ary ASK, M-ary FSK, M-ary PSK, M-ary QAM; Minimum Shift Keying (MSK), Gaussian MSK (GMSK), probability of bit-error & symbol-error calculations for digital modulation techniques under AWGN channel. Error Control Coding: Need and types of error control coding, Linear block-code: generation and syndrome decoding, Convolutional code: generation and Viterbi decoding. Familiarization with standard: IEEE 1007-1991, IEEE P802.11.				
Laboratory Work: Practical/experiments based on the hardware using communication kits, and simulation with the help of available simulation packages.				
Course Learning Objectives (CLO) The students will be able to: 1. Evaluate the performance of Analog modulation techniques. 2. Analyze the concept of different pulse modulation techniques. 3. Evaluate PAM signals, baseband shaping for ISI reduction, and correlative coding. 4. Perform statistical analysis of the transmitted and received modulated waveforms from estimation and detection point of view.				

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5. Evaluate the performance of digital modulation techniques of digital communication system operating over AWGN channel.
6. Analyze the concepts of error control coding.

Text Books

1. John G. Proakis, Masoud Salehi, *Communication System Engineering*, PHI, 2nd Edition, 2002
2. John G Proakis, *Digital Communications*, McGraw-Hill, Third Edition (1994)
3. Simon Haykin, *Digital Communications*, Wiley, Student Edition (1988)
4. Bernard Sklar, *Digital Communications: Fundamentals and Applications*, Prentice Hall (2001)

Reference Books

1. Taub & Schilling, *Principles of Communication Systems*, McGraw-Hill Publications, Second Edition (1998)
2. Simon Haykin, *Communication Systems*, Wiley, Fourth Edition (2006)
3. B.P. Lathi, *Modern Analog and Digital Communication Systems*, Oxford University Press, Third Edition (1998)

UEC502: DIGITAL SIGNAL PROCESSING				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The subject of discrete-time signal processing constitutes an important part of communication and computer engineering. The signals are processed to generate a sequence of numbers that represent samples of a continuous variable in a domain such as time, space, or frequency. Major objective is to apply various mathematical and computational algorithms to continuous-time and discrete-time signals to produce a modified signal that's of higher quality than the original signal, which in turn improves the efficiency of underlying systems. Its utility to analyze the vital characteristics of signals and systems in time- and frequency-domain makes it an inevitable module of engineering practice.				
Syllabus Brief Review of Transforms: Introduction to sampling theorem, Concept of frequency in continuous-time and discrete-time signals, Brief details about Laplace-transform, z-transform, CTFT, DTFT and DFT, decimation-in-time and decimation-in-frequency FFT algorithms. Discrete-time Signals' and Systems' Frequency Response Analysis: Power density spectrum of periodic signals, Energy density spectrum of aperiodic signals, Cepstrum, Concept of bandwidth, LTI systems as frequency-selective filters (LPF, HPF, BPF, digital resonators, notch filters, comb filters and all pass filters), inverse systems and deconvolution. Implementation of Discrete-time Systems: LTI systems characterized by constant-coefficient difference equations and their impulse response attributes. Structures for FIR systems, Structures for IIR systems, Recursive and nonrecursive realizations, Linear filtering methods based on DFT, and Goertzel algorithm. Design of FIR Filters: Causality and its implications, Characteristics of practical frequencyselective filters, Symmetric and antisymmetric FIR filters, Design of linear-phase FIR filters using window method (Hamming, Hanning, Kaiser etc.), Design of FIR filters using frequencysampling method. Design of IIR Filters: Characteristics of commonly used analog filters, Design of IIR filters from analog filters by approximation of derivatives, Design by impulse invariance, Design by bilinear transformation. Multirate Signal Processing: Decimation by a factor D, Interpolation by a factor I, Sampling rate conversion by a rational factor I/D, Polyphase filter structures for decimation and interpolation, sampling rate conversion with cascaded integrator comb filters, Introduction to digital filter banks. Optimum Filtering and Equalization: Wiener filtering, Linear prediction and Concept of equalization. Familiarization with Standards: IEEE 265-1966 , IEEE 1057-2007, ISO/IEC 11172-3.				
Laboratory Work: MATLAB software based lab practicals related to DSP and its applications.				
Course Learning Objectives (CLO)				

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The students will be able to:

1. Analyze the frequency-domain properties of signals and systems.
2. Apply linear time-invariant systems for discrete-time signal processing.
3. Design and implement FIR frequency-selective filters.
4. Design and implement IIR frequency-selective filters.
5. Simplify engineering system problems using multirate-signal processing and optimum-filtering approaches

Text Books

1. *D.G. Manolakis and J.G. Proakis, Digital Signal Processing: Principles, Algorithms, and Applications. 4th ed., India: Pearson Education (Paperback), 2007.*
2. *A.V. Oppenheim, R.W. Schaffer, and J.R. Buck, Discrete-time Signal Processing. 2nd ed., India: Pearson Education (Paperback), 2007.*

Reference Books

1. *V.K. Ingle and J.G. Proakis, Digital Signal Processing using MATLAB. 2nd ed., India: Thomson Learning, 2007.*
2. *S. Salivahanan and A. Vallavaraj, Digital Signal Processing. 2nd ed., New York, U.S.A.: Tata McGraw-Hill, 2011*

UEC512: LINEAR INTEGRATED CIRCUITS AND APPLICATIONS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: To enhance comprehension capabilities of students through understanding of operational amplifiers, frequency response, various applications of operational amplifiers, active filters, oscillators, analog to digital and digital to analog converters and few special function integrated circuits.				
Syllabus Introduction to Differential Amplifiers: Differential Amplifier, configurations of differential amplifier, Analysis of single input balanced output, single input unbalanced output, dual input balanced output and dual input unbalanced output differential amplifiers Operational amplifier: various characteristics of op-amp, CMRR, PSRR, Internal structure of Op-amp, Ideal Op-amp. Inverting and Non-Inverting Configuration, Ideal Open-Loop and Closed-Loop Operation of Op-Amp, Feedback Configurations: Voltage-Series Feedback Amplifier, Voltage-Shunt Feedback Amplifier, Differential Amplifiers with One & Two Op-Amps Frequency Response of an Op-Amp: Introduction to Frequency Response, Compensating Networks, Frequency Response of Internally Compensated Op-Amp, Frequency response of Non-compensated Op-Amp, Closed-Loop Frequency Response. General Applications: DC & AC Amplifiers, Peaking Amplifier, Summing, Scaling and Averaging amplifier, Instrumentation Amplifier, The Integrator, The Differentiator, Log and Antilog Amplifier, Comparator, Zero Crossing Detector, Schmitt Trigger, Sample and Hold Circuit, Clippers and Clampers etc. Active Filters and Oscillators: Butterworth Filters, Band-Pass Filters, Band Reject Filters, All Pass Filters, Phase Shift Oscillator, Wien Bridge Oscillator, Voltage-Controlled Oscillator (VCO), Square Wave Generator. Specialized IC Applications: Introduction, The 555 Timer, Monostable and Astable Multivibrator using IC 555, Phase-Locked Loop (PLL), Voltage Regulators.				
Familiarization with Standards: IEEE 165-1977.				
Laboratory Work: Inverting and Non-Inverting Characteristics of an Op-Amp, Measurement of Op-amp parameters, Op-amp as integrator & differentiator, comparator, Schmitt trigger, Converter (ADC, DAC), square wave generator, Sawtooth waveform generator, precision half wave and full wave rectifiers, log-antilog amplifier, 555 as an astable, monostable and bi-stable multivibrators, active filters.				
Course Learning Objectives (CLO) The student will be able to: 1. Analyse differential amplifier and its configurations. 2. Explain various op-amp parameters and configurations. 3. Analyse frequency response of op-amps 4. Design various applications using op-amps 5. Implement circuits using Linear IC (555)				
Text Books 1. Ramakant A. Gayakwad, 'OP-AMP and Linear IC's', Prentice Hal, 1999. 2. Sergio Franco, 'Design with operational amplifiers and analog integrated circuits', McGraw-Hill, 2002.				

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Reference Books

1. *D. Roy Choudhry, Shail Jain, "Linear Integrated Circuits", New Age International Pvt. Ltd., 2000.*
2. *J. Michael Jacob, 'Applications and Design with Analog Integrated Circuits', Prentice Hall of India, 2002.*

UEC610: COMPUTER ARCHITECTURE				
	L	T	P	Cr
	3	0	0	3
Course Objective: To introduce the concept of parallelism followed in the modern RISC based computers by introducing the basic RISC based DLX architecture. To make the students understand and implement various performance enhancement methods like memory optimization, Multiprocessor configurations, Pipelining and interfacing of I/O structures using interrupts and to enhance the student's ability to evaluate performance of these machines by using evaluation methods like CPU time Equation, MIPS rating and Amdahl's law.				
Syllabus Fundamentals of Modern Computer Design: Historical Evolution of Microprocessors, Basic Functional Units of a Computer and its operational Concepts. Computer Types - Von-Neuman & Harvard Architecture, CISC and RISC architectures. Performance metrics- CPU Time Equation, MIPS Rating, Amdahl's Law and Measure of system Availability (MTTF, MTTR), Control Unit- Hardwired and micro-programmed Control unit. Instruction Set Principles: Classification of Instruction set architectures- Stack, Register and Accumulator based, Addressing modes of DLX architecture, Instruction format , DLX ISA, Coding Examples Effectiveness of DLX. Pipelining and Parallelism: Idea of pipelining, The basic pipeline for DLX processor, the MIPS pipeline Design issues of Pipeline Implementation, Pipeline Hazards-Structural Hazards, Data hazards, Control Hazards, Techniques to remove—Structural Hazards, Overcoming data hazards with Forwarding/ Bypassing and dynamic scheduling, overcoming Control Hazards –Static Techniques – Freeze the pipeline, Prediction Techniques , Advanced DLX hardware Dynamic techniques – 1/2/n bit predictors , Branch delay Slot , Pipeline Scheduling and Loop Unrolling and Scheduling. Multicycle operations, Superscalar DLX Architecture, The VLIW Approach. Memory Hierarchy Design: Introduction to Multilevel Memories, Principle of Temporal and Spatial Locality, Cache memory, Cache Addressing, Cache Organization, Write Policies, Reducing Cache Misses, Cache Associatively Techniques, Reducing Cache Miss Penalty, Reducing Hit Time, Main Memory Technology, Virtual memory and its addressing, Fast Address Translation, Page tables, Multilevel Page table, Translation Lookaside buffer, Crosscutting issues in the design of Memory Hierarchies. Multiprocessors: Characteristics of Multiprocessor Architectures, Flynn's multiprocessor architecture, Multiple Centralized Shared Memory Architectures- Benes and Butterfly network, Distributed Shared Memory Architectures, Cache Coherence, Synchronization, Models of Memory Consistency-Snoopy Protocols. Input/ Output Organization and Buses: Accessing I/O Devices, Interrupts, Handling Multiple Devices, Controlling device Requests, Exceptions, Direct Memory Access, Bus arbitration policies, Synchronous and Asynchronous buses, Parallel port, Serial port, Standard I/O interfaces, Peripheral Component Interconnect (PCI) bus and its architecture, SCSI Bus, Universal Synchronous Bus (USB) Interface. Familiarization with standards: IEEE P802.11, ANSI/IEEE standard 754.				

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Laboratory Work:**Course Learning Objectives (CLO)**

After the completion of the course, the students will be able to:

1. Explain the major differences between a RISC and a CISC based processor.
2. Evaluate the performance of a processor with an enhancement applied and analyze it for future application.
3. Identify the concept of parallelism and pipelining and hazard removal in context to the same.
4. Investigate the Memory Hierarchy design, Memory Addressing and Address translation ,
5. Identify Multiprocessor Configuration, Memory Interconnections- Coherence and Input/output devices interfacing to a processor.

Text Books

1. Hennessy, J. L., and D. A. Patterson. *Computer Architecture: A Quantitative Approach*, 2nd ed. San Mateo, CA: Morgan Kaufman, 1995. ISBN: 1558603727
2. Hennessy, J. L., Patterson, D. A., *Computer Architecture: A Quantitative Approach*, Elsevier (2009) 4th ed.
3. Hamacher, V., Carl, Vranesic, Z.G. and Zaky, S.G., *Computer Organization*, McGraw-Hill (2002) 2nd ed.

Reference Books

1. Murdocca, M. J. and Heuring, V.P., *Principles of Computer Architecture*, Prentice Hall (1999) 3rd ed.
2. Stephen, A.S., Halstead, R. H., *Computation Structure*, MIT Press (1999) 2nd ed.

UEC750: MOS CIRCUIT DESIGN				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The course aims to present the principles and techniques of both MOS based digital and analog circuit design, connecting digital circuits, logic design, and analog components with the fundamental device physics, processing techniques and transistor level characteristics of Silicon integrated circuits, both in theoretical and practical aspects.				
Syllabus MOS Transistor Theory: MOS Structure and its operation, I-V Characteristics, Threshold Voltage Equation, Body Effect, Second Order Effects, Scaling Theory and Limitations of Scaling, Short-Channel Effects, MOS Capacitors, MOS switch, Noise in MOS transistors. NMOS & CMOS Process technology: Evolution of ICs. Masking sequence of NMOS and CMOS Structures, Latch up in CMOS, Electrical Design Rules, Stick Diagram, Layout Design. Circuit Characterization: Resistive Load & Active Load MOS Inverters, NMOS Inverters, CMOS Inverters: Static Characteristics, Switching Characteristics, Interconnect Parasitics, Propagation Delay, Static and Dynamic Power Dissipation, Noise Margin, Logic Threshold Voltage, Logical effort, Driving large loads. Combinational Circuits: MOS Logic Circuits with Depletion NMOS loads, CMOS Logic Circuits, CMOS logic Styles, Realization of simple gates, Complex logic circuits, Pass Gate, Transmission Gate. Operation of MOS Circuits: Small Signal equivalent model and operation of MOS Transistor, MOS as an Amplifier, Calculation of the DC Bias Point, Voltage Gain, Transconductance, Modeling the Body Effect, Biasing of Discrete MOS Amplifiers and Integrated Circuit MOS Amplifiers. Familiarization with standard: IEEE 1181-1991.				
Laboratory Work: Familiarization with Circuit design/simulation tools (Cadence/Mentor/Tanner Tools) for schematic and layout entry, Circuit simulation using SPICE. DC transfer Characteristics of Inverters, Transient response, Calculating propagation delays, rise and fall times, Circuit design of inverters, Complex gates with given constraints.				
Course Learning Objectives (CLO) The students will be able to: 1. Understand MOS transistor operation. 2. Illustrate the fabrication process for NMOS and CMOS circuits. 3. Design and analyze NMOS and CMOS inverters. 4. Implement the various combinational circuits using different logic styles. 5. Analyze MOS transistor as an amplifier.				
Text Books 1. Kang, Sung-Mo (Steve) & Leblebici, Yusuf., <i>CMOS Digital Integrated Circuits Analysis & Design</i>, McGraw Hill, (1999) 2nd ed. 2. A. S. Sedra and K. C. Smith, <i>Microelectronic Circuits</i>. 4th ed. New York, NY: Oxford				

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University Press, 1998.

Reference Books

1. *Gregorian, R. and Temes, G.C., Analog MOS Integrated Circuits for Signal Processing, John Wiley (2004).*
2. *Jan Rabaey, A. Chandrakasan & Nikolic, B., Digital Integrated Circuits – A Design Perspective, Pearson, (2003) 2nd ed.*
3. *CMOS VLSI Design: A Circuits and Systems Perspective, 4th ed., Neil Weste and David Harris, Pearson Addison Wesley, 2011.*
4. *Pucknell D. A., & Eshraghian, K., Basic VLSI Design, Prentice Hall of India, (2007) 3rd ed.*

UTA025: INNOVATION AND ENTREPRENEURSHIP				
	L	T	P	Cr
	1	0	2	3.0
Course Objective: This course aims to provide the students with a basic understanding in the field of entrepreneurship, entrepreneurial perspectives, concepts and frameworks useful for analyzing entrepreneurial opportunities, understanding eco-system stakeholders and comprehending entrepreneurial decision making. It also intends to build competence with respect business model canvas and build understanding with respect to the domain of startup venture finance.				
Syllabus Introduction to Entrepreneurship: Entrepreneurs; entrepreneurial personality and intentions - characteristics, traits and behavioral; entrepreneurial challenges. Entrepreneurial Opportunities: Opportunities- discovery/ creation, Pattern identification and recognition for venture creation: prototype and exemplar model, reverse engineering. Entrepreneurial Process and Decision Making: Entrepreneurial ecosystem, Ideation, development and exploitation of opportunities; Negotiation, decision making process and approaches, - Effectuation and Causation. Crafting business models and Lean Start-ups: Introduction to business models; Creating value propositions - conventional industry logic, value innovation logic; customer focused innovation; building and analyzing business models; Business model canvas, Introduction to lean startups, Business Pitching. Organizing Business and Entrepreneurial Finance: Forms of business organizations; organizational structures; Evolution of organization, sources and selection of venture finance options and its managerial implications. Policy Initiatives and focus; role of institutions in promoting entrepreneurship.				
Laboratory Work:				
Course Learning Objectives (CLO) Upon successful completion of the course, the students should be able to: 1. Explain the fundamentals behind the entrepreneurial personality and their intentions 2. Discover/create and evaluate opportunities. 3. Identify various stakeholders for the idea and develop value proposition for the same. 4. Describe various Business Models and design a business model canvas. 5. Analyse and select suitable finance and revenue models for start-up venture.				
Text Books 1. Ries, Eric(2011), <i>The lean Start-up: How constant innovation creates radically successful businesses</i>, Penguin Books Limited. 2. Blank, Steve (2013), <i>The Startup Owner's Manual: The Step by Step Guide for Building a Great Company</i>, K&S Ranch. 3. S. Carter and D. Jones-Evans, <i>Enterprise and small business- Principal Practice and Policy</i>, Pearson Education (2006)				

Reference Books

1. T. H. Byers, R. C. Dorf, A. Nelson, *Technology Ventures: From Idea to Enterprise*, McGraw Hill (2013)
2. Osterwalder, Alex and Pigneur, Yves (2010) *Business Model Generation*.
3. Kachru, Upendra, *India Land of a Billion Entrepreneurs*, Pearson
4. Bagchi, Subroto, (2008), *Go Kiss the World: Life Lessons For the Young Professional*, Portfolio Penguin
5. Bagchi, Subroto, (2012). *MBA At 16: A Teenager's Guide to Business*, Penguin Books
6. Bansal, Rashmi, *Stay Hungry Stay Foolish*, CIIE, IIM Ahmedabad
7. Bansal, Rashmi, (2013). *Follow Every Rainbow*, Westland.
8. Mitra, Sramana (2008), *Entrepreneur Journeys (Volume 1)*, Booksurge Publishing
9. Abrams, R. (2006). *Six-week Start-up*, Prentice-Hall of India.
10. Verstraete, T. and Laffitte, E.J. (2011). *A Business Model of Entrepreneurship*, Edward Elgar Publishing.
11. Johnson, Steven (2011). *Where Good Ideas comes from*, Penguin Books Limited.
12. Gabor, Michael E. (2013), *Awakening the Entrepreneur Within*, Primento.
13. Guillebeau, Chris (2012), *The \$100 startup: Fire your Boss, Do what you love and work better to live more*, Pan Macmillan
14. Kelley, Tom (2011), *The ten faces of innovation*, Currency Doubleday
15. Prasad, Rohit (2013), *Start-up sutra: what the angels won't tell you about business and life*, Hachette India.

SEMESTER-VI

UEC867: MODERN CONTROL THEORY				
	L	T	P	Cr
	2	1	0	2.5
Course Objective: This course provides the insight of the fundamentals of modern control theory by analysing time and frequency response of open and closed loop systems. Furthermore, the concept is extended to advanced concepts of modern control theory - centred on the system stability and state space methods. Emphasis is placed on concepts of controllability and observability of control systems.				
Syllabus Mathematical Models, Block Diagrams and Signal Flow Graphs of Systems: Introduction of mathematical models and transfer function, construction and reduction of block diagram and signal flow graphs, application of Mason's gain formula. Time-Domain Analysis of Control Systems: Transient and steady state response, time response of first and second-order systems, steady-state errors, types of systems and error constants. System Stability: Conditions for stability of linear systems, algebraic stability criteria, Routh criterion, root locus techniques, frequency domain analysis, correlation between frequency response and transient response, stability analysis of the control systems using various methods. Classical Controllers: General aspects of the closed-loop control design problem, P, PD, PI and PID controllers. State Variable Analysis: Introduction, state variable representation, conversion of transfer function model to state variable model, conversion of state variable model to transfer function model, state equations, concepts of controllability and observability,				
Laboratory Work:				
Course Learning Objectives (CLO) The students will be able to: 1. Interpret closed-loop and open-loop control system representations using block diagrams and signal flow graphs. 2. Evaluate the time and frequency response of control systems and establish the relationship between them. 3. Analyze the stability of the control systems and learn various methods to judge the stability criterion. 4. Understand the fundamentals of designing of P-I-D controllers. 5. Demonstrate an understanding of state-space analysis concepts, and apply the principles of controllability and observability to both classical and digital control systems.				
Text Books 1. Nagrath, I. J., and Gopal, M., <i>Control Systems Engineering</i>, New Age International Publishers, 2006, 4th ed. 2. Benjamin C. Kuo, <i>Automatic Control Systems</i>, Pearson education, 2003 3. Ogata, Katsuhiko, <i>Modern Control Engineering</i>, Prentice-Hall, (2010) 5th ed.				

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Reference Books

1. Warwick, Kevin, *An Introduction to Control Systems*, World Scientific Publishing Co. Ptv. Ltd, (1996) 2nd ed.
2. Levine, W. S., *Control System Fundamentals*, CRC Press, (2000) 3rd ed.
3. Mutambara, Arthur G. O., *Design and Analysis of Control Systems*, CRC Press, (1999) 2nd ed.

UEC640: IOT based Systems				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The objective of this course is to impart necessary and practical knowledge of components of Internet of Things and develop skills required to build real-life IoT based projects.				
Syllabus Introduction to IoT: Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals- Devices and gateways, Data management, Business processes in IoT, Everything as a Service(XaaS), Role of Cloud in IoT, Security aspects in IoT. Elements of IoT: Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, Basics of Sensor Networks. I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for Communication Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP. IoT Application: Development Solution framework for IoT applications- Implementation of Device integration, Data acquisition and integration. IoT Case Studies: IoT case studies and mini projects based on Industrial automation, Transportation, Agriculture, Home Automation.				
1. Laboratory Work: a) Familiarize with Arduino UNO and interface LED/Buzzer with Arduino and write a program to turn ON LED for 1 sec after every 2 seconds. b) To interface the Push button with Arduino, write a program to turn ON the LED when a push button is pressed. 2. a) To interface the LDR sensor with Arduino, write a program to print LDR readings on a serial monitor. b) According to the LDR readings, write a program to turn ON & OFF multiple LEDs. 3. a) To interface the DHT11 sensor with Arduino, write a program to print temperature and humidity readings on a serial monitor. b) According to the DHT readings, write a program to turn ON & OFF multiple LEDs. 4. To interface the HC-05 Bluetooth module with Arduino, write a program to turn ON & OFF multiple LEDs with a smartphone using Bluetooth. 5. Write a program to send sensor data to a smartphone using the HC-05 Bluetooth module with Arduino. 6. Familiarize with the ESP32 Development board and write a program to set up the Bluetooth and Wi-Fi features of ESP32 MCU. 7. Write a program to control multiple LEDs using ESP32 and mobile devices with the help of MQTT Protocol. 8. Familiarization with BLYNK Cloud Platform to set up BLYNK Cloud dashboard and control multiple LEDs using a mobile device using ESP32				

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9. Upload sensor data to the cloud and retrieve it on any mobile device using the BLYNK Cloud Platform. 10. Write a Program in ESP32 to control LEDs/Home appliances with the help of Google Assistant/Amazon Alexa. Project Work based on various applications of IoT.
Course Learning Objectives (CLO) Upon successful completion of the course, the students should be able to: 1. Analyse the design aspects of hardware and software components of IoT. 2. Design Interface for I/O devices, sensors & communication modules. 3. Analyse and process data from sensors. 4. Apply IoT knowledge to Implement basic IoT applications on embedded platform.
Text Books 1. Pethuru Raj and Anupama C. Raman, “The Internet of Things: Enabling Technologies, Platforms, and Use Cases”, CRC Press 2. Vijay Madisetti, Arshdeep Bahga, Internet of Things, “A Hands on Approach”, University Press
Reference Books 1. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, “Introduction to Internet of Things: A practical Approach”, ETI Labs 2. Adrian McEwen, “Designing the Internet of Things”, Wiley 3. Raj Kamal, “Internet of Things: Architecture and Design”, Mc Graw Hill 4. Cuno Pfister, “Getting Started with the Internet of Things”, O Reilly Media

UNC504: Artificial Intelligence				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: This course provides an in-depth introduction to the field of Artificial Intelligence (AI). Students will gain a comprehensive understanding of AI concepts, applications, and problem-solving approaches. The course covers various topics ranging from intelligent agents to advanced applications in natural language processing, computer vision, games, and robotics.				
Syllabus Introduction to Artificial Intelligence: Overview of AI and its applications, Intelligent agents and problem-solving approaches, Uninformed search algorithms (breadth-first, depth-first, etc.), Informed search algorithms (heuristics, A*, etc.) Constraint Satisfaction Problems and Adversarial Search: Introduction to constraint satisfaction problems, Constraint propagation and backtracking algorithms, Search and game-playing techniques, Expectimax search and utility evaluation Markov Decision Processes and Reinforcement Learning: Introduction to Markov decision processes (MDPs), Value iteration and policy iteration algorithms, Model-free reinforcement learning, Exploration-exploitation trade-off. Probabilistic Models and Advanced Applications: Introduction to probability theory in AI, Bayes' nets: representation, inference, sampling, Decision diagrams and the value of perfect information, Introduction to Neural Networks.				
Laboratory Work: Students will apply the fundamentals learned during the lecture classes for better understanding.				
Course Learning Objectives (CLO) By the end of the course, students will be able to: 1. Define the fundamental concepts and terminology associated with artificial intelligence. 2. Understand the problem solving approaches using AI. 3. Develop and apply AI algorithms using Markov decision process and reinforcement learning 4. Develop and apply machine learning algorithms for various tasks				
Text Books 1. "Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig 2. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy 3. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville				

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4. *"Pattern Recognition and Machine Learning" by Christopher Bishop*
5. *"Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron*

UHU005: HUMANITIES FOR ENGINEERS				
	L	T	P	Cr
	2	0	2	3.0
Course Objectives: The objective of the course is to understand the interplay between, psychological, ethical and economic principles in governing human behaviour. The course is designed to help the students to understand the basic principles underlying economic behaviour, to acquaint students with the major perspectives in psychology to understand human mind and behavior and to provide an understanding about the how ethical principles and values serve as a guide to behavior on a personal level and within professions.				
Syllabus UNIT I: PSYCHOLOGICAL PERSPECTIVE Introduction to Psychology: Historical Background, Psychology as a science. Different perspectives in Psychology. Perception and Learning: Determinants of perception, Learning theories, Behavior Modification. Motivational and Affective basis of Behaviour: Basic Motives and their applications at work. Components of emotions, Cognition and Emotion. Emotional Intelligence. Group Dynamics and Interpersonal relationships. Development of self and personality. Transactional Analysis. Culture and Mind. Laboratory work: 1. Experiments on learning and behaviour modification. 2. Application of Motivation Theories: Need based assessment. 3. Experiments on understanding Emotions and their expressions. 4. Personality Assessment. 5. Exercises on Transactional analysis. 6. Role plays, case studies, simulation tests on human behaviour. UNIT II: HUMAN VALUES AND ETHICAL PERSPECTIVE Values: Introduction to Values, Allport-Vernon Study of Values, Rokeach Value Survey, Instrumental and Terminal Values. Value Spectrum for a Good Life: Role of Different Types of Values such as Individual, Societal, Material, Spiritual, Moral, and Psychological in living a good life. Moral and Ethical Values: Types of Morality, Kant's Principles of Morality, Factors for taking ethical decisions, Kohlberg's Theory of Moral Development. Analyzing Individual human values such as Creativity, Freedom, Wisdom, Love and Trust. Professional Ethics and Professional Ethos, Codes of Conduct, Whistle-blowing, Corporate Social Responsibility. Laboratory Work: Practical application of these concepts by means of Discussions, Role-plays and Presentations, Analysis of Case studies on ethics in business and CSR. UNIT III: ECONOMIC PERSPECTIVE Basics of Demand and Supply Production and cost analysis				

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Market Structure: Perfect and Imperfect Markets.

Investment Decisions: capital Budgeting, Methods of Project Appraisal.

Macroeconomic Issues: Gross domestic product (GDP), Inflation and Financial Markets.

Globalisation: Meaning, General Agreement on Trade and tariffs (GATT), World Trade Organisation (WTO). Global Liberalisation and its impact on Indian Economy.

Laboratory Work:

The practicals will cover numerical on demand, supply, market structures and capital budgeting, Trading games on financial markets, Group discussions and presentations on macroeconomic issues. The practicals will also cover case study analysis on openness and globalisation and the impact of these changes on world and Indian economy.

Micro Project: Global Shifts and the impact of these changes on world and Indian economy.

Course Learning Outcomes (CLO):

Upon the successful completion of this course, students will be able to:

1. Improve the understanding of human behavior with the help of interplay of professional, psychological and economic activities.
2. Able to apply the knowledge of basic principles of psychology, economics and ethics for the solution of engineering problems.

Explain the impact of contemporary issues in psychology, economics and ethical principles on engineering.

Text Books:

1. Morgan, C.T., King, R.A., Weisz, J.R., & Schopler, J. *Introduction to Psychology*, McGraw Hill Book Co(International Student (1986).
2. A. N. Tripathi, *Human Values*, New Age International (P) Ltd (2009).
3. Krugman, Paul and Wells Robin, *Economics*, W.H. Freeman & Co Ltd. Fourth Edition (2015).
4. Rubinfeld Pindyck. *Microeconomic Theory and application*, Pearson Education New Delhi (2012).
5. Samuelson, Paul, A. and Nordhaus, William, D. *Economics*, McGraw Hill, (2009).
6. Mankiw, Gregory N. *Principles of Macroeconomics*, South-Western College Pub., (2014).
7. Gregory, Paul R. and Stuart, Robert C. *The Global Economy and Its Economic Systems*, 2013South-Western College Pub (2013).

Reference Books:

1. Atkinson, R.L., Atkinson, R.C., Smith, E.E., Bem, D.J. and Nolen-Hoeksema, S. (2000). *Hilgard's Introduction to Psychology*, New York: Harcourt College Publishers.
2. Berne, Eric (1964). *Games People Play – The Basic Hand Book of Transactional Analysis*. New York: Ballantine Books.
3. Ferrell, O. C and Ferrell, John Fraedrich *Business Ethics: Ethical Decision Making & Cases*, Cengage Learning (2014).
4. Duane P. Schultz and Sydney Ellen Schultz, *Theories of Personality*, Cengage Learning, (2008).
5. Saleem Shaikh. *Business Environment*, Pearson (2007).
6. Chernilam, Francis *International Buisness-Text and Cases*, Prentice Hall (2013).
7. Salvatore, Dominick, Srivastav, Rakesh., *Managerial Economics: Principles with Worldwide Applications*, Oxford, 2012.
8. Peterson H. Craig. and. Lewis, W. Cris. *Managerial Economics*, Macmillan Pub Co; (1990).

SEMESTER-VII

UEC719: 5G WIRELESS COMMUNICATION SYSTEM				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: To impart knowledge about wireless communication systems and related design parameters to undergraduate students. To inculcate ability in students to design wireless communication systems, which can provide high data rate to a large number of users. The main goal is to explore the fundamentals of wireless communication systems, including the evolution from 2G to 5G, to understand the principles and technologies behind 5G networks and standards.				
Syllabus Introduction to Wireless and Mobile Communication: Evolution of wireless Communication Standards from 2G to 5G, Cellular architecture and fundamental of 2G, 3G, and 4G, Introduction to 5G Usage scenarios, Spectrum for 5G, Spectrum access modes and sharing scenarios, carrier aggregation. Wireless Communication Channel Models: Free Space Propagation Model, Large-Scale Path Loss model, Overview of outdoor and Indoor propagation models, Small-Scale Fading and Multipath Propagation, fading channel Parameters, Rayleigh, Rician and Nakagami-m Fading Channel Model, 5G Channel Model Physical Layer: Introduction to diversity and combining techniques, Introduction to MIMO, Alamouti scheme, and channel capacity. Waveform in 4G: OFDM, (transmitter and receiver structure at baseband level), numerology of OFDM in LTE and IEEE 802.11, Waveform in 5G, 5G New Radio (NR), Numerology in 5G waveform, Frame structure in 5G NR, Frequency Ranges, Bandwidths, and Bands for 5G NR, Channel Bandwidth versus UE Channel Bandwidth, Overview of waveform beyond 5G, NOMA. 5G Network Architecture: Core network architecture, RAN architecture, Main Functional Entities of the 5G Core, High-Level Features of 5G Core , standalone and non-standalone mode, Service data adaptation protocol (SDAP), open RAN, multi-access edge computing (MEC), network function virtualization ,Network Slicing, QoS Interworking with Non-3GPP Access Technologies Familiarization with standards: <i>IEEE 802.11 to 802.11ac, 802.15, LTE:3GPP TS 21.201, 3GPP TS 36, 3Gpp standard for 5G: 5G NR,</i>				
Laboratory Work: 5G lab is funded by Ministry of Communication and Department of Telecom, India.				
Experiment	Brief Description			Hours
1.	Introduction to 4G VoLTE Smart Phone Section observe the Transmitted/Received RF signals and (I & Q) constellations. Study and			2

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	measure Battery voltages the Battery charging, power management unit in 4G LTE Smart Phone and SIM Interface Section.	
2.	Study of 4G ATCommands	2
3.	To find the bite error rate of EGC (Equal Gain Combining) diversity technique in Rayleigh fading channels. To find the bite error rate of SC (Selection Combining) diversity technique in Rayleigh fading channels.	2
4.	To find the bite error rate of MIMO System (Alamouti's Code) in Rayleigh fading channels.	2
5.	To find the bite error rate of OFDM (Orthogonal Frequency Division Multiplexing) technique	2
6.	Demonstration of working of 5G RAN and CORE	4
7.	Demonstration of concept of network slicing in 5G architecture	4

Course Learning Objectives (CLO)

The student will be able to:

1. Understand the fundamental principles of different generations of mobile communication systems.
2. Explore key aspects of 5G New Radio (NR), including waveform numerology, frame structure, and spectrum allocation.
3. Analyse the 5G NR layer using techniques such as OFDM, diversity, and MIMO.
4. Evaluate and analyze various fading channels based on their parameters across different scenarios.
5. Understand the high-level architecture of 5G technology.

Text Books:

1. *Rappaport, T.S., Wireless Communication-Principles and Practice, Pearson, (2000) 2nd Edition.*
2. *Haykin S & Moher M., Modern Wireless Communication, Pearson, (2005) 3rd Edition.*
3. *Goldsmith, Andrea. Wireless communications. Cambridge university press, 2005.*

Reference Books:

1. *Lee, William C. Y., Mobile Communication Design and Fundamentals, (1999) 4th Edition.*
2. *Pandya, R., Mobile and Personal Communication System, PHI (2002) 5th Edition.*
3. *Fundamentals of 5g Communications: Connectivity for Enhanced Mobile Broadband and Beyond, Wanshi Chen, Peter Gaal, Juan Montojo, Haris Zisimopoulos , McGrawHill, 2021*
4. *5G Mobile Communications, Wei Xiang, Kan Zheng, Xuemin (Sherman) Shen, Springer International Publishing, January 2017*
5. *IEEE Journal on Selected Areas in Communications IEEE Communications Magazine*

UEC720: ANTENNA AND WAVE PROPAGATION				
	L	T	P	Cr
	2	0	2	4.0
Course Objective: Students will be able to understand vector theory, antenna basic parameters, linear wire antennas, antenna arrays and their patterns, folded dipole, Yagi Uda, loop and Microstrip antenna, wave propagation over ground, through troposphere and ionosphere.				
Syllabus Introduction to Basic Antenna parameters: Radiation pattern, Radiation intensity, Beam width, Gain, Directivity, Polarization, Bandwidth, Efficiency, Side lobes, Side lobe level, Antenna Vector Effective Length and Equivalent Areas, Maximum Directivity and Maximum Effective Area, Friss Transmission Equation. Radiation Integrals and Auxiliary Potential Functions: Retarded vector and scalar potential, Vector Potential A for an Electric Current Source J, Vector Potential F for a Magnetic Current Source M, Electric and Magnetic Fields for Electric (J) and Magnetic (M) Current Sources. Linear Wire Antennas: Radiation from an infinitesimal small current element, Radiation from an elementary dipole (Hertzian dipole), Small Dipole, Finite length dipole, half wave dipole. Antenna Arrays: Two-Element Array, Broadside arrays, End fire arrays. N-Element Linear Array: Uniform Amplitude and Spacing, N-Element Linear Array: Directivity, N-Element Linear Array: Uniform Spacing, Non uniform Amplitude, Binomial Array, Chebyshev Arrays, Principle of pattern multiplication. Propagation of Radio Waves: Different modes of propagation: Ground waves, Space waves, Space wave propagation over flat and curved earth, Wave propagation in the Ionosphere, Critical frequency, Maximum usable frequency (MUF), Skip distance, Virtual height. Familiarization with Standards: <i>IEEE 145-2013, IEEE 1502-2007 and IEEE 211-1997</i> Laboratory Work: Drive antenna by voltage, Radiation pattern of half wave dipole, Radiation pattern of monopole, Effective height of antenna, Radiation pattern of capacitance and inductive loaded antenna, Directional radiation from two composite antennas, Radiation from conducting sheet with slot, Matching stub in antenna, Measure the SWR, Radiation polar diagram of directional antenna.				
Course Learning Objectives (CLO) The students will be able to: 1. Apply basic vector theory in solving antenna problems. 2. Examine various antenna parameters and apply those to the practical scenario. 3. Analyze wire antennas and asses their performance in real-world problems 4. Design and analyze antenna arrays for practical applications. 5. Identify and examine the characteristics of radio wave propagation for communication applications and effect on human health.				
Text Books 1. <i>Antenna Theory, Ballanis, John Wiley & Sons, 2003.</i> 2. <i>Antennas and Radio Propagation, Collins, R. E, McGraw-Hill, 1987</i>				

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Reference Books

1. *Antennas, Kraus and Ronald Marhefka, John D., Tata McGraw-Hill, 2002.*
2. *Microwave & RF Design, Michael Steer, Sci.Tech Publishing, 2009.*

SEMESTER-VIII

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

UNC801: DATA SCIENCE FOR ENGINEERS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: The objective of the Data Science for Engineers course is to equip engineering students with the foundational knowledge and practical skills needed to analyze, interpret, and leverage data for informed decision-making and problem-solving in engineering contexts..				
Syllabus Introduction to R: Variables and Datatypes in R, Data frames, Recasting and Joining of Dataframes, Arithmetic, Logical and Matrix Operations in R, Advanced Programming in R : Functions, Control Structures Data Visualization in R Basic Graphics Linear Algebra for Data science Solving Linear Equations, Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors Statistical Modelling Random Variables and Probability Mass/Density Functions Sample Statistics, Hypotheses Testing Optimization for Data Science Unconstrained Multivariate Optimization , Gradient (Steepest) Descent (OR) Learning Rule, Multivariate Optimization With Equality Constraints, Multivariate Optimization With Inequality Constraints, Introduction to Data Science, Solving Data Analysis Problems - A Guided Thought Process Predictive Modelling Linear Regression, Model Assessment, Diagnostics to Improve Linear Model Fit, Simple Linear Regression Model Building, Simple Linear Regression Model Assessment, Multiple Linear Regression, Cross Validation, Multiple Linear Regression Modelling Building and Selection, Classification, Logistic Regression, Performance Measures, Logistic Regression Implementation in R Algorithms K - Nearest Neighbors (kNN), K - Nearest Neighbors implementation in R, K - means Clustering, K - means Implementation in R				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Apply statistical methods and probability theory to analyze and interpret data using R language. 2. Create and interpret various types of data visualizations to communicate insights effectively. 3. Understand and apply basic machine learning algorithms to engineering problems. 4. Apply data science techniques to solve real-world engineering problems, such as predictive maintenance, optimization, and system modeling.				

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Text Books

1. *APPLIED STATISTICS AND PROBABILITY FOR ENGINEERS – BY DOUGLAS MONTGOMERY*

Reference Books

1. *INTRODUCTION TO LINEAR ALGEBRA - BY GILBERT STRANG*

UEC714: FIBER OPTIC COMMUNICATION

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	L	T	P	Cr
	2	0	2	3.0

Course Objective: To understand fiber optic communication system, transmitter section, medium- the optical fiber, receiver section, analyze system based on important parameters for characterizing optical fiber, optical source, detector and amplifier, fundamentals and advances in lasers, LEDs, photodiodes.

Syllabus

Optical Fibers and Their Characteristics: Introduction to high frequency communication, nature of light, advantages of optical communication, fiber structures, wave guiding, basic optical laws and definitions, optical fiber modes and configuration, mode theory for circular waveguides, single mode fibers, graded index fiber, fiber materials, fabrication and mechanical properties, fiber optic cables; joints, splices, connectors, attenuation, signal distortion, dispersion and polarization mode dispersion in optical fibers, mode coupling, design optimization of single mode fibers.

Optical Sources and Amplifiers: Light emitting diodes, semiconductor laser, various configurations of semiconductor laser, performance parameters of LEDs and semiconductor lasers, light source linearity, modal partition and reflection noise, reliability consideration; power launching and coupling, optical amplifiers: erbium doped fiber amplifier, semiconductor optical amplifier, Raman amplifier.

Photodetectors: Operating principle and physical properties of photodiodes, pin and avalanche photodiodes, photodetector noise, response time, avalanche multiplication noise, temperature effect on avalanche gain, photodiode material.

Optical Communication Systems: Optical receiver operation- fundamental receiver operation, digital receiver performance calculation, preamplifier types, analog receivers. digital transmission systems- point to point links, line coding, noise effects on system performance. analog system: overview of analog links, carrier to noise ratio, multichannel transmission techniques, WDM: basics and components, LAN, coherent optical fiber communication- classification of coherent system, requirements on semiconductor lasers, modulation techniques.

Familiarization with standards: IEEE 404-1977, IEEE 812-1984, ITU G.651-G.657, ANSI Z136.2.

Laboratory Work: Basic optical communication link experiments (analog & digital), measurement of numerical aperture, splicing, multiplexing experiments, bending losses, measurement with OTDR, design and performance analysis using simulation tools.

Course Learning Objectives (CLO)

The students will be able to:

1. Identify and formulate the types, basic properties and transmission characteristics of optical fibers.
2. Analyse different types of optical sources and amplifiers for efficient optical fiber communication.
3. Analyse and formulate pin and avalanche photodetectors in optical fiber communication systems.
4. Realize the analog and digital fiber optic communication systems and networks with

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different modulation techniques.
Text Books 1. Keiser, Gred, <i>Optical Fiber Communications</i>, Tata McGraw-Hill, (2013) 5th ed. 2. Senior, John M., and Yousif Jamro, M., <i>Optical fiber communications: principles and practice</i>, Prentice Hall, (2009) 3rd ed.
Reference Books 1. Ajoy Kumar Ghatak and K. Thyagarajan, <i>Optical Electronics</i>, Cambridge University Press (2012) 2nded. 2. Bahaa E. A. Saleh, Malvin C. Teich, <i>Fundamentals of Photonics</i>, John Wiley & Sons, (2019) 3rd ed.

ELECTIVE-I

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

UEC520: Machine Learning				
	L	T	P	Cr
	2	0	2	4
Course Objective: The aim of this course is to build fundamental understanding of Machine Learning (ML) and its application for Signal Processing. The course will cover a range of mathematical techniques essential in Machine Learning, empowering students to create and enhance their own models effectively.				
Syllabus Introduction: Introduction to real world signals - text, speech, image, video. Data as Signal, time series, and possible representations, Introduction to machine learning, its types. Data Pre-processing: Feature extraction and front-end signal processing, Time-domain features (mean, variance, etc.), Frequency-domain features (spectral analysis). Data scaling and pre-processing, Normalization, Data partitioning into training, test and validation sets. Supervised Learning: Classification and regression, Decision trees, Random forests, Support vector machines (SVM), Multilayer perceptrons and back propagation, Linear regression, Logistic regression, Evaluation metrics. Unsupervised learning: Clustering, K-means clustering, Hierarchical clustering, Dimensionality reduction, Singular Value Decomposition (SVD), Principal component analysis, Self-organizing Maps (SOM), Reinforcement learning using Markov Decision Process. Model selection techniques: Cross-validation, Bias-variance trade-off, Hyper parameter tuning, Applications of Machine Learning for Signal Processing.				
Laboratory Work: Basic of Python and programs related to Machine Learning applications.				
Course Learning Objectives (CLO) The students will be able to: 1. Understand Machine learning and apply various feature extraction methods to signals. 2. Apply various data pre-processing techniques and find ways of selecting suitable model parameters for different supervised machine learning models. 3. Develop and apply machine learning algorithms for unsupervised learning tasks. 4. Implement basic machine learning techniques for Signal processing.				
Text Books 1. 'Introduction to Machine Learning' by Ethem Alpaydin 2. 'Machine Learning: A Probabilistic Perspective' by Kevin P. Murphy. 3. 'Pattern Recognition and Machine Learning' by Christopher M. Bishop				
Reference Books 1. 'Neural Networks and Learning Machines' by Simon Haykin's 2. 'Machine Learning: A Bayesian and Optimization Perspective' by Sergios Theodoridis and Konstantinos Koutroumbas.				

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UNC515: OPERATING SYSTEMS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To understand the role, responsibilities, and the algorithms involved for achieving various functionalities of an Operating System.				
Syllabus Introduction and System Structures: Computer-System Organization, Computer-System Architecture, Operating-System Structure, Operating-System Operations, Process Management, Memory Management, Storage Management, Protection and Security, Computing Environments, Operating-System Services, User and Operating-System Interface, System Calls, Types of System Calls, System Programs, Operating-System Design and Implementation, Operating-System Structure. Process Management: Process Concept, Process Scheduling, Operations on Processes, Interprocess Communication, Multi-threaded programming: Multicore Programming, Multithreading Models, Process Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Algorithm Evaluation. Deadlock: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock. Memory Management: Basic Hardware, Address Binding, Logical and Physical Address, Dynamic linking and loading, Shared Libraries, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table, Virtual Memory Management: Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing, Allocating Kernel Memory. File Systems: File Concept, Access Methods, Directory and Disk Structure, File-System Mounting, File Sharing, Protection, File-System Structure, File-System Implementation, Directory Implementation, Allocation Methods, Free-Space Management. Disk Management: Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap-Space Management, RAID Structure. Protection and Security: Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of the Access Matrix, Access Control, Revocation of Access Rights, Capability-Based Systems, The Security Problem, Program Threats, System and Network Threats, User Authentication, Implementing Security Defenses, Firewalling to Protect Systems and Networks. Concurrency: The Critical-Section Problem, Peterson's Solution, Synchronization Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization, Monitors.				
Laboratory Work: To explore detailed architecture and shell commands in Linux/Unix environment, and to simulate CPU scheduling, Paging, Disk-scheduling and process synchronization algorithms.				
Course Learning Objectives (CLO) <i>Approved by the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.</i>				

The students will be able to:

1. Explain the structure, architecture, and operations of computer systems, including system calls and resource management.
2. Analyze scheduling algorithms, interprocess communication, and multithreading models for efficient process management.
3. Implement memory allocation techniques, including paging, segmentation, and virtual memory, for efficient resource utilization.
4. Develop strategies for file system implementation, disk scheduling, RAID structures, and swap-space management to optimize storage performance.
5. Apply methods to handle deadlocks and design synchronization techniques to ensure secure and concurrent system operations.

Text Books

1. Silberschatz A., Galvin B. P. and Gagne G., *Operating System Concepts*, John Wiley & Sons Inc (2013) 9th ed.
2. Stallings W., *Operating Systems Internals and Design Principles*, Prentice Hall (2018) 9th ed.

Reference Books

1. Bovet P. D., Cesati M., *Understanding the Linux Kernel*, O'Reilly Media (2006), 3rd ed.
2. Kifer M., Smolka A. S., *Introduction to Operating System Design and Implementation: The OSP 2 Approach*, Springer (2007).

UEC606: DATA COMMUNICATION				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To introduce basic concepts of Data communication with different models. Enumerate the physical layer, Data Link Layer, Network Layer, Transport Layer and Application Layer, explanation of the function(s) of each layer. Understanding of switching concept and different types of switching techniques.				
Syllabus Overview of Data Communication and Networking: Data communications, Networks, The Internet, Protocols and standards, Layered tasks, OSI model, TCP /IP protocol Architecture. Physical layer: Analog and digital, Analog signals, Digital signals, Analog versus digital, Data rate limit, Transmission impairments, Line coding, Block coding, Sampling, Transmission mode, Modulation of digital data, Telephone modems, Modulation of analog signal, FDM, WDM, TDM, Guided media, Unguided media, Circuit switching, Telephone networks, DSL technology, Cable modem. Data link layer: Types of errors, Detection, Error correction, Flow and error control, Stop and wait ARQ, go back n ARQ, Selective repeat ARQ, HDLC, point to point protocol, PPP stack, Random access, Controlled access, Channelization, 802.15.4; FDDI, Bluetooth; Telephone networks, DSL technology, Cable modem, SONET/SDH. Connecting devices, Backbone network, Virtual LAN, Cellular telephony, Satellite networks. Network layer: Internetworks, Addressing, Subnetting, Routing, ARP, IP, ICMP, IPV6, Unicast routing, Unicast routing protocol, Multicast routing, Multicast routing protocols. Transport layer: Process to process delivery, User datagram protocol (UDP), Transmission control protocol (TCP), Data traffic, Congestion, Congestion control, Quality of service, Techniques to improve QOS, Integrated services, Differentiated services, QOS in switched networks. Application layer: Client server model, Socket interface, Name space, Domain name space, Distribution of name space, DNS in the internet, Resolution, DNS messages, DDNS, Encapsulation, Electronic mail, File transfer, HTTP, World wide web (WWW), Digitizing audio and video, Audio and video compression, streaming stored audio/video, Streaming live audio/video, Real time interactive audio/video, Voice over IP. Switching: Introduction to Virtual circuit switching including frame relay, ATM and Softswitch Architecture, Packet switching, Message Switching. Familiarization with standards: IEEE 802.3 to 802.6 and 802.11, IEEE/ISO/IEC 8802-15-6-2017, IEEE 802.1CS-2020.				
Course Learning Objectives (CLO) The students will be able to: 1. Understand the layered architecture of Internet's reference models: OSI & TCP/IP and basis of physical layer and media. 2. Analyze the design issues, framing, error detection and correction, channel allocation techniques and link layer protocols. 3. Evaluate the network layer, internetworking, IP and IP addressing mechanism. 4. Apply various routing algorithms, elements of transport protocols, congestion control, and QOS in networks. 5. Understand various communication applications like email, web browser, and familiarization with cryptography and network security.				

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Text Books

1. Ferouzan, Behrouz A., *Data Communications and Networking with TCPIP Protocol Suite*, TATA McGraw Hill (2022) 6th Edition.
2. Tanenbaum, Andrew S., *Computer Networks*, PHI (2013) 5th Edition.

Reference Books

1. *Microwave & RF Design*, Michael Steer, Sci.Tech Publishing, 2009.
2. Stallings William, *Data and Computer Communication*, Pearson Education (2017) 10th Edition.
3. James F. Kurose, *Computer networking: A top-down approach*, Pearson Education (2017), 6th Edition.

UEC521: DESIGN USING SYSTEM VERILOG				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The objective of this course is to provide methods and techniques to design and verify the functionality of digital electronic systems using System Verilog.				
Syllabus Data types: Built-in data types, fixed-size arrays, dynamic arrays, queues, associative arrays, array methods, creating new types with typedef, creating user-defined structures, enumerated types, constants, strings. Design of Combinational and Sequential Circuits: HDL modelling of combinational and sequential circuits: multiplexer, decoder, adder, flip flops and registers, simple design examples, testbench for combinational and sequential circuits. Mealy and Moore FSMs, Design Examples. Procedural statements and routines: procedural statements, tasks, functions, and void functions, routine arguments. Connecting the test bench and design: Separating the test bench and design, the interface construct, stimulus timing, interface driving and sampling, program – module interactions, system Verilog assertions. Basic OOP: Introduction, OOP terminology, creating objects, object deallocation, static variables vs. global variables, class methods, defining methods outside of the class, using one class inside another, understanding dynamic objects, copying objects, public vs. local, building a layered test bench. Randomization: Randomization in System Verilog, Constraint, solution probabilities, controlling multiple constraint blocks, pre_randomize and post_randomize functions, random number functions, constraints tips and techniques, common randomization problems, iterative and array constraints. Threads and interprocess communication: Threads, disabling threads, interprocess communication, events, semaphores, mailboxes and building a test bench with threads & interprocess communication (IPC). Familiarization with standards: IEEE 1364-2005, IEEE 1800-2017.				
Laboratory Work: Modelling and simulation of digital blocks, their verification by using System Verilog.				
Course Learning Objectives (CLO) The students will be able to: 1. Illustrate different data types used in System Verilog. 2. Design and test the functionality of digital system. 3. Apply the concepts of object oriented programming in System Verilog. 4. Apply randomization methods in System Verilog.				
Text Books 1. Chris Spear, SystemVerilog for Verification A Guide to Learning the Testbench Language Features, Springer, (2008) 2nd Ed.				

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| <p>2. <i>Stuart Sutherland, Simon Davidmann and Peter Flake, SystemVerilog for Design Second Edition: A Guide to Using SystemVerilog for Hardware Design and Modeling, Springer (2006); 2nd ed.</i></p> |
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Reference Books

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| <p>1. <i>Bhaskar, J., System Verilog Primer, Bsp (2013).</i></p> |
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ELECTIVE-II

UEC754: IMAGE PROCESSING AND COMPUTER VISION				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To make students understand image fundamentals and how digital images can be processed, Image enhancement techniques and its application, Image compression and its applicability, fundamentals of computer vision, geometrical features of images, object recognition and application of real time image processing.				
Syllabus Introduction: Digital image representation, fundamental steps in image processing, elements of digital image processing systems digitization. Digital Image fundamentals: A Simple Image Model, Sampling and Quantization, Relationship between Pixel, Image Formats, Image Transforms. Image Enhancement: Histogram processing, image subtraction, image averaging, smoothing filters, sharpening filters, enhancement in frequency and spatial domain, low pass filtering, high pass filtering. Image Compression: Fundamentals, Image Compression Models, Elements of Information Theory, Error-Free Compression, Lossy Compression, Recent Image Compression Standards. Computer Vision: Imaging Geometry; Coordinate transformation and geometric warping for image registration, Hough transforms and other simple object recognition methods, Shape correspondence and shape matching, Principal Component Analysis, Shape priors for recognition. Familiarization with standards: <i>IEEE 610.4-1990</i>.				
Laboratory Work: 1. Introduction to image processing on MATLAB. 2. Image effects based on image quantization. 3. Image enhancement algorithms for histogram processing, filtering. 4. Fourier transform of images and filtering in frequency domain. 5. Realization of any one image compression algorithm. 6. Introduction to computer vision tools. Minor Project: Image Compression and Facial Feature Detection with FPGA/ASIC/ARM/DSP Processors.				
Course Learning Objectives (CLO) The students will be able to: 1. Illustrate the fundamentals of image processing. 2. Apply image enhancement techniques in spatial and frequency domain. 3. Analyze the mathematical modeling of image restoration and compression. 4. Demonstrate real-time image processing with computer vision.				
Text Books				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

1. *Gonzalez, R.C., and Woods, R.E., Digital Image Processing, Dorling Kingsley (2009) 3rd ed.*
2. *Jain A.K., Fundamentals of Digital Image Processing, Prentice Hall (2007).*
3. *Sonka M., Image Processing and Machine Vision, Prentice Hall (2007) 3rd ed.*
4. *D. Forsyth and J. Ponce, Computer Vision - A modern approach, Prentice Hall.*
5. *B. K. P. Horn, Robot Vision, McGraw-Hill.*
6. *E. Trucco and A. Verri, Introductory Techniques for 3D Computer Vision, Prentice Hall.*
7. *Richard Szeliski, Computer Vision: Algos and Applications, Springer.*

Reference Books

1. *Tekalp A.M., Digital Video Processing, Prentice Hall (1995).*
2. *Ghanbari M., Standard Codecs: Image Compression to Advanced Video Coding, IET Press (2003).*

UEC832: RADAR AND REMOTE SENSING				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To introduce the basic functioning of a radar system and to make the students understand this by taking a specific example of MTI and PULSE Doppler radar. Implement the usage of these systems with the help of specific sensors for gaining knowledge about inaccessible areas (remote sensing).				
Syllabus Introduction to Radar: Basic Radar – The simple form of the Radar Equation – Radar Block Diagram – Radar Frequencies – Applications of Radar – The Origins of Radar The Radar Equation: Introduction – Detection of Signals in Noise – Receiver Noise and the Signal-to-Noise Ratio – Probability Density Functions – Probabilities of Detection and False Alarm – Integration of Radar Pulses – Radar Cross Section of Targets – Radar cross Section Fluctuations – Transmitter Power – Pulse Repetition Frequency – Antenna Parameters – System losses – Other Radar Equation Considerations MTI and Pulse Doppler Radar: Introduction to Doppler and MTI Radar – Delay – Line Cancelers – Staggered Pulse Repetition Frequencies – Doppler Filter Banks – Digital MTI Processing – Moving Target Detector – Limitations to MTI Performance – MTI from a Moving Platform (MTI) – Pulse Doppler Radar – Other Doppler Radar Topics – Tracking with Radar – Monopulse Tracking – Conical Scan and Sequential Lobing – Limitations to Tracking Accuracy - Low-Angle Tracking – Tracking in Range – Other Tracking Radar Topics – Comparison of Trackers – Automatic Tracking with Surveillance Radars. Fundamentals of Remote Sensing: Definition of terms, Concepts and types of remote sensing; evolution of remote sensing Technology, stages in remote sensing technology, spatial data acquisition, interdisciplinary nature and relation with other disciplines, applications of remote sensing, advantages of RS over conventional methods of survey and inventorying. Basic Principles of Remote Sensing: Types of remote sensing with respect to wavelength regions; Definition of radiometry; Black body radiation; Spectral characteristics of solar radiation; EMR Interaction with Earth materials; Spectral signature concepts spectral reflectance and emittance specular reflection and nonspecular reflectance Albedo of materials EMR interaction with rocks, minerals, vegetation and water -Factors affecting spectral reflectance of materials. Instruments used to study the spectral reflectance spectrophotometer spectro-radiometer. Sensors - Types of sensors- passive sensors and active sensors; imaging systems, photographic sensors; Sensor resolution- spectral, spatial, radiometric and temporal; Imaging sensors and non-imaging radiometers; photograph v/s image, Panchromatic, Multispectral, hyperspectral, stereo images, Optical mechanical line scanner; Pushbroom scanner; Imaging spectrometer; spaceborne imaging sensors, active and passive microwave sensors; Thermal sensors; Atmospheric sensors; Sonar; Laser, Radar, hyperspectral sensors. Platforms - Principles of satellite Missions; Types of platforms- airborne remote sensing, space borne remote sensing; Orbital elements of satellite; satellites for Land, Ocean, and atmospheric studies IRS, Landsat, SPOT, Radarsat, quick bird, Ikonos and ESA satellite series. Image Interpretation and Analysis - Fundamentals of satellite image interpretation;				

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Types of imaging, elements of interpretation; Techniques of visual interpretation; Generation of Thematic maps.

Introduction to advanced Remote Sensing Technologies: Synthetic Aperture Radar; Side Looking Airborne Radar; Hyper spectral Imaging Spectrometer; Lidar; Thermal Imaging System; Advanced Laser Terrain Mapping

Familiarization with standards: *IEEE P2936, IEEE 686-1997.*

Laboratory Work: NA.

Course Learning Objectives (CLO)

The students will be able to:

1. Understand the basic working of a RADAR.
2. Understand the working of a Moving target Indicator (MTI) on the basis of Doppler shift.
3. Recognize the advantages of RS over conventional methods of survey and inventorying.
4. Distinguish among various sensors for specific Remote sensing applications.
5. Understand the types of sensing and interpret the results obtained.

Text Books

1. Skolnik, Merrill I., *Introduction to Radar Systems*, Tata McGraw-Hill (2003) 3rd ed.
2. Peebles, Peyton Z., *Radar Principles*, John Wiley, (2004) 2nd ed.
3. *Microwave Radar and Radiometric Remote Sensing* by F.T. Ulaby, D.G. Long University of Michigan Press, 2013, ISBN 0472119354.

Reference Books

1. Toomay, J.C., *Principles of Radar*, PHI, (2004) 2nd ed.
2. *Microwave Remote Sensing: Active and Passive, Volume I: Microwave remote sensing fundamentals and radiometry* by F. Ulaby, R. Moore, A. Fung Artech House, 1981, ISBN 0890061904.

UEC638: VLSI Testing and Verification				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: In this course students will learn test economics, fault modelling, logic and fault simulation, and ATPG concepts for combinational and sequential circuits. Students will learn about the memory testing and the verification concepts.				
Syllabus Introduction: Role of testing in VLSI design, Issues in test and verification of complex chips, VLSI test process and equipment, Test economics, Yield analysis and product quality. Fault modelling and fault simulation: Physical faults and their modelling, Stuck-at faults, Bridging faults, Fault collapsing, Fault simulation, Deductive, Parallel and Concurrent fault simulation, Combinational and sequential SCOAP measures. ATPG for combinational circuits: D-Algorithm, Boolean Difference, PODEM, Random, Exhaustive and Weighted Test Pattern Generation. ATPG for sequential circuits: ATPG for Single-Clock Synchronous Circuits, Time frame expansion Method. Memory testing and BIST: Permanent, Intermittent and pattern-sensitive faults, March test notion, Memory testing using March tests, PLA testing, Ad-Hoc DFT methods, Scan design, Memory BIST. Verification: Design verification techniques based on simulation, Analytical and formal approaches, Functional verification, Timing verification, Formal verification, Basics of equivalence checking and Model checking, Hardware emulation.				
Laboratory Work: NA.				
Course Learning Objectives (CLO) The students will be able to: 1. Illustrate fault modelling and collapsing. 2. Make use of combinational and sequential Automatic Test Pattern Generation Techniques. 3. Analyze different memory faults and their testing methods. 4. Develop the verification plan and test benches for VLSI designs.				
Text Books 1. M. Bushnell and Vishwani Agrawal, <i>Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits</i>, Springer, ISBN 978-0792379911. 2. Chris Spear, <i>System Verilog for Verification</i>, Springer, ISBN 978-1-4614-0714-0				
Reference Books 1. M. Abramovici, M. Breuer, and A. Friedman, <i>Digital System Testing and Testable Design</i>, IEEE Press, 1994 2. Diraj K. Pradhan, "Fault Tolerant Computer System Design", Prentice Hall.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

3. L. T. Wang, C. W. Wu, and X. Wen, *VLSI Test Principles and Architectures*, Morgan Kaufmann, 2006, ISBN-13: 978-0-12-370597-6, ISBN-10: 0-12-370597-5.
4. *System-on-a-Chip Verification-Methodology and Techniques*, P. Rashinkar, Paterson and L. Singh, Kluwer Academic Publishers, 2001
5. Janick Bergeron, *“Writing test benches functional verification of HDL models”* Kluwer Academic Publishers, New York, Boston, Dordrecht, London, Moscow, 2002.

UEC707: NETWORK VIRTUALIZATION AND SOFTWARE DEFINED NETWORKING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To have a deep understanding of two important, emerging network technologies: Software Defined Networking (SDN) and Network Functions Virtualization (NFV). Use SDN emulator (Mininet) to set up and test network topologies.				
Syllabus Software Defined Network: History of programmable networks and Evolution of Software Defined Networking (SDN), IETF Forces, Active Networking. Separation of Control and Data Plane - Concepts, Advantages and Disadvantages, OpenFlow, protocol, 4D network architecture. Traditional Networking versus SDN. Control & Data Plane: Overview, distributed and centralized control plane & data plane. Control plane: Existing SDN Controllers including Floodlight and Open Daylight projects. Customization of Control Plane: Switching and Firewall. Data Plane: Software-based and Hardware-based; Programmable Network Hardware. Network Virtualization: Concepts, Applications, Existing Network Virtualization Framework (VMWare and others), Network Functions Virtualization (NFV) and Software Defined Networks: Concepts, Implementation and Applications. Network Programmability: Introduction, Northbound Application Programming Interface, Current Languages and Tools. Data Center Networks: Packet, Optical and Wireless Architectures Network Topologies Use Cases of SDNs: Data Centers, Internet Exchange Points, Backbone Networks, Home Networks, Traffic Engineering. Familiarization with standard: <i>IEEE P1916.1, P1915.1, P1917.1.</i>				
Laboratory Work: 1. Set up and get familiar with SDN emulator – Mininet and set up a virtual network. 2. Basic mininet operations. 3. Manually control the switch. 4. Move the rules to SDN controller. 5. Set different forwarding rules for each switch in the controller.				
Course Learning Objectives (CLO) The students will be able to: 1. Analyze the SDN architecture and advantages of programmable networks over traditional networks. 2. Analyse the impact of decoupled control plane and data plane on network.				

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3. Identify network virtualization components and their applications.
4. Design and implement networking problems using SDN-friendly network emulator.

Text Books

1. *SDN: Software Defined Networks, An Authoritative Review of Network Programmability Technologies*, By Thomas D. Nadeau, Ken Gray Publisher: O'Reilly Media, August 2013, ISBN: 978- 1-4493-4230-2, ISBN 10:1-4493-4230-2.
2. *Software Defined Networks: A Comprehensive Approach*, by Paul Goransson and Chuck Black, Morgan Kaufmann, June 2014, Print Book ISBN: 9780124166752, eBook ISBN : 9780124166844.

Reference Books

1. *Network Innovation through OpenFlow and SDN: Principles and Design*, Edited by Fei Hu, CRC Press, ISBN-10: 1466572094, 2014.
2. *Doherty, Jim. SDN and NFV simplified: a visual guide to understanding software defined networks and network function virtualization*. Addison-Wesley Professional, 2016.

ELECTIVE-III

UEC643: AUDIO AND VIDEO PROCESSIG				
	L	T	P	Cr
	2	0	2	3
Course Objective: To provide students with the knowledge of basic characteristics of speech and video signal. To describe basic algorithms of speech analysis common to many applications. To make students acquainted with state-of-the-art video processing techniques, their technical details and challenges.				
Syllabus				
Part A				
Introduction: Introduction to speech processing, Characteristics of speech signals, Digitization of speech signal.				
Digital Models for Speech signals: Human Speech production, Acoustic Phonetics. Uniform Tube Modeling of Speech Production.				
Speech Recognition: Automatic speech recognition system, architecture, features extraction, effect of windowing, spectrogram, cepstral analysis, Hidden Markov Models (HMM), Viterbi algorithm, Vector Quantization (VQ).				
Speaker recognition: Speaker verification/authentication vs. speaker identification, Automatic Speaker Verification system, pattern matching (e.g. dynamic time warping (DTW), VQ, HMM).				
Part B				
Introduction: Video formats, Capturing of video signals, Color space, Quality				
Video Motion Estimation: general methodologies-Block matching algorithm, Deformable blocks matching algorithm, Region based motion estimation, Multi-resolution motion estimation.				
Video Coding: Transform and Quantization, Content dependent video coding, Region based video coding, Object based video coding, Knowledge based video coding, Semantic video coding, Scalable Video Coding.				
Video Compression: inter-frame and intra-frame compression, Lossy and Loss less compression techniques, 3D video compression standard (H.264).				
Familiarization with standards: IEEE 1857-2013, IEEE 1857.5-2015, IEEE H.264/AVC video coding standard.				
Laboratory Work: Students have to write MATLAB /Python programs to gain basic understanding of speech signal, to apply the fundamental concepts of signal processing on speech signals, extraction of fundamental frequency, dividing raw video into frames, compression, coding and reframing the video.				
Course Learning Objectives (CLO)				
The students will be able to:				
1. Model speech signals and represent them digitally.				
2. Analyze speech and speaker recognition systems.				

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3. Explain video formats, color spaces and compression techniques.
4. Implement algorithms for video motion estimation and coding.

Text Books

1. Rabiner, L. and Schafer, R., *Digital Processing of Speech Signals*. Signal Processing, Prentice-Hall (1978).
2. Deller, J., Proakis, J. and Hansen, J., *Discrete-Time Processing of Speech Signals*, IEEE (1993).
3. M. Tekalp, *Digital Video Processing*, Prentice Hall, 2nd Edition, 2015.
4. Iain E. Richardson, *THE H.264 ADVANCED VIDEO COMPRESSION STANDARD*, John Wiley and Sons, Ltd., 2003.

Reference Books

1. Borden, G. and Harris, K., *Speech Science Primer*, 2nd Edition, Williams and Wilkins, (2006).
2. Furui, S., *Digital Speech Processing, Synthesis and Recognition*, CRC (2001).
3. Alan C. Bovik, *The Essential Guide to Video Processing*, Academic Press; 2009
4. J. W. Woods, *Multidimensional Signal, Image, and Video Processing and Coding*, Academic Press, 2011.

UNC602: DATABASE MANAGEMENT SYSTEMS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To become familiar with different types of databases and their applications and learn different types of techniques and strategies.				
Syllabus Introduction: Data, data processing requirement, desirable characteristics of an ideal data processing system, traditional file based system, its drawback, concept of data dependency, Definition of database, database management system, 3-schema architecture, database terminology, benefits of DBMS. Relational Database: Relational data model: Introduction to relational database theory: definition of relation, keys, relational model integrity rules, concept of Relational algebra and Relational Calculus. Database Analysis: Conceptual data modeling using E-R data model -entities, attributes, relationships, generalization, specialization, specifying constraints, Conversion of ER Models to Tables, Practical problems based on E-R data model. Relational Database Design: Functional Dependency, types of functional dependencies, Armstrong axioms, closure of fd's, canonical cover, Normalization- 1NF, 2NF, 3NF, BCNF, 4NF and 5NF and practical problems based on these forms. Transaction Management and Concurrency control: Concept of Transaction, States of Transaction and its properties, Need of Concurrency control, concept of Lock, Two phase locking protocol. Database Implementation: Introduction to SQL, DDL aspect of SQL, DML aspect of SQL – update, insert, delete & various form of SELECT- simple, using special operators, aggregate functions, group by clause, sub query, joins, co-related sub query, union clause, exist operator. PL/SQL - cursor, stored function, stored procedure, triggers, error handling, and package. Laboratory work: Students will perform SQL commands to demonstrate the usage of DDL and DML, joining of tables, grouping of data and will implement PL/SQL constructs. They will also implement one project. Project: It will contain database designing & implementation, should be given to group of 2-4 students. While doing projects emphasis should be more on back-end programming like use of SQL, concept of stored procedure, function, triggers, cursors, package etc. Project should have continuous evaluation and should be spread over different components.				
Course Learning Objectives (CLO) The students will be able to: 1. Analyze the Information Systems as socio-technical systems, its need and advantages as compared to traditional file-based systems. 2. Design database using E-R data model by identifying entities, attributes and relationships. 3. Apply and create Relational Database Design process with Normalization of data.				

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4. Implement the concepts of transaction management, concurrence control and recovery management.
5. Demonstrate use of SQL and PL/SQL to implementation database applications.

Text Books

1. *Database System Concepts*, Silberschatz A., Korth F. H. and Sudarshan S., Tata McGraw Hill, 6th ed, 2010.
2. *Fundamentals of Database Systems*, Elmasri R. and Navathe B. S., Pearson, 7th ed, 2016.

Reference Books

1. *SQL, PL/SQL the Programming Language of Oracle*, Bayross I., BPB Publications, 4th ed, 2009.
2. *Modern Database Management*, HofferJ., Venkataraman, R. and Topi, H., Pearson, 12th ed 2016.

UEC644: ANALOG IC DESIGN				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The goal is to achieve a basic understanding and knowledge of the driving and limiting factors in circuit performance, of circuit design techniques, and of technology issues important to integrated amplifier circuits. To familiarize the design and analysis of basic analog integrated circuits i.e. single ended amplifiers, differential amplifiers, current sources and mirrors, reference circuits, etc. in a standard flow with consideration of performance and power. The course will also familiarize with the issues like noise analysis, OP-Amp design, stability and compensation				
Syllabus Basic MOS Device Physics: MOS IV Characteristics, Second order effects, Short-Channel Effects, MOS Device Models, Review of Small Signal MOS Transistor Models. Single Stage Amplifiers: Common Source Stage, Source Follower, Common Gate Stage, Cascode, Folded Cascode. Differential Amplifier: Single ended and Differential Operation, Qualitative and Quantitative Analysis of Differential pair, Common Mode response. Current Sources and Mirrors: Current Sources, Basic Current Mirrors, Cascode Current Mirrors, Large Signal and Small-Signal analysis. Frequency Response of Amplifiers: Miller Effect, Association of Poles with nodes, Frequency Response of all single stage amplifiers. Voltage References: Different Configurations of Voltage References, Major Issues, Supply Independent Biasing, Temperature-Independent References. Feedback: General Considerations, Topologies, Effect of Loading. Operational Amplifier: General Considerations, Theory and Design, Performance Parameters, Single-Stage Op Amps, Two-Stage Op Amps, Design of 2-stage MOS Operational Amplifier, Gain Boosting, Comparison of various topologies, slew rate, Offset effects, PSRR. Stability and Frequency Compensation: General Considerations, Multi-pole systems, Phase Margin, Frequency Compensation, Compensation Techniques. Familiarization with standard: <i>IEEE 802.15.4.</i>				
Laboratory Work: Review of Mentor Tools; Analysis of Various Analog Building Blocks such as, Current Sources, Current Mirrors, Differential Amplifier, Output Stages; Design and Analysis of Op-Amp (Closed loop and open loop) and its Characterization, Switched-Capacitor Integrator.				
Course Learning Objectives (CLO) The students will be able to: 1. Utilize the MOS transistor in analog circuits.				

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2. Analyze and design the single-stage amplifiers.
3. Analyze and design current sources/sinks/mirrors.
4. Analyze the characteristics of OP-AMP.

Text Books

1. Razavi, B., *Design of Analog CMOS Integrated Circuits*, Tata McGraw Hill (2008).
2. Gregorian, R. and Temes, G.C., *Analog MOS Integrated Circuits for Signal Processing*, John Wiley (2004).

Reference Books

1. Allen, P.E. and Holberg, D.R., *CMOS Analog Circuit Design*, Oxford University Press (2002) 2nd ed.
2. Johns, D.A. and Martin, K., *Analog Integrated Circuit Design*, John Wiley(2008).
3. Gray, P.R., Hurst, P.J., Lewis, S.H., and Meyer, R.G., *Analysis and Design of Analog Integrated Circuits*, John Wiley (2001) 5th ed.

UEC645: MICROWAVE ENGINEERING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To enhance student's comprehensive capabilities in Microwave engineering through understanding of electromagnetic wave generation, transmission and measurements theory and technology by study of microwave transmission medium, media and microwave devices and components.				
Syllabus Wave Guides and Resonators: TE, TM Modes in rectangular & Circular wave guides, characteristics impedance of waveguides, Excitation of wave guides. Microwave Components: Waveguide Microwave Junctions, Scattering matrix and their properties, Microwave T junctions – H Plane Tee, E Plane Tee, Directional coupler – Two hole directional coupler and scattering matrix of a directional coupler, Waveguide joints, Bends, Corners, Transition & twists, Waveguide terminations, Ferrite devices – faraday rotation in devices, Circulator & isolator. Microwave Tubes and Circuits: High frequency limitations of conventional tubes, Klystrons - two cavity klystron amplifier & oscillator, Reflex klystron, Travelling wave tube, Microwave cross-field tube magnetron – Operation and MW characteristics. Microwave Measurements: General measurement setup, Microwave bench, Power measurement – low, Medium & high, Attenuation measurement, Measurement of VSWR, Measurement of dielectric constant, Measurement of Impedance: using Smith Chart, Measurement with spectrum analyzer, Scalar & vector network analyzer operation. Microwave Solid State Devices & Their Applications: P-I-N devices, GUNN Diode, IMPATT. Familiarization with standards: IEEE 521-2002, IEEE 147-1979, IEEE 470-1972 and IEEE 544-1975, IRE 2.S1-1955				
Laboratory Work: To study the performance of mode characteristics of reflex klystrons circulator, Characteristics of Gunn diode, Directional coupler, Attenuator, Sliding screw tuner, Verify the relation of wavelength, finding unknown impedance, VSWR measurement, E-plane, H-plane, Magic Tee, Computer based simulation experiments.				
Course Learning Objectives (CLO) The students will be able to: 1. Analyse different propagating modes in a waveguide 2. Examine various microwave components and their operation. 3. Explain the basic principles of devices used in microwave generation. 4. Evaluate the microwave measurement techniques for various microwave components.				
Text Books 1. Liao, S.Y., <i>Microwave Devices & Circuits</i>, Tata McGraw Hill (2006) 2nd edition. 2. David M. Pozar, <i>Microwave Engineering</i>, Wiley-India(2011) 3rd edition 3. Collins, Robert, <i>Foundation of Microwave Engineering</i>, McGraw Hill (2005) 3rd edition.				

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Reference Books

1. Wolf E.A., and kaul, R., *Microwave Engineering & Systems Applications*, Wiley Interscience (2002) 4th edition.
2. Sze, S. M., *Physics of Semiconductor Devices*, Wiley Eastern (2003) 2nd edition.
3. Sarvate, V.V., *Electromagnetic Fields & Waves*, John Wiley & Sons (2004) 3rd edition.

ELECTIVE-IV

UEC751: DSP PROCESSORS

UEC751: DSP PROCESSORS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To enhance comprehension capabilities of students through fundamentals of DSP processors with its architecture, memory architecture along with instruction set, interrupts, pipelining and different types of DSP processors.				
Syllabus				
An Introduction to DSP Processors: Advantages of DSP, characteristics of DSP systems, classes of DSP applications, DSP processor embodiment and alternatives, Fixed and floating point number representation, IEEE 754 format representation, Fixed Vs Floating point processors.				
DSP Architecture: An introduction to Harvard Architecture, Differentiation between Von-Neumann and Harvard Architecture, Quantization and finite word length effects, Bus Structure, Central Processing Unit, ALU, Accumulators, Barrel Shifters, MAC unit, compare, select, and store unit (CSSU), data addressing and program memory addressing.				
Memory Architecture: Memory structures, features for reducing memory access required, wait states, external memory interfaces, memory mapping, data memory, program memory and I/O memory, memory mapped registers.				
Addressing and Instruction Set: Various addressing modes - implied addressing, immediate data addressing, memory direct addressing, register direct and indirect addressing, and short addressing modes, Instruction types, various types registers, orthogonality, assembly language and application development.				
Interrupts and Pipelining: Interrupts, pipelining and performance, pipelining depth, interlocking, interrupt effects, instruction pipelining.				
Processors: Architecture and instruction set of TMS320C3X, TMS320C5X, Status register 0 (ST0), Status register 1 (ST1), Circular buffer control register (CBCR), Processor mode status register (PMST), some example programs, An introduction to TMS320C6713 and TMS320C5517 Kit, <i>IEEE 754, ANSI standard TMS320</i>				
Laboratory Work: NA				
Course Learning Objectives (CLO)				
The students will be able to:				
1. Differentiate between generalised processor and DSP processor.				
2. Analyze special characteristics and features of generalized DSP processors.				
3. Understand the concept of memory architectures and addressing modes for generalized DSP processors.				
4. Understand pipelining and programming concept of TMS320C3X and TMS320C5X.				
5. Understand detailed architectures, different registers and instruction sets of TMS320C3X and TMS320C5X.				
Text Books				
1. Lapslev, P., Bier, J., Shoham, A. and Lee, E.A., <i>DSP Processor Fundamentals:</i>				

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|---|
| <p><i>Architecture and Features, IEEE Press Series on Signal Processing, IEEE (2000).</i></p> <p>2. Venkataramani, B. and Bhaskar, M., <i>Digital Signal Processor: Architecture, Programming and Applications</i>, Tata McGraw Hill (2003).</p> <p>3. TI DSP reference set (www.ti.com).</p> |
| <p>Reference Books</p> <p>1. Padmanabhan, K., Ananthi, S. and Vijayarajeswaran, R., <i>A practical Approach to Digital Signal Processing</i>, New Age International Pvt. Ltd (2001).</p> <p>2. Babast, J., <i>Digital Signal Processing Applications using the ADSP-2100 family</i>, PHI (1992).</p> |

UEC630: DEEP LEARNING AND APPLICATIONS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To introduce students to fundamentals of deep learning and their applications				
Syllabus Introduction: Introduction to Deep Learning, Deep Supervised Learning, Deep Supervised Learning. Deep Learning Fundamentals: Optimization Methods and Regularization: Gradient Descent, Stochastic Gradient Descent (SGD), Regularization; Neural Network Fundamentals, Convolutional Neural Networks: History of Convolutional Networks, Convolutional Networks and Computer Vision, Audio and Other Domains, Structural Prediction and Natural Language Processing Learning with Memory: Recurrent Neural Network Basics, Advanced Recurrent Neural Networks, Sequential Data Modeling, Embedding Methods for NLP: Unsupervised and Supervised Embeddings, Embedding Methods for NLP: Embeddings for Multi-relational Data, Deep Natural Language Processing Applications: LeNet: Recognizing Handwritten Digits, MiniVGGNet: Going Deeper with CNNs.				
Laboratory Work: Introduction to Python, Development of a neural network, Apply gradient descent algorithm, Explore LeNet, Use pertained models for recognition of MINST/CIFAR-10.				
Course Learning Objectives (CLO) The students will be able to: 1. Explain fundamentals of deep learning 2. Apply optimisation techniques on neural networks. 3. Design and implement CNN architectures 4. Implement and evaluate deep learning models for real-world applications.				
Text Books 1. <i>Deep Learning by Aaron Courville, Ian Goodfellow, and YoshuaBengio</i> 2. <i>Deep Learning with Python by François Chollet</i> 3. <i>Deep Learning: A Practitioner's Approach by Adam Gibson and Josh Patterson</i>				

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UEC752: IC FABRICATION TECHNOLOGY				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To gain knowledge about crystal growth and wafer preparation techniques. Subsequently, the thoroughly understanding of different integral steps needed for IC components fabrication mainly bipolar and field effect transistors. To acquire knowledge of various packaging techniques.				
Syllabus Integrated Circuits: Introduction, Impact of ICs on Industry, Advantages over discrete components, Monolithic and Hybrid ICs, Scales of integration and related issues. Crystalline and non-crystalline materials used in IC fabrication. Growth of Single Crystals wafers: Si crystal structure, Si crystal growth using Czochralski's method, Float Zone method and Bridgeman technique, characteristics and crystal evaluation, Wafer Shaping operations, Slicing, polishing and etching, Wafer cleaning. Epitaxy Film Formation: Importance of epitaxial layer growth, Types of epitaxy: VPE, MBE, MOCVD, Defects in epitaxial layers and their removal. Doping: Introduction, Basic concept of impurity diffusion in a semiconductor crystal. Fick's Laws, Analytical solution of diffusion equation, Gaussian solution near the surface and infinite medium. Ion-implantation, doping measurement methods. Subsequent Processes: Thermal Oxidation, Photolithography, Electron beam and X-Ray lithography, Positive and negative photo resist, dry and wet etching, Metallization using PVD, and clean room: Standards. MOSFET Technology: Design of junction diode, BJT, MOSFET and CMOS. Packaging of I.C's: Importance of packaging, Mountings in packages using Dual-in-line (DIP) or TO packages. Packages using surface-mount-technology (SMT). Familiarization with standards: IEEE 62659-2015.				
Laboratory Work:				
Course Learning Objectives (CLO) The students will be able to: 1. Explain crystal growth and wafer preparation techniques. 2. Understand different fabrication processes used in IC industry. 3. Apply IC fabrication technology to fabricate different semiconductor devices. 4. Explain various packaging techniques.				
Text Books 1. Sze, S. M., <i>VLSI Technology</i>, Wiley Eastern, USA (1999) 2nd ed. 2. J.D. Plummer, M.D. Deal and P.B. Griffin, <i>Silicon Vlsi Technology: Fundamentals</i>				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Practice and Modeling, Pearson Education Inc,(2009), 1st ed.

Reference Books

1. *Pucknell and Eshraghian, Basic VLSI Design, (2000) 2nd edition*
2. *Nagchoudhri, D., Principles of Microelectronics Technology (2002) 4th edition.*
3. *Sze, S. M., Semiconductor Devices, Physics & Technology, (2001) 3rd ed.*

UEC812: WIRELESS SENSOR NETWORKS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To provide a succinct introduction to the field of wireless sensor networks by introducing the fundamentals of network architectures, protocols and deployment methods. To familiarise with various networks platforms and tools for wireless sensor networks				
Syllabus Introduction and Overview of Wireless Sensor Networks: Background of Sensor Network Technology, Application of Sensor Networks, Challenges for Wireless Sensor Networks, Enabling Technologies for Wireless Sensor Networks. Sensor Node Hardware and Network Architecture: Single-node Architecture: Hardware Components, Operating Systems and Execution Environments, Network Architecture: Sensor Network Scenarios, Optimization Goals and Figures of Merit, Gateway Concepts. Network Protocols: MAC Protocols: Requirement and design constraints for MAC Protocols, Important classes of MAC Protocols, MAC Protocols for Wireless Sensor Networks, Routing Protocols: Classification of Routing Protocols, Energy-Efficient Routing, Geographic Routing. Deployment and Configuration: Localization and Positioning, Single-hop Localization, Positioning in Multi-hop environments, Coverage and Connectivity, Naming and Addressing in Sensor Networks, Assignment of MAC addresses. Sensor Network Platforms and Tools: Sensor Node Hardware – Berkeley Motes, Programming Challenges, Node-level Software Platforms, Node-level Simulators. Familiarization with standards: <i>IEEE 1451.0 and IEEE 1451.5-802.11.</i> Micro Project: WSN based monitoring of Temperature				
Laboratory Work: Introduction to Python, Development of a neural network, Apply gradient descent algorithm, Explore LeNet, Use pertained models for recognition of MINST/CIFAR-10.				
Course Learning Objectives (CLO) The students will be able to: 1. Define the fundamental concepts of Wireless Sensor Networks (WSNs) and list their applications in various domains. 2. Explain the various architectures of Wireless Sensor Networks, its related hardware and protocols 3. Familiarise with deployment and configuration methods. 4. Acquainted to Node-level Software Platforms.				
Text Books 1. Holger Karland Andreas Willig, <i>Protocols and Architectures for Wireless Sensor Networks</i>, JohnWiley,2005. 2. Feng Zhao & Leonidas J. Guibas, <i>Wireless Sensor Networks - An Information Processing Approach</i>, Elsevier,2007.				

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Reference Books

1. *Kazem Sohraby, Daniel Minoli, and Taieb Znati, Wireless Sensor Networks-Technology, Protocols and Applications, John Wiley, (2007).*
2. *Raghavendra, Cauligi S, Sivalingam, Krishna M., Zanti Taieb, Wireless Sensor Network, Springer 1st Ed, (2004) (ISBN: 978-4020-7883-5).*
3. *Anna Hac, Wireless Sensor Network Designs, John Wiley, (2003).*

GENERIC ELECTIVE

UPH064 : NANOSCIENCE AND NANOMATERIALS				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: To introduce the basic concept of Nanoscience and advanced applications of nanotechnology.				
Syllabus Fundamental of Nanoscience: Features of Nanosystem, Free electron theory and its features, Idea of band structures, Density of states in bands, Variation of density of state and band gap with size of crystal, Quantum Size Effect: Concepts of quantum effects, Schrodinger time independent and time dependent equation, Electron confinement in one-dimensional well and three-dimensional infinite square well, Idea of quantum well structure, Quantum dots and quantum wires, Nano Materials: Classification of Nano Materials their properties, Basic concept relevant to application, Fullerenes, Nanotubes and nano-wires, Thin films chemical sensors, Gas sensors, Vapour sensors and Bio sensors, Synthesis and processing: Sol-gel process, Cluster beam evaporation, Ion beam deposition, Chemical bath deposition with capping techniques and ball milling, Cluster assembly and mechanical attrition, Sputtering method, Thermal evaporation, Laser method, Characterization: Determination of particle size, XRD technique, Photo luminescence, Electron microscopy, Raman spectroscopy, STEM, AFM, Applications: Photonic crystals, Smart materials, Fuel and solar cells, Opto-electronic devices.				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon completion of the course, Students will be able to 1. discriminate between bulk and nano materials, 2. establish the size and shape dependence of Materials' properties, 3. correlate 'quantum confinement' and 'quantum size effect' with physical and chemical properties of nanomaterials, 4. uses top-down and bottom-up methods to synthesize nanoparticles and control their size and shape 5. characterize nanomaterials with various physico-chemical characterization tools and use them in development of modern technologies				
Recommended Books: 1. Booker, R., Boysen, E., Nanotechnology, Wiley India Pvt, Ltd, (2008) 2. Rogers, B., Pennathur, S., Adams, J., Nanotechnology, CRS Press (2007) 3. Bandyopadhyay, A.K., Nano Materials, New Age Int., (2007) 4. Niemeyer, C. N., and Mirkin, C, A., Nanobiotechnology: Concepts, Applications and				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Perspectives, Wiley VCH, Weinheim, Germany (2007)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

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UEN006 : TECHNOLOGIES FOR SUSTAINABLE DEVELOPMENT				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: To provide acquaintance with modern cleaner production processes and emerging energy technologies; and to facilitate understanding the need and application of green and renewable technologies for sustainable development of the Industry/society.				
Syllabus Concepts of Sustainability and Industrial Processes: Industrialization and sustainable development; Cleaner production (CP) in achieving sustainability; Source reduction techniques - Raw material substitution; Process modification and equipment optimization; Product design or modification; Reuse and recycling strategies; Resources and by-product recovery from wastes; Treatment and disposal; CDM and Pollution prevention programs; Good housekeeping; CP audits, Green Design: Green buildings - benefits and challenges; public policies and market-driven initiatives; Effective green specifications; Energy efficient design; Passive solar design; Green power; Green materials and Leadership in Energy and Environmental Design (LEED) Renewable and Emerging Energy Technologies: Introduction to renewable energy technologies- Solar; wind; tidal; biomass; hydropower; geothermal energy technologies; Emerging concepts; Biomolecules and energy; Fuel cells; Fourth generation energy systems.				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon completion of the course, the students will be able to: 1. comprehend basic concepts in source reduction, waste treatment and management 2. Identify and plan cleaner production flow charts/processes for specific industrial sectors 3. examine and evaluate present and future advancements in emerging and renewable energy technologies				
Recommended Books: 1. Kirkwood, R,C, and Longley, A,J, (Eds.), <i>Clean Technology and the Environment</i>, Chapman & Hall, London (1995), 2. World Bank Group; <i>Pollution Prevention and Abatement Handbook – Towards Cleaner Production</i>, World Bank and UNEP; Washington DC (1998), 3. Modak, P,, Visvanathan, C, and Parasnis, M,, <i>Cleaner Production Audit, Course Material on Cleaner Production and Waste Minimization</i>; United Nations Industrial Development Organization (UNIDP) (1995), 4. Rao, S, and Parulekar, B,B,, <i>Energy Technology: Non-conventional; Renewable and Conventional</i>; Khanna Pub,(2005) 3rd Ed,				

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Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

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UHU017 : INTRODUCTION TO COGNITIVE SCIENCE				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: This course provides an introduction to the study of intelligence, mind and brain from an interdisciplinary perspective, It encompasses the contemporary views of how the mind works, the nature of reason, and how thought processes are reflected in the language we use, Central to the course is the modern computational theory of mind and it specifies the underlying mechanisms through which the brain processes language, thinks thoughts, and develops consciousness.				
Syllabus Overview of Cognitive Science: Newell's big question, Constituent disciplines, Interdisciplinary approach, Unity and diversity of cognitive science, Philosophy: Philosophy of Mind, Cartesian dualism Nativism vs, empiricism, Mind-body problem, Functionalism, Turing Test, Modularity of mind, Consciousness, Phineas Gage, Physicalism. Psychology: Behaviorism vs, cognitive psychology, The cognitive revolution in psychology, Hardware/software distinction , Perception and psychophysics, Visual cognition, Temporal dynamics of visual perception, Pattern recognition, David Marr's computational theory of vision, Learning and memory, Theories of learning, Multiple memory systems, Working Memory and Executive Control, Memory span, Dissociations of short- and long-term memory, Baddeley's working memory model. Linguistics: Components of a grammar, Chomsky, Phrases and constituents, Productivity, Generative grammars, Compositional syntax, Productivity by recursion, Surface- and deep structures, Referential theory of meaning, Compositional semantics, Semantics, Language acquisition, Language and thought. Neuroscience: Brain anatomy, Hierarchical functional organization, Decorticate animals, Neuroimaging, Neurophysiology, Neuron doctrine, Ion channels, Action potentials, Synaptic transmission, Synaptic plasticity, Biological basis of learning, Brain damage, Amnesia, Aphasia, Agnosia, Parallel Distributed Processing(PDP), Computational cognitive neuroscience, The appeal of the PDP approach, Biological Basis of Learning, Cajal's synaptic plasticity hypothesis, Long-term potentiation (LTP) and depotentiation (LTD), NMDA receptors and their role in LTP, Synaptic consolidation, Vertical integration, The Problem of representation, Shannon's information theory. Artificial Intelligence: Turing machines, Physical symbol systems, Symbols and Search Connectionism, Machine Learning,, Weak versus strong AI, Subfields, applications, and recent trends in AI, Turing Test revisited, SHRDLU, Heuristic search, General Problem Solver (GPS), Means-ends analysis. Cognitive architectures: Tripartite architecture, Integration, ACT-R Architecture Modularity.				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon completion of the course, the students will be able to: 1. Identify cognitive science as an interdisciplinary paradigm of study of cross-cutting areas such as Philosophy, Psychology, Neuroscience, Linguistics, Anthropology, and				

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Artificial Intelligence.
2. Explain various processes of the mind such as memory and attention, as well as representational and modelling techniques that are used to build computational models of mental processes;
3. Acquire basic knowledge of neural networks, linguistic formalism, computing theory, and the brain.
4. Apply basic Artificial Intelligence techniques to solve simple problems.
Recommended Books:
1. Bermúdez, J.L., <i>Cognitive Science: An Introduction to the Science of the Mind</i> (2nd Ed.), Cambridge, UK: Cambridge (2014).
2. Friedenberg ,J,D, and Silverman,G, <i>Cognitive Science: An Introduction To The Study Of Mind</i> , Sage Publications:, London (2014)
3. Thagard, P., <i>Mind: An introduction to Cognitive Science</i> , MIT Press, (2005)
Thagard, P., (1998) <i>Mind Readings: Introductory Selections on Cognitive Science</i> , MIT Press, Cambridge, Mass

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

UHU018: INTRODUCTION TO CORPORATE FINANCE				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: This course aims to provide the students with the fundamental concepts, principles and approaches of corporate finance, enable the students to apply relevant principles and approaches in solving problems of corporate finance and help the students improve their overall capacities.				
Syllabus Introduction to corporate finance: Finance and corporate finance. Forms of business organizations, basic types of financial management decisions, the goal of financial management, the agency problem; the role of the financial manager; basic types of financial management decisions. Financial statements analysis: Balance sheet, income statement, cash flow, fund flow financial statement analysis Computing and interpreting financial ratios; conducting trend analysis and Du Pont analysis. The time value of money: Time value of money, future value and compounding, present value and discounting, uneven cash flow and annuity, discounted cash flow valuation. Risk and return: Introduction to systematic and unsystematic risks, computation of risk and return, security market line, capital asset pricing model. Long-term financial planning & Financial Decisions: Various sources of long-term financing, the elements and role of financial planning, financial planning model, percentage of sales approach, external financing needed. Cost of capital, financial leverage, operating leverage. Capital structure, theories of capital structure net income, net operating income & M&M proposition I and II. Short-term financial planning and management: Working capital, operating cycle, cash cycle, cash budget, short-term financial policy, cash management, inventory management, credit management. Capital budgeting: Concepts and procedures of capital budgeting, investment criteria (net present value, payback, discounted payback, average accounting return, internal rate of return, profitability index), incremental cash flows, scenario analysis, sensitivity analysis, break-even analysis, Dividend policy: Dividend, dividend policy, Various models of dividend policy (Residual approach, Walter model, Gordon Model, M&M, Determinants of dividend policy. Security valuation: Bond features, bond valuation, bond yields, bond risks, stock features, common stock valuation, and dividend discount & dividend growth models. Common stock yields, preferred stock valuation.				
Laboratory Work: NA				
Course Learning Objectives (CLO)				
Recommended Books: 1. Brealey, R. A., Myers. S.C., Allen, F., Principles of Corporate Finance (9th edition), The McGraw-Hill, London, (2006). 2. Ehrhardt, M.C., Brigham, E.F., Financial Management: Theory and Practice (10th				

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edition) South Western-Cengage, New York (2011)

3. *Van Horne, J.C., Wachowicz, J.M., Kuhlemeyer, G.A., 2005, Fundamentals of Financial Management, Pearson, Vancouver (2010)*
4. *Pandey, I. M., Financial management, Vikas Publishing House Pvt. Ltd., Noida (2011)*
5. *Elton, E.J. and Gruber, M.J., Modern Portfolio Theory and Investment Analysis, (7th Edition), John Wiley and Sons, New York (2007)*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

UMA069: GRAPH THEORY AND APPLICATIONS				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: The objective of the course is to introduce students with the fundamental concepts in graph Theory, with a sense of some its modern applications. They will be able to use these methods in subsequent courses in the computer, electrical and other engineering.				
Syllabus Introduction: Graph, Finite and infinite graph, incidence and degree, Isolated vertex, Pendant vertex and null graph, Isomorphism, Sub graph, Walks, Paths and circuits, Euler circuit and path, Hamilton path and circuit, Euler formula, Homeomorphic graph, Bipartite graph, Edge connectivity, Computer representation of graph, Digraph. Tree and Fundamental Circuits: Tree, Distance and center in a tree, Binary tree, Spanning tree, Finding all spanning tree of a graph, Minimum spanning tree. Graph and Tree Algorithms: Shortest path algorithms, Shortest path between all pairs of vertices, Depth first search and breadth first of a graph, Huffman coding, Cuts set and cut vertices, Warshall's algorithm, topological sorting. Planar and Dual Graph: Planner graph, Kuratowski's theorem, Representation of planar graph, five-color theorem, Geometric dual. Coloring of Graphs: Chromatic number, Vertex coloring, Edge coloring, Chromatic partitioning, Chromatic polynomial, covering. Application of Graphs and Trees: Konigsberg bridge problem, Utilities problem, Electrical network problem, Seating problem, Chinese postman problem, Shortest path problem, Job sequence problem, Travelling salesman problem, Ranking the participant in a tournament, Graph in switching and coding theory, Time table and exam scheduling, Applications of tree and graph in computer science.				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon completion of the course, the students will be able to: 1) understand the basic concepts of graphs, directed graphs, and weighted graphs and able to present a graph by matrices. 2) understand the properties of trees and able to find a minimal spanning tree for a given weighted graph. 3) understand Eulerian and Hamiltonian graphs. 4) apply shortest path algorithm to solve Chinese Postman Problem. 5) apply the knowledge of graphs to solve the real-life problem.				
Recommended Books: 1. Deo, N., <i>Graph Theory with Application to Engineering with Computer Science</i>, PHI, New Delhi (2007) 2. West, D. B., <i>Introduction to Graph Theory</i>, Pearson Education, London (2008) 3. Bondy, J. A. and Murty, U.S.R., <i>Graph Theory with Applications</i>, North Holland Publication, London (2000)				

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4. Rosen, K. H., *Discrete Mathematics and its Applications*, Tata-McGraw Hill, New Delhi (2007)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

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UMA070 : ADVANCED NUMERICAL METHODS				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: The main objective of this course is to motivate the students to understand and learn various advanced numerical techniques to solve mathematical problems governing various engineering and physical problems.				
Syllabus Non-Linear Equations: Methods for multiple roots, Muller's, Iteration and Newton-Raphson method for non-linear system of equations and Newton-Raphson method for complex roots. Polynomial Equations: Descartes' rule of sign, Birge-vieta, Giraffe's methods. System of Linear Equations: Cholesky and Partition methods, SOR method with optimal relaxation parameters. Eigen-Values and Eigen-Vectors: Similarity transformations, Gerschgorin's bound(s) on eigenvalues, Given's and Rutishauser methods. Interpolation and Approximation: Cubic and B – Spline and bivariate interpolation, Least squares approximations, Gram-Schmidt orthogonalisation process and approximation by orthogonal polynomial, Legendre and Chebyshev polynomials and approximation. Differentiation and Integration: Differentiation and integration using cubic splines, Romberg integration and multiple integrals. Ordinary differential Equations: Milne's, Adams-Moulton and Adam's Bashforth methods with their convergence and stability, Shooting and finite difference methods for second order boundary value problems.				
Laboratory Work: NA				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1) find multiple roots of equation and apply Newton -Raphson's method to obtain complex roots as well solution of system of non - linear equations. 2) learn how to obtain numerical solution of polynomial equations using Birge - Vitae and Giraffe's methods. 3) apply Cholesky, Partition and SOR methods to solve system of linear equations. 4) understand how to approximate the functions using Spline, B- Spline, least square. approximations 5) learn how to solve definite integrals by using cubic spline, Romberg and initial value problems and boundary value problems numerically.				
Recommended Books: 1. Gerald, C.F. and Wheatley, P.O., <i>Applied Numerical Analysis</i>, Pearson Education (2008) 7th ed. 2. Gupta, S.R., <i>Elements of Numerical Analysis</i>, MacMillan India (2009). 3. Atkinson, K.E., <i>An introduction to Numerical Analysis</i>, John Wiley (2004) 2nd ed. 4. S.D. Conte, S.D. and Carl D. Boor, <i>Elementary Numerical Analysis: An Algorithmic</i>				

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Approach, Tata McGraw Hill (2005).

Jain M. K., Iyengar. S.R.K. and Jain, R.K. Numerical Methods for Scientific and Engineering Computation, New Age International (2008) 5th ed

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

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UHU016: INTRODUCTORY COURSE IN FRENCH				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: The objectives of the course is to introduce to the students: 1. The basics of French language to the students. It assumes that the students have minimal or no prior knowledge of the language. 2. To help them acquire skills in writing and speaking in French, comprehending written and spoken French. 3. The students are trained in order to introduce themselves and others, to carry out short conversation, to ask for simple information, to understand and write short and simple messages, to interact in a basic way. 4. The main focus of the students will be on real life language use, integration of French and francophone culture, & basic phrases aimed at the satisfaction of needs of concrete type. 5. During class time the students are expected to engage in group & pair work.				
Syllabus Communicative skills: Greetings and Its Usage, Asking for and giving personal information, How to ask and answer questions, How to talk over the phone, Exchange simple information on preference, feelings etc. Invite, accept, or refuse invitation, Fix an appointment, Describe the weather, Ask for/give explanations, Describe a person, an object, an event, a place. Grammar : Pronouns: Pronom sujets (Je/ Tu/Il/Elle/Nous/Vous/Ils/Elles), Nouns: Genders, Articles: Definite article and Indefinite articles, Verbs: Regular verbs (-er, -ir ending) Irregular verbs (-re ending), Auxiliary verbs (avoir, être, aller). Adjective: Description, Adjective possessive, Simple Negation, Tense: Present, Future, Questions, Singular & plural. Vocabulary: Countries and Nationalities, Professions, Numbers (ordinal, cardinal), Colours, Food and drinks, Days of the week, Months, Family, Places. Phonetics: The course develops the ability, to pronounce words, say sentences, questions and give orders using the right accent and intonation. To express surprise, doubt, fear, and all positive or negative feelings using the right intonation. To distinguish voiced and unvoiced consonants. To distinguish between vowel sounds.				
Laboratory Work: NA				
Upon the completion of the course: 1. The students begin to communicate in simple everyday situations acquiring basic grammatical structure and vocabulary. 2. The course develops oral and reading comprehension skills as well as speaking and writing. 3. Students can demonstrate understanding of simple information in a variety of authentic materials such as posters, advertisement, signs etc. 4. Discuss different professions, courses and areas of specialisation. 5. Write simple messages, letters, composition and dialogues. Complete simple forms				

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and documents.

6. Express feelings, preferences, wishes and opinions and display basic awareness of francophone studies.
7. Units on pronunciation and spelling expose students to the different sounds in the French language and how they are transcribed.

Recommended Books:

1. *Alter ego-1 : Méthode de français* by Annie Berthet, Catherine Hugot, Véronique M. Kizirion, Beatrix Sampsonis, Monique Waendendries, Editions Hachette français langue étrangère.
2. *Connexions-1 : Méthode de français* by Régine Mérieux, Yves Loiseau, Editions Didier
3. *Version Originale-1: Méthode de français* by Monique Denyer, Agustin Garmendia.
4. *Marie-Laure Lions-Olivieri*, Editions Maison des Langues, Paris 2009
5. *Latitudes-1 : Méthode de français* by Régine Mérieux, Yves Loiseau, Editions Didier
6. *Campus-1 : Méthode de français* by Jacky Girardet, Jacques Pécheur, Editions CLE International.
7. *Echo-1 : Méthode de français* by J. Girardet, J. Pécheur, Editions CLE International.

UBT510: BIOLOGY FOR ENGINEERS				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: To learn about living world and basic functioning of biological systems. The course encompasses understanding of origin of life, its evolution and some of its central characteristics. It also aims to familiarize engineering students to some of the intricate biological phenomena and mechanisms.				
Syllabus Characteristics of life: Living versus non-living organisms, origin of life, theory of evolution, diversity of life, classification of life into animals, plants, fungi, protists, archaea and bacteria. Phylogenetics and its relationship with evolution. Introduction to biological systems: Cell as basic unit of life, cellular organelles and their functions, important biomacromolecules (carbohydrates, lipids, proteins and nucleic acids) and their properties. Cell membrane: Membrane structure, selective permeability, transport across cell membrane, active and passive transport, membrane proteins, type of transport proteins, channels and pumps, examples of membrane transport in cell physiology. Classical and molecular genetics: Heredity and laws of genetics, genetic material and genetic information, Structure and properties of DNA, central dogma, replication of genetic information, universal codon system, encoding of genetic information via transcription and translation.				
Laboratory Work: NA				
After completion of this course the students will be able to: 1. Describe living-systems and differentiate them from non-living systems 2. Explain the theory of evolution and apply it non-living world 3. Apply properties of nucleic acids in molecular recognition-based diagnostics 4. Familiarized with various transport mechanisms across cell membranes Explain how genetic information is stored, replicated and encoded in living				
Recommended Books: 1. Nelson, D.L., Cox, M.M., Lehninger: Principles of Biochemistry, WH Freeman (2008) 5th ed. 2. Dhami, P.S., Srivastava, H.N. Chopra, G., A Textbook of Biology, Pradeep Publications (2008). 3. Das, H.K., Textbook of Biotechnology, John Wiley & Sons (2004) 3rd Edition. 4. Gardner, E.J., Simmons, M., Peter, S.D., Principles of Genetics, John Wiley & Sons (2008) 5. Albert, B., Essential Cell Biology, Taylor & Francis, London (2009)				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

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UCS002: INTRODUCTION TO CYBER SECURITY				
	L	T	P	Cr
	2	0	0	2.0
Course Objective: In this course, the student will learn about the essential building blocks and basic concepts around cyber security such as Confidentiality, Integrity, Availability, Authentication, Authorization, Vulnerability, Threat and Risk and so on.				
Syllabus Introduction: Introduction to Computer Security, Threats, Harm, Vulnerabilities, Controls, Authentication, Access Control, and Cryptography, Authentication, Access Control, Cryptography Programs and Programming: Unintentional (Non-malicious) Programming Oversights, Malicious Code—Malware, Countermeasures Web Security: User Side, Browser Attacks, Web Attacks Targeting Users, Obtaining User or Website Data, Email Attacks Operating Systems Security: Security in Operating Systems, Security in the Design of Operating Systems, Rootkit Network Security: Network Concepts, Threats to Network Communications, Wireless Network Security, Denial of Service, Distributed Denial-of-Service Strategic Defenses: Security Countermeasures, Cryptography in Network Security, Firewalls, Intrusion Detection and Prevention Systems, Network Management Cloud Computing and Security: Cloud Computing Concepts, Moving to the Cloud, Cloud Security Tools and Techniques, Cloud Identity Management, Securing IaaS Privacy: Privacy Concepts, Privacy Principles and Policies, Authentication and Privacy, Data Mining, Privacy on the Web, Email Security, Privacy Impacts of Emerging Technologies, Where the Field Is Headed Management and Incidents: Security Planning, Business Continuity Planning, Handling Incidents, Risk Analysis, Dealing with Disaster Legal Issues and Ethics: Protecting Programs and Data, Information and the Law, Rights of Employees and Employers, Redress for Software Failures, Computer Crime, Ethical Issues in Computer Security, Incident Analysis with Ethics Emerging Topics: The Internet of Things, Economics, Computerized Elections, Cyber Warfare.				
Laboratory Work: NA				
Course Learning Objectives (CLO) After completion of this course, the students will be able to: 1. Understand the broad set of technical, social & political aspects of Cyber Security and security management methods to maintain security protection 2. Appreciate the vulnerabilities and threats posed by criminals, terrorist and nation states to national infrastructure 3. Understand the nature of secure software development and operating systems 4. Recognize the role security management plays in cyber security defense and legal and social issues at play in developing solutions.				

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Recommended Books:

1. Pfleeger, C.P., *Security in Computing*, Prentice Hall, 5th edition (2010)
2. Schneier, B., *Applied Cryptography*, Second Edition, John Wiley & Sons (1996)
3. Rhodes-Ousley, M., *Information Security: The Complete Reference, Second Edition, Information Security Management: Concepts and Practice*. New York, McGraw-Hill, (2013).
4. Whitman, M.E. and Herbert J. M., *Roadmap to Information Security for IT and Infosec Managers*, Course Technology, Boston, MA (2011).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

INDUSTRY COLLABORATIVE ELECTIVES

UCS539: Finance, Accounting and Valuation				
	L	T	P	Cr
	2	0	2	3
Course Objective: Understanding relationship of finance, accounting and valuation of securities.				
Syllabus Introduction to Accounting: Meaning of accounting, the accounting process, fundamental equation, types of accounts, accounting statements, recording of transactions, conceptual framework Summary Statements: Types of summary statements, preparation of the statements, relationship between the statements, introduction to financial statement analysis Basics of Finance: Meaning of finance, process of financial decision making, types of financial decisions, capital structure decisions. Time Value of Money: Meaning, principle, calculations, interest rates, importance of interest rates, importance of different types of interest rates and returns. Valuation: Introduction to valuation, valuation of stocks, valuation of bonds, methods and techniques				
Laboratory Work: To gain an understanding of proprietary software for international derivatives 1. Introduction to proprietary software. 2. Detailed understanding of basic features of proprietary software. 3. Understanding order types and their implementation 4. Regular practice to understand execution of basic strategies. 5. Introduction to trading and investment. 6. Introduction to analytical methods				
Course Learning Objectives (CLO) The students will be able to: 1. Explain the basic accounting concepts and apply the fundamental equation in basic business transactions 2. Explicate and apply the techniques learnt for doing financial statement analysis 3. Explain various financial decisions and evaluate some of them 4. Explicate the principle of time value of money (TVP) and importance of interest rates in TVP 5. Apply the methods learnt for valuation of securities				
Recommended Prerequisites: Basics of Microsoft Excel				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Reference Books

1. *Jamie Pratt. (8th Edition). Financial Accounting in an Economic Context.*
2. *Ross, Westerfield, Jaffe & Jordan. Corporate Finance: Core Principles and Applications.*

UCS537: SOURCE CODE MANAGEMENT				
	L	T	P	Cr
	2	0	2	3
Course Objective: The objective of the course is to teach techniques to combine software development and IT operations using DevOps. It helps to understand faster software development practices with higher quality.				
Syllabus Traditional Software Development: The Advent of Software Engineering, Waterfall method, Developers vs IT Operations conflict. Rise of Agile methodologies: Agile Vs Waterfall Method, Iterative Agile Software Development, Individual and team interactions over processes and tools, working software over comprehensive documentation, Customer collaboration over contract negotiation, responding to change over following a plan Definition and Purpose of DevOps: Introduction to DevOps, DevOps and Agile, Minimum Viable Product, Application Deployment, Continuous Integration, Continuous Delivery CAMS (Culture, Automation, Measurement and Sharing): CAMS – Culture, Automation, Measurement, Sharing, Test-Driven Development, Configuration Management, Infrastructure Automation, Root Cause Analysis, Blamelessness, Organizational Learning. Typical Toolkit for DevOps: Introduction to continuous integration and deployment, Version control system Source Code Management History and Overview: Examples - SVN, Mercury and Git, History - Linux and Git by Linus Torvalds, Version Control System: Version control system vs Distributed version control system: Local repository, Advantages of distributed version control system, The Multiple Repositories Models, completely resetting local environment, Revert - cancelling out changes.				
Laboratory Work: Basic structure and Implementation of various distributed version control systems for source code management.				
Course Learning Objectives (CLO) After completion of this course, the students will be able to: 1. Identify the need for migrating from traditional software development to Agile model and then to DevOps. 2. Define and understand the basic principles and need of DevOps and Continuous Delivery. 3. Understand the history and overview of Source Code Management, along with real-time examples. 4. Differentiate between centralized and distributed version control systems and basic operations in version control systems and Demonstrate the use of various version control systems.				
Text Books 1. <i>The DevOps Handbook</i> - Book by Gene Kim, Jez Humble, Patrick Debois, and Willis				

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Willis.

2. *Pro Git* – Book by Scott Chacon and Ben Straub (available at <https://git-scm.com/book/>).

Reference Books:

1. *What is DevOps?* - by Mike Loukides.

UCS660: Continuous Integration and Continuous Deployment				
	L	T	P	Cr
	2	0	2	3
Course Objective: The objective of the course is to teach techniques to automate the process of integration and deployment software product. It covers prerequisites, anatomy and framework/tools used for the automated process of continuous integration and continuous deployment.				
Syllabus DevOps Automation: Phases in software development-delivery pipeline, components of automated software delivery, RAD model and model driven architecture. Automation Benefits: advantages of automation, Time and efforts saving scenarios, error preventing scenarios. Continuous Integration and Continuous Deployment Introduction: Overview and practices of continuous integration, working mechanism and benefits of continuous integration; continuous delivery's introduction and pipeline. Prerequisites and benefits, introduction and business drivers of continuous deployment, benefits of continuous deployment. Stages and Anatomy of CI CD: Core continuous integration process and advanced continuous integration process, release process, continuous delivery engineering practices, continuous testing & promotion of builds, continuous monitoring of delivery pipeline, understanding continuous feedback process. Testing, Debugging and Refactoring: Understanding test-driven development (TDD), categories of TDD, Junit framework, need for code refactoring, its process and strategies. Understanding Framework and Tools: Common frameworks and code architectures, third party code, IDEs (Eclipse, Netbeans and IntelliJ), common mistakes and avoiding them, issues with making code IDE dependent. Laboratory Work: Setting up Jenkins, Jenkins job, parameters, build, post-build actions and pipeline; Jenkins plugins, using Jenkins as a continuous integration server; Configuring Jenkins with git plugin; Jenkins pipeline to poll the feature branch. Course Learning Objectives (CLO) After completion of this course, the students will be able to: 1. Understand the phases of software development-delivery pipeline and automation benefits. 2. Identify and apply continuous integration and deployment prerequisites, process and benefits. 3. Understand and implement the continuous delivery engineering practices and release process. 4. Identify & use the test-driven deployment and various tools/frameworks used for continuous integration and delivery in DevOps Text Books 1. Gene Kim, Jez Humble, Patrick Debois, John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Technology Organizations”, IT revolution Press (2016) 1st ed.

Reference Books:

1. Sander Rossel, “Continuous Integration, Delivery, and Deployment: Reliable and Faster Software Releases with Automating Builds, Tests, and Deployment”, Packt Publishing (2017) 1st ed.
2. Online material available at:<https://digitallearn.xebiaacademyglobal.com/>

UCS659: BUILD AND RELEASE MANAGEMENT				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course includes theory and lab. The course comprises four modules. The main objective of this course to help participants understand the process of build and release management.				
Syllabus Introduction to Build and Release Management: Introduction to build, understanding different phases of build and release management, introduction to release management, best practices for build and release management, concept of build abstraction and dependency abstraction. Dependency Management: Introduction to dependency management, how to use source code repositories, managing transitive dependencies, dependency scope and discussion of various tools like Ant, Maven and Gradle. Document and Reporting: Introduction to build document and reporting, different types of documentation, understanding site life cycle, advance site configurations and reports, generation of unit test reports, generation of code coverage reports, code coverage tools, code coverage pros and cons. Release Cycle: To understand project release life cycle, different stages of release lifecycle, source code repositories, how to install and configure source code repositories and deploying build to production goals- prepare, perform, clean and rollback. Laboratory Work: Setting up Maven environment and understanding POM hierarchy, creation of a project using Maven and its configurations.				
Course Learning Objectives (CLO) After completion of this course, the students will be able to: 1. Explain the basics of build and release management by learning build abstraction and declarative dependency management. 2. Describe dependency management and the associated concepts like repositories, dependency identification and scope, transitive dependencies, and the examples for build tools. 3. Discuss the process of documentation and reporting, using site life cycle, site configuration and generation of unit testing and code coverage reports 4. Define release cycle and the phases of release, preparing, cleaning and performing goals.				

UCS675: Financial Markets and Portfolio Theory				
	L	T	P	Cr
	2	0	2	3
Course Objective: Understanding various financial markets and their interrelationships.				
Syllabus Banks and Financial Institutions: Types of financial institutions, evolution of financial system, flow of money, creation of money Monetary System: Monetary authority, monetary policy framework, policy tools, comparison of different countries Risk and Return: Meaning of risk, meaning of return, estimation of risk and return Capital and Money Markets: Meaning, types, capital and money market instruments Portfolio Theory: Meaning of portfolio, theoretical principles, choices, optimal weights, optimal portfolio choice, introduction to pricing models				
Laboratory Work: To use the proprietary software in live international derivatives 1. Introduction to an international derivatives product 2. Introduction to technical analysis 3. Practice of analytical methods on the proprietary software 4. Introduction to evaluation methods				
Recommended Prerequisites: Course – Finance, Accounting and Valuation;				
Course Learning Objectives (CLO) After the completion of the course, the student will be able to: 1. Explain the role of the financial system and the process of creation of money 2. Explicate various monetary policy tools 3. Explain relationship between risk and return 4. Explicate the types of capital markets and money markets including the market instruments 5. Apply the portfolio theory to choose an optimal portfolio				
Reference Books: 1. <i>Financial Markets and Institutions</i> – Anthony Saunders & Marcia Millon Cornett 2. <i>Ross, Westerfield, Jaffe & Jordan. Corporate Finance: Core Principles and Applications.</i>				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

UCS658: Derivatives Pricing, Trading and Strategies				
	L	T	P	Cr
	2	0	2	3
Course Objective: Understanding methods of valuation and strategies of trading derivative instruments.				
Syllabus Law of One Price: Meaning, implication of the law of one price, no arbitrage model, usage in pricing of securities and derivative instruments Pricing and Valuation: Basic principles, building blocks, assumptions, difference between price and value, pricing and valuation of basic derivative instruments Basics of Option Pricing: Meaning of options, types of options, difference between options and basic derivative instruments				
Laboratory Work: To use the proprietary software in live international derivatives 1. Introduction to derivatives strategies 2. Introduction to an additional international derivatives product 3. Learning and creating trading strategies 4. Practicing the strategies on the proprietary software 5. Understanding the role of derivatives in risk reduction				
Recommended Prerequisites: Courses Finance, Accounting and Valuation + Financial Markets and Portfolio Theory.				
Course Learning Objectives (CLO) After the completion of the course, the student will be able to: 1. Explain and apply the law of one price in pricing models 2. Explicate and apply formulae to calculate price and value of a generic forward contract 3. Explain and apply formulae to calculate price and value of basic derivative instruments 4. Explicate the meaning and types of options 5. Apply the basic methods to calculate option prices				
Reference Books: 1. <i>Fundamentals of Futures and Options Markets</i> - John C. Hull 2. Ross, Westerfield, Jaffe & Jordan. <i>Corporate Finance: Core Principles and Applications</i>.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

UMC743: Quantitative and Statistical Methods for Finance				
	L	T	P	Cr
	2	0	2	3
Course Objective: Understanding quantitative and statistical methods used for finance and derivatives.				
Syllabus Refresher on Statistics: Correlation, OLS regression, probability distributions and moments, using Microsoft Excel for statistical calculations and interpretations Option Pricing Introduction to option pricing models, formulae and derivation, option Greeks, risk management using options Financial Time Series: Introduction to time series, types, univariate and multivariate time series models, autocorrelation, AR models, MA models, ARMA models, ARIMA models, stationary series, unit-root Volatility: Meaning of volatility, types, methods of calculation, volatility models, estimation				
Laboratory Work: To use the proprietary software in live international derivatives 1. Introduction to other derivative products 2. Refining trading strategies created in previous courses. 3. Practicing the strategies on the proprietary software. 4. Application of the quantitative and statistical methods 5. Introduction to algorithmic trading on the proprietary software				
Recommended Prerequisites: Courses – Finance, Accounting and Valuation + Financial Markets and Portfolio Theory + Derivatives Pricing, Trading and Strategies				
Course Learning Objectives (CLO) After the completion of the course, the student will be able to: 1. Explain and apply the time series models learnt for financial data 2. Explicate and apply PCA for financial data 3. Explain and apply option pricing models to calculate prices of options 4. Explicate types of volatility and volatility models 5. Apply the basic methods learnt to estimate volatility				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Reference Books:

- 1. Introduction to Time Series Analysis and Forecasting - Douglas C. Montgomery, Cheryl L. Jennings*
- 2. Fundamentals of Futures and Options Markets - John C. Hull*

UCS758: System Provisioning and Configuration Management				
	L	T	P	Cr
	2	0	2	3
Syllabus Understanding Containers: Transporting goods analogy and its problems, Containerization platform, images and runtime, comparison with virtual machine, chroot system call, FreeBSD Jails, Linux containers (LXC), Docker. Introduction to Containerization: Docker architecture, different environments (Dev, QA and Prod), overcoming issues with different environments, virtual machine for dev/deployments, containers for dev/deployments, advantages and drawbacks of containerization. Orchestration Tools: Orchestration: its definition and need, Docker swarm and Kubernetes, AWS (ECS and EKS), Kubernetes on cloud, monitoring containers and its process. Introduction to Provisioning: Basic and software definition, provisioning concepts, reason for exclusive provisioning, configuration management definition and tools, difference between provisioning and configuration management, provisioning tools, test machines for provisioning, deployment and its relationship with provisioning. On Premise Provisioning: Understanding and Defining On Premise, On Premise provisioning infrastructure, Templating, server templating and its challenges. Provisioning on Cloud: defining cloud provisioning, types of cloud provisioning, life-cycle of provisioning on cloud, On Premise cloud mitigation strategies, network security enablement from On Premises to cloud, micro-services management in cloud. Provisioning and Configuration Management: State of tools in provisioning and configuration, definition and need for configuration management, its benefits and drawbacks in DevOps, need for monitoring in DevOps, reasons for using provisioning and configuration tools, automation, preventing errors and tracking changes, examples of tools and their capabilities.				
Laboratory Work: System Provisioning: Automation of infrastructure, AWS configuration for Terraform, create IAM User, security group, spinning up with EC2 instance, variables, resources, modules, state management, VPC, IAM policy, S3 bucket and its variables. Containers Lab: Playing with Vagrant and understanding its file, Docker machine, Dockerfile, Docker extras, DTR, Docker compose and swarm, Kubernetes -Minikube, deploying Pods and services on Minikube.				
Course Learning Objectives (CLO) After the completion of the course the student will be able:				

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1. Understand the concept of Virtualization and Containerization.
2. Familiarize with Orchestration and System provisioning.
3. Analyze and demonstrate the infrastructure automation and state management in the cloud environment.
4. Understand and demonstrate the need for configuration management

Text Book:

Course material by Xebia, Xebia Academy: Log in to the site (learnhub.xebia.com).

Gerardus Blokdyk, *Server Provisioning and Configuration Management Complete Self-Assessment Guide*, 5STARCooks, 2018.

Reference Book:

Erika Heidi, *An Introduction to Configuration Management, Tutorials*, 2019.

UCS542: UI & UX SPECIALIST

	L	T	P	Cr
	2	0	2	3

Course Objective: The main objective of this course is to impart concepts related with web technology which are essential for the development of web applications. The key technology components include web markup languages, client and server side programming.

Syllabus

HTML& CSS: Introduction to HTML, Introduction, HTML Page Structure, Create HTML document, Understand the various elements available in HTML, HTML Use, Attributes in HTML, Need of Attributes, Common Attributes, HTML forms, Apply validations to the form elements, Creating web pages with HTML5, HTML5 introduced features, HTML5 form validate/no validate, HTML5 canvas, embedding audio, and video in a webpage, drag and drop, HTML5 Local Storage, HTML5 web workers and server sent events. What is CSS, how to insert CSS in HTML, How CSS adds value to HTML, Difference between Semantic and HTML mark-up, CSS3, CSS Selectors, Buttons, CSS float and clear, CSS align - horizontal and center, CSS Padding, CSS Links, CSS Lists, CSS Tables.

JavaScript: What is JavaScript, Importance of JavaScript, What can JavaScript Do?, JavaScript with HTML Attributes, JavaScript with CSS, Operators, JavaScript Syntax, JavaScript Data Types, JavaScript Functions, Setting up Environment, Variables, Control flow, if. Else, switch, loops, JavaScript HTML DOM Elements, JavaScript Syntax, Operators, Data Types, JavaScript String Methods, JavaScript Functions, Arrays, Sorting, Joins, Reduce map.

Frontend Architecture: Introduction to Frontend Development, History, MVC, MVP, MVVM& Web Apps, Development of AJAX, Introduction to DOM, Basic DOM Manipulation, Reactive Programming, Refreshing ES6 Specifications and Features ECMA Script, ES6 let and const, The arrow functions, New Literal Syntax, Classes, Inheritance using extends, Default Parameter Values, Spread Operator (...), Iterators and Generators, Features of React, Practical Application, Why need React, How React Works, Leveraging Virtual DOM, Setting up React.

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REST API, JSX: Why JSX, Embedding JavaScript Expression in JSX, JSX Attributes, JSX Comments, Styling and Representation as Object, The State of the Component, Changing the State, Props of Component, Using Props, Props Validation in React, Similarities Between State & Props
RESTAPI : Intro to API, History of API Development, Development of AJAX, CRUD; GraphQL; HTTP ,HTTP 1.1,HTTP/2, Stream prioritization, Introduction to React Native, Setting up React Native, The Expo Client, Working up on the First Project, Style, Flexbox Layout.
Node.js: Introduction to Node.js, History, Why Node.js, Node.js Architecture, Features, Working of Node.js, Installation & Setting Up Node, setting up React, REPL Environment, REPL Commands, Variable, Components of Node.js, Local Modules, Module Exports:Export Object, Export Class, Loading Module from Separate Folder, Operating System, File Systems.

Laboratory Work:

Course Learning Objectives (CLO)

After the completion of the course the student will be able:

1. To acquire knowledge and skills for creation of full web framework considering both client and server side programming.
2. To develop a dynamic and interactive WebPages by the use of HTML, XHTML, CSS, Java script and DHTML.
3. To build web apps with fully functional frontend with database connectivity.
4. To undertake transfer of multimedia content over the Internet between the multi platforms web applications through communication protocols and the learner will also be able to build web API.

Text Books

1. M. Srinivasan, *Web Technology: Theory and Practice*, Pearson (2012).
2. A.Kumar, *Web Technology: Theory and Practice*, CRC Press (2018).

Reference Books:

1. *Web programming with HTML, XHTML and CSS, 2e*, Jon Duckett, Wiley India.
2. *Web technologies: A Computer Science Perspective*, Jeffrey C. Jackson ,1st ed., Pearson (2007).

UCS677: DATA ENGINEERING				
	L	T	P	Cr
	2	0	2	3
Course Objective: Basic concepts of database, Mongo DB, SQL, and Java script.				
Syllabus Getting started with MongoDB: No SQL Databases, Features of MongoDB, Installation overview, Documents, Collections, Databases, What is the NoSQL approach? Why Use the NoSQL Approach, Benefits of No SQL, Types of Databases, Key-Value Stores, Wide-column Stores/ Columnar Databases, Document/Document-store/Document-oriented Databases, Graph-based Databases, Starting and stopping MongoDB Javascript in MongoDB: Javascript in MongoDB, Execution of a JavaScript file in MongoDB, Making the output of find readable in shell, Complementary Terms, Installation, Basic commands on mongo shell, HelloWorld, Create, Update Delete, Read, Update of embedded documents, more update operators, Updating multiple documents. Collections: List all collections in the database, List all databases, Find(), FindOne(), limit, skip, sort and count the results of the find() method, Query Document – Using AND, OR and IN Conditions, find() method with Projection, Find() method with Projection, \$set operator to update specified field(s) in document(s), Insert a document, Create a Collection, Drop Collection, Aggregation Indexes: Indexes, Index Creation Basics, Dropping/Deleting an Index, Sparse indexes and Partial indexes, Get Indices of a Collection, Compound, Unique Index, Single field, Delete, List, Mongoas Shards Sharding Environment Setup: Managing Database for Availability and Performance, Database Scaling, Database Distribution Models, Database Replication, Types of Database Replication, Master-Slave Replication, Peer-to-Peer Replication, Advantages and Disadvantages of Peer-to-Peer Replication, Introduction to Sharding, Why Sharding, The Lookup Strategy, Basic configuration with three nodes, Mongo as a Replica Set, Mongoose.				
Laboratory Work:				
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Learn NoSQL, Mongo DB, and Javascript. 2. Apply basic concepts of indexing and collections using MongoDB. 3. Formulate the basic usage of database such as graph-based database, creating and dropping of collections, and indexing and solving real-world problems. 4. Able to handle and update multiple documents using Mongo DB. 5. Able to manage database for availability and performance, and scaling of database.				

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Text Books

1. Andreas Kretz, *The Data Engineering Cook Book*, 6th ed. (2019)
2. Alex Petrov, *Database Internals: A deep dive into how distributed data systems work*, O'REILLY Publication (2021).

Reference Books:

1. S. Bradshaw, Eoin Brazil, and Kristina Chodorow, *MongoDB: The definite guide: Powerful and Scalable Data Storage*, O'REILLY Publication (2021).
2. Dan Sullivan, *NoSQL for Mere Mortals*, O'REILLY Publication (2021).

UCS662: TEST AUTOMATION				
	L	T	P	Cr
	2	0	2	3
Course Objective: The course provides understanding of software testing and how to use various tools (like Selenium and TestNG etc.) used for automation of software testing.				
Syllabus Introduction to Software Testing: Seven principles of Software Testing, SDLC vs STLC, Testing Life Cycle, Usability Testing, why do we need Usability Testing, how to do Usability testing, Advantages & Disadvantages, Functional Testing, End to End Testing, Methods, Advantages & Disadvantages, Compatibility Testing, Types GUI testing, Techniques API testing, Advantages Test Automation: Selenium: Selenium components, Selenium Architecture, TestNG: Installing TestNG in Eclipse, TestNG Annotations – Understanding usage, setting priority of execution for test cases, Hard Assertion, Soft Assertion, TestNG Reports, ANT-Downloading & Configuring, XSLT report generation using TestNG and Ant. Introduction to Selenium 3.x: Describe Selenium 3.x advantages and implementation, Define drivers for Firefox, IE, chrome, iPhone, Android etc. Analyze first Selenium Code, differentiate between Close and Quit, Describe Firepath and firebug Add-ons installation in Mozilla, inspect elements in Mozilla, Chrome and IE, Identifying Web Elements using id, name, class, Generate own CSS Selectors. Differentiate between performance of CSS Selectors as compared to Xpaths, define class attribute, Handle Dynamic objects/ids on the page, Analyze whether object is present on page or not Manual Testing: Manual Testing, Manual Testing – How to Approach? Manual Testing – Myth and fallacy, Defect Life Cycle, Qualities of a good Manual Tester, Manual Testing Vs Automation Testing, Types, System Testing, Acceptance Testing, Unit Testing, Techniques, Integration Testing, Smoke- Sanity Testing. Introduction to Test Design: Test Scenario, Test Case Design, Test Basis Traceability Matrix				
Laboratory Work:				
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Understand the concepts related to software testing and test automation. 2. Take into account the different considerations when planning automated tests vs. manual tests 3. Architect the test project and fit it to the architecture of the tested application 4. Design and implement highly reliable automated tests 5. Understand how different types of automated tests will fit into your testing strategy, including unit testing, load and performance testing, visual testing, and more				
Text Books				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

1. Axelrod Arnon, *Complete Guide to Test Automation: Techniques, Practices, and Patterns for Building and Maintaining Effective Software Projects*, A press (2018).
2. Gundecha U. and Cocchiaro C., *Learn Selenium: Build data-driven test frameworks for mobile and web applications with Selenium Web Driver 3*, Packt (2019).

Reference Books:

1. Diego Molina, *Selenium Fundamentals*, Packt (2018).
2. Aditya P. Mathur, *Foundations of Software Testing*, Pearson Education(2008).

UCS745: CLOUD & DEVOPS				
	L	T	P	Cr
	2	0	2	3
Course Objective: The objective of the course is to teach techniques to automate the process of integration and deployment software product.				
Syllabus Introduction to DevOps: Definition of DevOps, Challenges of traditional IT systems & processes, History and emergence of DevOps, DevOps definition and principles governing DevOps, DevOps and Agile, The need for building a business use case for DevOps, Purpose of DevOps, Application Deployment, Automated Application Deployment, Application Release Automation (ARA), Components of Application Release Automation (ARA), Continuous Integration, Best Practices of CI, Benefits of CI, Continuous Delivery Typical Toolkit for DevOps: DevOps, An Overview, Achieving DevOps, Continuous Practices, Continuous Integration (CI), How does CI Work? Continuous Integration Practices, Benefits of Continuous Integration A Quick Recap of Continuous Delivery, Continuous Delivery Process, Benefits of Continuous Delivery, Continuous Deployment Source Code Management: History of Version Control Systems (VCS), Basic operations in a VCS, Examples of version control systems, Subversion (SVN), Features and Limitations, Mercurial, Git, Overview, History - Linux and Git by Linus Torvalds, Advantages of Git, Explain how local version control works, Centralized Version Control Systems (CVCS), Distributed Version Control Systems (DVCS), advantages of DVCS, Private Workspace, Easier merging, Easy to scale horizontally, List the disadvantages of DVCS, Explain how CVCS and DVCS compare with each other, Describe the working of the multiple repositories model Unit IV Application Containerization Understanding Containers: Transporting Goods Analogy, Problems in Shipping Industry before Containers, Shipping Industry Challenges, Container: Virtualization Introduction, Hypervisor, Scope of Virtualizations, Containers vs Virtual Machines, Understanding Containers, Containerizations Platform, Runtime and Images, Container Platform, Container Runtime, The Chroot System, FreeBSD Jails, Linux Containers (LXC), Docker Introduction to Containerization: Docker architecture, Docker Daemon (Container Platform), Docker Rest API, CLIDifferent environments: (Dev, QA and Prod), Overcoming issues with different environments, Development Environment Docker Swarm and Kubernetes, Architecture, AWS (ECS, EKS), AWS Elastic Container Services Architecture, Azure Kubernetes Services, Openshift, Kubernetes on cloud, Monitoring of container				
Laboratory Work:				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively.

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Understand the benefits of DevOps over other software development processes.
2. Understand the phases of software development-delivery pipeline and automation benefits.
3. Identify and apply continuous integration and deployment prerequisites, process and benefits.
4. Understand and implement the continuous delivery engineering practices and release process.
5. Identify & use the test-driven deployment and various tools/frameworks used for continuous integration and delivery in DevOps.
6. Demonstrate the different DevOps Tools like Git, Docker, and Kubernetes etc.

Text Books

1. Arundel, J., & Domingus, J., *Cloud Native DevOps with Kubernetes: building, deploying, and scaling modern applications in the cloud*. O'Reilly Media, (2019).
2. Kim, G., Humble, J., Debois, P., & Willis, J., *The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations*. IT revolution Press 1st ed(2016).
3. Bass, L., Weber, I., & Zhu, L., *DevOps: A software architect's perspective*. Addison-Wesley Professional (2015).

Reference Books:

1. Fox, A., Patterson, D. and Joseph, S., *Engineering Software as a Service: An Agile Approach Using Cloud Computing*, 1st Edition (2013).
2. Rossel, S., *Continuous Integration, Delivery, and Deployment: Reliable and Faster Software Releases with Automating Builds, Tests, and Deployment*. Packt Publishing, 1st ed (2017).

Note: Color coding for NVIDIA baskets:

Teaching Kit (Audio or PPT only) / NVIDIA DLI (Online Courses)

Manual Teaching (NVIDIA/Fore Solutions)

Self-Learn From Online Material (Open Source Reading/Lectures)

UCS551: Conversational AI: Accelerated Data Science				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with fundamental and advanced methods of data science: from Data Collection to Analytics and Machine Learning with RAPIDS on Text and Graphical problems.				
Syllabus Introduction: Fundamentals of Data Science, GPU Acceleration, RAPIDS Framework Data Collection: Collecting Data, Scraping Data, Popular Scraping libraries, Data Annotation Data Pre-processing (ETL): Introduction to Data-preprocessing, Data Cleaning & Statistical Preprocessing, Data Cleaners: OpenRefine and Wrangler, Feature Selection: Introduction to Filter Methods, Introduction to Model- based methods, Feature Reduction: PCA. Introduction to Machine Learning - Supervised: Introduction to Supervised Learning, Linear Model, RAPIDS acceleration: Linear Regression, Overfitting and Cross Validation, Decision Tree, Visualizing Classification: {ROC, AUC, Confusion Matrix}, Bagging, Random Forests, RAPIDS Acceleration: Random Forest, Boosting, RAPIDS acceleration: K- NN, XGBoost. Introduction to Machine Learning - Unsupervised: Introduction to Unsupervised Learning, Kmeans & Hierarchical Clustering, RAPIDS acceleration: K-Means, DBSCAN, PCA, t-SNE, UMAP, Visualizing Clusters, RAPIDS acceleration: PCA [t-SNE], UMAP, DBSCAN. Graph Analytics: How to Represent & Store Graphs, Graph Power Laws, Centralities: Degree, Betweenness, Clustering Coefficient, PageRank & Personalized PageRank, Interactive Graph Exploration, RAPIDS Acceleration: Graphistry & cuXFilter. Introduction to Deep Learning: Introduction to Artificial Neural Networks (ANNs), Artificial Neurons, Layers, Perceptron, Multilayer Perceptron, Advanced Deep Neural Networks (DNNs), Batch Normalization, Hyperparameter tuning, Activation Functions, Metrics, Optimization, Regularization.				
Laboratory Work: 1. Introduction to Dockers & Containers, Introduction to NVIDIA GPU Cloud (NGC). 2. Practical on Traditional Data Science packages (Numpy, Pandas, Scipy, Scikit-Learn). 3. Accelerated Data Science framework RAPIDS: Introduction to RAPIDS and cuDF. 4. Data Collection via API/Web Scraping. 5. Decision Tree Classification Clustering in RAPIDS.				

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6. Random Forest Classification in RAPIDS. 7. KMeans Clustering Implementation in RAPIDS. 8. Dimensionality Reduction and Visualization in RAPIDS. 9. Graph Analytics with cuGraph. 10. Latent semantic indexing for text via singular value decomposition(cuML). 11. Accelerating Workloads using RAPIDS 12. Introduction to DL Frameworks: PyTorch, and Tensorflow (Keras)
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. In depth understanding of Data Analytics, Pre-processing and Visualization Toolkits. 2. Comprehend and apply different classification and clustering techniques. 3. Understand GPU computing for building advanced data science applications. 4. Understand the concept of Neural Networks and its implementation using deep learning frameworks.
Text Books 1. Mitchell M., T., <i>Machine Learning</i>, McGraw Hill (1997) 1st Edition. 2. Alpaydin E., <i>Introduction to Machine Learning</i>, MIT Press (2014) 3rd Edition. 3. Vijayvargia Abhishek, <i>Machine Learning with Python</i>, BPB Publication (2018).
Reference Books: 1. Bishop M., C., <i>Pattern Recognition and Machine Learning</i>, Springer-Verlag (2011) 2nd Edition. 2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., <i>Machine Learning, Neural and Statistical Classification</i>. Overseas Press (1994).

UCS664: Conversational AI: Natural Language Processing				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course provides a broad introduction to deep learning and natural language processing. It offers some of the most cost-effective approaches to automated knowledge acquisition in the emerging field of natural language understanding using deep learning and GPU Computing.				
Syllabus Introduction: Natural Language Processing and its Applications, Introduction to Deep Learning, NVIDIA Toolkits, SDKs and platforms for Deployment. Introduction to Natural Language Processing: What is NLP, Principles and Traditional Methods, Linguistics, Why Machine Learning and Deep Learning. Introduction to NLP using Deep Learning: Embeddings: Feedforward NN, Word2Vec, GloVE, Contextualization (ELMo etc.), Deep Recurrent Models: RNNs, GRUs, LSTMs. Advanced NLP using Deep Learning: Introduction to NeMo, Self-Attention, Transformer Networks: BERT and its Variants, Megatron etc, Working with open source datasets: GLUE Benchmarks. Applications of NLP: Exploring NLP Problem Statements- Information Retrieval, Intent Slot Filling, Machine Translation, Punctuation & Capitalization, Question and Answering Machine Machine, Relation Extraction, Sentiment Analysis, Token Classification in NeMo. Introduction to NVIDIA Toolkits and SDKs: Transfer Learning Toolkit, TensorRT Optimization, Triton Inference Server for Inferencing and Deployments, Various Visualization Tools, Kubernetes Deployment.				
Laboratory Work: •Introduction to DL Frameworks: TLT, PyTorch, and Tensorflow (Keras). •Binary Classification with Perceptron and Logistic Regression. •Neural Modules (NeMo) for Training Conv AI Models, Exploring NeMo Fundamentals, Exploring NeMo Model Construction, Nemo Swap App Demo. •Sentiment Analysis & Text Classification with NeMo. •Intent Slot Filling for ChatBot using Joint Bert Model with NeMo. •Machine Translation with NeMo. •Question & Answering Machine with NeMo. •Information Retrieval, Punctuation & Capitalization, Relation Extraction, Sentiment Analysis, Token Classification with NeMo. •Hands-on practical on TensorRT Optimization, Triton Inference Server.				
Course Learning Objectives (CLO)				

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After the completion of the course the student will be able to:

1. Understand the concept of Neural Networks and its implementation in the context of Machine Learning.
2. Comprehend and apply recurrent neural networks on various NLP applications.
3. Understand the concept of basic transformer networks and its variants.
4. Apply transformer-based networks and its variants for NLP applications like text classification, question-answering and machine translation systems.

Text Books

1. Schmidhuber, J. (2015). "Deep Learning in Neural Networks: An Overview". *Neural Networks* 61: 85-117.
2. Bengio, Y., LeCun, Y., and Hinton, G. (2015). "Deep Learning". *Nature* 521: 436-44.
3. Allen, James, *Natural Language Understanding, Second Edition*, Benjamin/Cumming, 1995.
4. Bengio, Y., Courville, A., and Vincent, P. (2013). "Representation learning: A review and new perspectives", *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35 (8): 1798-1828.
5. Deep Natural Language Processing course offered at the University of Oxford: <https://github.com/oxford-cs-deepnlp-2017/lectures>
6. "The Unreasonable Effectiveness of Recurrent Neural Networks" by Andrej Karpathy: <https://karpathy.github.io/2015/05/21/rnn-effectiveness/>
7. Manning, Christopher and Heinrich, Schutze, *Foundations of Statistical Natural Language Processing*, MIT Press, 1999.

Reference Books:

1. Arel, I., Rose, D.C., and Karnowski, T.P. (2010). "Deep Machine Learning - A New Frontier in Artificial Intelligence Research". *IEEE*.
2. *Computational Intelligence Magazine* 5 (4): 13-18. Bengio, Y. (2009) "Learning deep architectures for AI". *Foundations and trends in Machine Learning* 2 (1): 1-127.
3. Radford, Andrew et. al., *Linguistics, An Introduction*, Cambridge University Press, 1999.
4. Christopher D. Manning and Hinrich Schütze. 1999. *Foundations of Statistical Natural Language Processing*. MIT Press.
5. "Understanding LSTM Networks" by Christopher Olah: <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

UCS749: Conversational AI: Speech Processing & Synthesis				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with the overall structure of the Conversational AI pipeline including Speech Processing, Recognition, and Synthesis and building end to end workflows using NeMo and Jarvis SDK.				
Syllabus Introduction: Fundamentals of Speech Processing, Applications of Speech Processing and Deploying NLP, ASR and TTS modules in Jarvis. Fundamentals of Speech Processing: Introduction to Statistical Speech Processing, HMMs for Acoustic Modeling, WFTS for Automatic Speech Recognition (ASR), Basics of Speech Production, Tied State HMMs, Introduction to NNs in Acoustic Modeling (Hybrid/TDNN/Tandem). {Papers} Automatics Speech Recognition (ASR): ASR - DNN models (Jasper, QuartzNet, Citrinet, Conformer-CTC), Open-source Datasets, Language Modelling: N-Gram, Neural Rescoring. {Survey , Jasper, QuartzNet, CitriNet , Nemo} Applications of Speech Processing: Speech Commands: Speech Commands Recognition using MatchboxNet. Overview of Noise Augmentation, Voice Activity Recognition and Speaker Recognition. {Survey, Nemo} Speech Synthesis: Text Normalization: Preparing Dataset and Text Normalization for input to Speech Synthesis model. Introduction to Text-to-Speech (TTS) Models:- Mel Spectrogram Generator: - Tacotron-2, Glow-TTS, Audio Generators:- WaveGlow, SqueezeWave. {Papers, Nemo} Jarvis Deployment: Introduction to Jarvis, Overview of Jarvis ASR, NLU and TTS APIs, Introduction to Jarvis Dialog Manager. Jarvis Deployment:- Nemo model deployment for ASR, NLP and TTS.				
Laboratory Work: • Practical Exercise on Statistical Speech Processing. {Traditional Signal Processing} • Automatic Speech recognition with NeMo on English Dataset. • Automatic Speech recognition with NeMo on Indic Language(Hindi) Dataset. • NeMo Speech Commands Recognition using MatchboxNet, Noise Augmentation, and Speaker Recognition. • Text to Speech using Tacotron-2 and WaveGlow with NeMo on English Dataset. • Text to Speech using Tacotron-2 and WaveGlow with NeMo on Indic Language (Hindi) Dataset. • End-to-End Conversational AI Model (Any Language): ASR/NLP/TTS with NeMo and Jarvis.				

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Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Understand Speech Processing pipeline for various applications with accelerated computing. ASR, Speaker Recognition etc.
2. Exploring Speech Synthesis on various data sets.
3. Deep practical hands-on experience from training to deployment of these applications using NVIDIA GPUs and Toolkits:- NeMo, and Jarvis.

Text Books

1. Jurafsky, Dan and Martin, James, *Speech and Language Processing, Second Edition*, Prentice Hall, 2008.
2. Daniel Jurafsky and James H. Martin, "*Speech and Language Processing*", 3rd edition draft, 2019 [JM-2019].
3. Mark Gales and Steve Young, *The application of hidden Markov models in speech recognition*, *Foundations and Trends in Signal Processing*, 1(3):195-304, 2008.

Reference Books:

1. Daniel Jurafsky and James H. Martin. 2009. *Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics*. 2nd edition. Prentice-Hall.
2. Geoffrey Hinton, Li Deng, Dong Yu, George E. Dahl, Abdel-rahman Mohamed, Navdeep Jaitly, Andrew Senior, Vincent Vanhoucke, Patrick Nguyen, Tara N. Sainath, and Brian Kingsbury, *Deep Neural Networks for Acoustic Modeling in Speech Recognition*, *IEEE Signal Processing Magazine*, 29(6):82-97, 2012.

UCS748: Generative AI				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course introduces students to the field of generative artificial intelligence with a focus on Large Language Models (LLMs). Students will learn the theoretical foundations behind LLMs and gain hands-on experience in training and fine-tuning these models for various generative tasks such as text generation, image generation, and more.				
Syllabus Introduction to Generative AI: Generative AI: Meaning, Capabilities and Potential, Applications of Generative AI, Tools for text, images, videos, audios and code generation. Generative AI Models: Introduction to Large Language Models (LLMs), History and evolution of LLMs, Transformer architecture: Attention mechanism, Pre-training and fine-tuning of LLMs, Language modeling objectives (e.g., Masked Language Modeling, Next Sentence Prediction), Data preprocessing for LLMs, Training strategies and best practices, Fine-tuning on custom datasets, Handling domain-specific data and tasks. Text Generation with LLMs: Text generation techniques, Conditional text generation, Sampling strategies, Evaluation metrics for text generation Image Generation with LLMs: Overview of image generation tasks, Generative Adversarial Networks (GANs) vs. LLMs for image generation, Fine-tuning LLMs for image generation, Evaluation metrics for image generation Beyond Text and Images: Multi-Modal Generation: Introduction to multi-modal generation, Combining LLMs with other generative models, Applications of multi-modal generation. Prompt Engineering: Meaning of Prompts and Prompt Engineering, Text-to-text prompt techniques: Interview Pattern Approach, Chain-of-Thought Approach, Tree-of-Thought Approach.				
Laboratory Work: 1. Pre-train a small language model on a text corpus and fine-tune it on a specific task or dataset 2. Fine-tune pre-trained language models on custom datasets for specific tasks like sentiment analysis or text classification. 3. Implement autoregressive decoding to generate text using a pre-trained language model and explore its limitations. 4. Build a conditional text generation model that takes input prompts or contexts to generate relevant responses. 5. Build a basic GAN architecture using a deep learning framework like TensorFlow or PyTorch. 6. Explore techniques to fine-tune pre-trained language models like GPT for image				

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generation tasks using frameworks like CLIP. 7. Experiment with combining LLMs like GPT with other generative models like GANs or Variational Autoencoders (VAEs) for multi-modal generation tasks. 8. Implement a multi-modal generation model that generates coherent captions for given images or generates images from textual descriptions. 9. Explore real-world applications of multi-modal generation such as image captioning, visual question answering (VQA), and generating visual explanations from textual input.
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Understand the theoretical foundations of Large Language Models (LLMs). 2. Apply LLMs for various generative tasks including text generation, image generation, and more. 3. Train and fine-tune LLMs on custom datasets. 4. Explore advanced topics and applications of LLMs in research and industry.
Text Books 1. <i>"Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville</i> 2. <i>"Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play" by David Foster.</i> 3. <i>"Dive into Deep Learning" by Aston Zhang.</i>
Reference Books: 1. <i>"Attention is All You Need" by Ashish Vaswani et al.</i> 2. <i>"Natural Language Processing with PyTorch" by Delip Rao and Brian McMahan</i> 3. <i>"GPT-3: Language Models are Few-Shot Learners" by Brown et al.</i>

UCS547: Edge AI and Robotics: Accelerated Data Science				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with fundamental knowledge of GPU Computing and Machine Learning and Deep Learning Primer.				
Syllabus Introduction: GPU Computing, Parallel Programming, Machine Learning with RAPIDS, Model Compression. Image Processing and Parallel Programming: GPU Programming, CUDA C/C++/Python, Accelerated Image Processing, nvJPEG, Numba. Introduction to Machine Learning - Supervised: Introduction to Supervised Learning, Linear Model, RAPIDS acceleration: Linear Regression, Overfitting and Cross Validation, Decision Tree, Visualizing Classification: {ROC, AUC, Confusion Matrix}, Bagging, Random Forests, RAPIDS Acceleration: Random Forest, Boosting, RAPIDS acceleration: K- NN, XGBoost. Deep Learning Model Compression : Introduction to model pruning, quantization and distillation Optimization Framework: Using TensorRT optimization, Deploying model on Triton Inference server.				
Laboratory Work: ● Practical on Traditional Data Science packages (Numpy, Pandas, Scipy, Scikit-Learn). ● CUDA C/C++ for Accelerated Computing. {DLI Online Course Section: Fundamentals of Accelerated Computing with CUDA C/C++} ● Numba to compile CUDA kernels for Numpy Acceleration in Python. {DLI Online Course Section: Fundamentals of Accelerated Computing with CUDA Python} ● Getting started with Accelerated Data Science with RAPIDS AI (cuPy, cuDF, cuSignal, cuML). ● Decision Tree Classification Clustering in RAPIDS. ● Random Forest Classification in RAPIDS. {DLI Online Course Section: Fundamentals of Accelerated Data Science with RAPIDS, Section 2: GPU-accelerated Machine Learning} Model Pruning, Post Training Quantization, Quantization Aware Training, Distillation				
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Demonstrate ability to deploy GPU accelerated code for image processing applications.				

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2. Apply model compression techniques while deploying deep learning architectures.
3. Implement Rapids AI framework for different machine learning tasks. .
4. Analyze and evaluate performance of deep learning-based inference models using TensorRT optimization and Triton Server .

Text Books

1. Mitchell M., T., *Machine Learning*, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., *Introduction to Machine Learning*, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, *Machine Learning with Python*, BPB Publication (2018)..

Reference Books:

1. Bishop M. C., *Pattern Recognition and Machine Learning*, Springer-Verlag (2011) 2nd Edition.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., *Machine Learning, Neural and Statistical Classification*. Overseas Press (1994).

UCS668: Edge AI and Robotics: Data Centre Vision				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with basic fundamental understanding and practical hands-on training of computer vision and deep learning models on data centre GPU servers.				
Syllabus Introduction: Introduction to Deep Learning, Formulating Computer Vision Problem Statements, Image Classification using CNN Architectures like VGG, Inception, ResNet(18/34/50/152). Working towards building Object detection and Segmentation pipelines. Introduction to Deep Learning: Introduction to Advanced Deep Neural Networks (DNNs), Batch Normalization, Hyperparameter tuning, Activation Functions, Metrics, Optimization, Regularization. Applications of Computer Vision (Image Classification): Introduction to NVIDIA Frameworks: {Transfer Learning using Transfer Learning Toolkit (TLT), Mixed Precision, DALI}, Image Classification using Deep CNN Architecture like VGG, ResNet18/34/50, re- training on custom dataset. Applications of Computer Vision (Object Detection & Segmentation): Introduction to Object Detection, Data Preprocessing, CNN Architecture like {SSD, YOLOv3, EfficientDet, Spinenet}, Metrics, Loss Functions, re-training on custom dataset, Segmentation: FCN- ResNet, Unet, MaskRCNN, Metrics and Loss functions. Graph Neural Network and Synthetic Data Generation: Introduction to Graph Neural Networks, Omniverse Replicator based synthetic data generation (SDG) using 3D assets.				
Laboratory Work: • Introduction to DL Frameworks: TLT, PyTorch, and Tensorflow (Keras). {DLI Online Course: Getting Started with Deep Learning} • {DLI Online Course: Deep Learning at Scale with Horovod} • Training Classification Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets. • Training Detection Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets. • Training Segmentation Models with and without Mixed Precision and Multi-GPU on Open & Custom Datasets. {DLI Online Course: Getting Started with Image Segmentation} • {DLI Online Course: Synthetic Data Generation for Training Computer Vision Models} • {DLI Online Course: Introduction to Graph Neural Networks}				
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Apply computer vision based techniques for problem solving in various domains 2. Demonstrate deployment of deep learning models, trained with mixed precision.				

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3. Analyze and evaluate performance of models for classification, detection and segmentation tasks.
4. Implementation synthetic image generation using generative models.

Text Books

1. Mitchell M., T., *Machine Learning*, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., *Introduction to Machine Learning*, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, *Machine Learning with Python*, BPB Publication (2018).
4. *Computer Vision: Algorithms and Applications*, R. Szeliski, Springer, 2011.
5. *Computer Vision: A Modern Approach*, D. Forsyth and J. Ponce, Prentice Hall, 2nd ed., 2011.

Reference Books:

1. Bishop M. C., *Pattern Recognition and Machine Learning*, Springer-Verlag (2011) 2nd Edition.
2. *Introductory techniques for 3D computer vision*, E. Trucco and A. Verri, Prentice Hall, 1998.
3. "Visualizing and Understanding Convolutional Networks" by Matthew D. Zeiler and Rob Fergus (2014)
4. "Convolutional Neural Networks for Visual Recognition" (Stanford course given by Fei-Fei Li, Andrej Karpathy, and Justin Johnson, 2016): <http://cs231n.github.io/>

UCS671: Edge AI and Robotics: Embedded Vision				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with advanced conceptual knowledge and practicals on various computer vision and deep learning applications and provide the overall environment for end-to-end pipeline development from data collection to deployment.				
Syllabus Introduction: Utilizing Jetpack SDK and other NVIDIA Toolkits to deploy CNN models on Jetson, Creating Jetbot kits and deploying various applications, Working with NVIDIA Robotics toolkit: Isaac SIM SDK and Gazebo for collision avoidance, path following. Introduction to Edge AI: AI at the Edge & IoT, Jetson Architecture, Getting Started with Jetpack, NGC Containers in Jetson, Getting started with NGC & Containers on Jetson. Introduction to NVIDIA Toolkits and SDKs: Transfer Learning Toolkit, Kubernetes Deployment, Deepstream SDK, Deploying Classification, Detection and Segmentation CNN Models on Jetson Devices. Perception & Autonomous Navigation: Building JetBot Kits, Introduction to basic motion on JetBot, Collision Avoidance: Stop/Go classifier (JetBot), freespace detection, Path Following: Recording user input/video + DNN training (DriveNet), Simulation: Gazebo & Isaac SIM. Advanced Vision & SLAM: Pose Recognition (Deploying Human pose model), Depth Estimation: Mono/Stereo depth and point extraction, Visual Odometry: Camera pose estimation from DNNs, SLAM on JetBot.				
Laboratory Work: • Setting up the Jetson Project kit. {DLI Online Course: Getting Started with AI on Jetson Nano.} • Deployment of Various Classification, Object Detection and Segmentation models in Jetson Nano. • Getting started building various Jetbot Kits. • Basic Motion with Jetbot • Collision Avoidance with Jetbot kit • Object following and Road following (DriveNet) with Jetbot. • Teleoperation with Jetbot. • Human Pose Estimation in Jetson Nano/JetBot. • Implementing SLAM on Jetbot.				
Course Learning Objectives (CLO)				

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After the completion of the course the student will be able to:

1. Demonstrate deployment of realtime vision based applications on edge devices
2. Implement kubernetes framework using Jetson Nano Devices.
3. Apply vision based pose estimation techniques.
4. Implement SLAM using Jetbot kits and vision sensors.

Text Books

1. *Computer Vision: Algorithms and Applications*, R. Szeliski, Springer, 2011.
2. *Computer Vision: A Modern Approach*, D. Forsyth and J. Ponce, Prentice Hall, 2nd ed., 2011.
3. Richard Szeliski, *Computer Vision: Algorithms and Applications*, Springer-Verlag London Limited 2011

Reference Books:

1. Richard Hartley and Andrew Zisserman, *Multiple View Geometry in Computer Vision*, Second Edition, Cambridge University Press, March 2004.
2. K. Fukunaga; *Introduction to Statistical Pattern Recognition*, Second Edition, Academic Press, Morgan Kaufmann, 1990.
3. R.C. Gonzalez and R.E. Woods, *Digital Image Processing*, Addison- Wesley, 1992.
4. Christopher M. Bishop; *Pattern Recognition and Machine Learning*, Springer, 2006.

UCS760: Edge AI and Robotics: Reinforcement Learning & Conversational AI				
	L	T	P	Cr
	2	0	2	3
Course Objective: This course will provide students with introduction to the basic mathematical foundations of Reinforcement Learning for building real world computer vision applications, and Conversational AI for developing Chatbots.				
Syllabus Introduction: GPU Computing, Implementing Behaviours of Robots such as Manipulation, and Task Learning, Fundamentals of Reinforcement Learning for Vision and Deploying Conversational AI pipelines in JetsonI. Manipulation: Overview of Manipulation in Robotics, Inverse Kinematics and Control, Gripping & Task Learning. Reinforcement Learning: Introduction to RL: RL agents, Dynamic Programming, Monte Carlo's and Temporal-Difference Methods, OpenAI Gym, RL in Continuous Spaces. {Added Lectures, Summaries} Conversational AI (NLP): Natural Language Processing: Introduction to NLP, BERT, Megatron, Applications of NLP: Information Retrieval, Intent Slot Filling, Machine Translation, Punctuation & Capitalization, Question and Answering Machine Machine, Relation Extraction, Sentiment Analysis, Token Classification in NeMo. Conversational AI (Speech Processing): Automated Speech Recognition: Introduction to ASR, Architectures: Jasper/QuartzNet/CitriNet, Text to Speech: TTS-Tacotron2/WaveGlow and Jarvis Deployment.				
Laboratory Work: ● Manipulation Lab: Building Pick-n-place. ● Manipulation Lab: Object Assembly. ● Game Agent: Open AI Gym (Jetbot in simulation). ● Conversational AI VoiceBot: Verbal JetBot commands/feedback, ect (optional mic/speaker).				
Course Learning Objectives (CLO) After the completion of the course the student will be able to: 1. Apply reinforcement learning for robot navigation. 2. Deploy computer vision models in various simulation environments. 3. Implement applications with conversational AI 4. Utilize programming and AI training & deployment tools for relevant model building in both edge hardware devices and simulation environments.				
Text Books 1. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation,				

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learning, and optimization 12 (2012): 3.

2. Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach." Pearson Education Limited, 2016.

3. Jurafsky, Dan and Martin, James, *Speech and Language Processing*, Second Edition, Prentice Hall, 2008.

4. Daniel Jurafsky and James H. Martin, "Speech and Language Processing", 3rd edition draft, 2019 [JM-2019].

Reference Books:

1. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning." MIT press, 2016.

2. Mark Gales and Steve Young, *The application of hidden Markov models in speech recognition*, *Foundations and Trends in Signal Processing*, 1(3):195-304, 2008.

3. Daniel Jurafsky and James H. Martin. 2009. *Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics*. 2nd edition. Prentice-Hall.

4. "Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto: <https://webdocs.cs.ualberta.ca/~sutton/book/the-book-2nd.html>

5. David Silver's course: <http://www0.cs.ucl.ac.uk/staff/d.silver/web/Teaching.html>

6. "Deep Reinforcement Learning: Pong from Pixels" by Andrej Karpathy: <https://karpathy.github.io/2016/05/31/rl/>

7. Talks on Deep Reinforcement Learning by John Schulman: https://www.youtube.com/watch?v=aUrX-rP_ss4 , and his Deep Reinforcement Learning course <http://rll.berkeley.edu/deeprlcourse/>

Volvo Specific Electives

Course Code UEC646	Network and Communications for Connected Vehicles	L	T	P	C
		3	0	0	3
Course Objectives 1. To attain a comprehensive overview of connected vehicles. 2. To understand the intra- and inter-vehicle communication. 3. Acquiring knowledge on VANET. 4. Able to tackle current problems in connected vehicle technology					
Syllabus					
Module 1	Introduction to Connected Vehicles				
Cyber Physical Systems in Transportation, Architecture and Components of Connected Cars, On-board Sensors and Platforms					
Module 2	Vehicle Communication Technologies				
Intra-Vehicle: LIN, CAN, FlexRay. Inter-Vehicle: Wi-Fi, GPS, GPRS, 5G. Ethernet Standards: Automotive Ethernet, SOME-IP, DO-IP					
Module 3	Network Architectures				
Overview of the Architecture- channel models- properties of V2V and V2I communication- performance of 802.11 in V2X.					
Module 4	VANET				
Vehicular AD hoc network operations- single-hop broadcasting and multi-hop broadcasting- mobile IP solution in VANET.					
Module 5	V2V Application				
Emergency vehicle approach, Blind spot lane change warning, Rear end collision warning, Emergency brake light, and Intersection movement assist.					
Module 6	V2I Application				
Traffic information system- Curve speed warning system- work zone warning system- weather impact warning system and Platoon detection.					
Module 7	Cyber Security Risk Management and Standards				
Secure Vehicular Communication Architecture- Secure and Privacy-enhancing Vehicular Communication- Data Trustworthiness- Future Challenges. SECOC for CAN protocol- MACSEC and TLS for Ethernet.					
Course Learning Outcomes 1. Understand the fundamentals of connected vehicles. 2. Comprehend network architectures, including intra-vehicle and inter-vehicle networks. 3. Familiarize with the VANET networking and V2X applications 4. Understand the Security and privacy-related issues of vehicular networks.					
Text Book(s) 1. Christoph Sommer, Falko Dressler, <i>Vehicular Networking</i> , Cambridge University Press, 2015. 2. Markus Meuck, Ingolf Karls, <i>Networking Vehicles to Everything</i> , DE Gruyter Press, 2018 3. Dominique Paret, Hassina Rebaine “ <i>Autonomous and Connected Vehicles Network Architectures from Legacy Networks to Automotive Ethernet</i> ” 2022 Wiley					
Reference Books 1. Peter Han Joo Chong “ <i>Vehicular Networks Applications, Performance Analysis and Challenges</i> ”, 2019, Nova Science Publishers.					

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2. Guojun Wang, Md Tabrez Nafis, Muhammad Arif, Mazin Abed Mohammed “*Vehicular Ad Hoc Networks Futuristic Technologies for Interactive Modelling, Dimensioning, and Optimization*” 2022, CRC Press
3. Jiajia Liu, Abderrahim Benslimane ·” *Intelligent and Connected Vehicle Security*”, 2022 River Publishers.

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